Notes on Lie groups and Lie algebras

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1 Matrix Group

1.1 Basics of group, ring and field

Let's first review the basics of group, ring and field.

• *Group*: a group is a set G with a binary operation "$\cdot$": $G \times G \to G$, such that the following conditions are satisfied:

1. Associativity: for all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. There exists a unique *identity element* $e \in G$, such that for any $a \in G$, $a \cdot e = e \cdot a = a$
3. For any $a \in G$, there exists a unique *inverse element* a^{-1} of a, such that $a \cdot a^{-1} = a^{-1} \cdot a = e$

• *Abelian group*: for all $a, b \in G$, they *commute*: $a \cdot b = b \cdot a$.

If there exists $a, b \in G$, they do not commute, then G is a *non-abelian* group.

• "Additive group": where "$\cdot$" is written as "$+$". "Multiplicative group": where "$\cdot$" is a written as a multiplication. They are essentially equivalent. We also often omit the "$\cdot$" in the group multiplication, and write gh for $g \cdot h$.

• *Subgroup* H of G: a subset $H \subset G$ such that the group operation $\cdot$ is also a group operation on H.

• *Normal subgroup* N of G: a subgroup that is invariant under the conjugation by the elements of G. For any $h \in N$, $g \in G$, $g^{-1}hg \in N$.

• *Quotient group* G/N, where N is a normal subgroup of G. The element of the set G/N are *cosets* $aN = \{a \cdot h | h \in N\}$. The group operation $G/N \times G/N \to G/N$ is defined as $(a_1 N) \cdot (a_2 N) = (a_1 \cdot a_2)N$.

• *Center* H of G is the set of elements that commute with every element in G. It is by definition a normal subgroup of G.

• *Direct product group* (Product group) $G \times H$: the group elements are (g, h), $g \in G$, $h \in H$. The group operation is $(g, h) \cdot (g', h') = (g \cdot g', h \cdot h')$.

• *Group homomorphism* $\phi : G \to H$ is a map satisfying

$$\phi(g) \cdot \phi(h) = \phi(g \cdot h)\,, \quad (\forall g, h \in G)\,. \tag{1.1}$$

From the above equation one can also derive $\phi(e_G) = e_H$, $\phi(g^{-1}) = \phi(g)^{-1}$.

ϕ is a *group isomorphism* if the inverse map $\phi^{-1} : H \to G$ is also a group homomorphism.

Remark: the subgroup statement $H \subset G$ is hold up to isomorphism of H. The statement $H \subset G$ is equivalent to: there exists an injective homomorphism $\phi : H \to G$.

• *Group action* of G on a set X: a map $\cdot : G \times X \to X$, such that for any $a, b \in G$, $x \in X$, (1) $e \cdot x = x$; (2) $a \cdot (b \cdot x) = (ab) \cdot x$.

• For a *finite group*, the *order* of group G, $|G|$, is defined as the total number of elements in the set G.

Examples: (1) Multiplicative group $\mathbb{Z}_N$: identity element $e = 1$, $\mathbb{Z}_N$ generated by the element a, $a^N = 1$. The group elements are

$$\{1, a, a^2, \ldots, a^{N-1}\}, \tag{1.2}$$

and the order of group is $|\mathbb{Z}_N| = N$. The group operation is $a^p \cdot a^q = a^{p+q \bmod N}$.

Let us apply the concepts in group theory to the case of $\mathbb{Z}_N$.

a) $\mathbb{Z}_N$ is an abelian group.

b) $\mathbb{Z}_N$ is the center of itself.

c) If $N = m \cdot n$, $\mathbb{Z}_m$ and $\mathbb{Z}_n$ are subgroups of $\mathbb{Z}_N = \mathbb{Z}_{mn}$. Denote by a_m the generator of $\mathbb{Z}_m$, and a_{mn} the generator of $\mathbb{Z}_{mn}$, there is an injective group homomorphism $\phi : \mathbb{Z}_m \to \mathbb{Z}_{mn}$ as $\phi(a_m) = a_{mn}^n$.

d) If $\gcd(m, n) = 1$, $\mathbb{Z}_{mn}$ is isomorphic to $\mathbb{Z}_m \times \mathbb{Z}_n$, e. g. $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$. Otherwise they are not isomorphic, e. g. $\mathbb{Z}_4 \neq \mathbb{Z}_2 \times \mathbb{Z}_2$.

(2) Permutation group S_n, whose elements are the permutations

$$\begin{pmatrix} 1 & 2 & 3 & \ldots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \ldots & \sigma(n) \end{pmatrix} \tag{1.3}$$

The order of S_n is $n!$. When $(n \geq 3)$, S_n is a non-abelian group, and the center of S_n is trivial. When $n = 2$, $S_2 = \mathbb{Z}_2$.

(3) Additive group $\mathbb{Z}$ of integers: the identity element $e = 0$, generated by the element 1. The group elements are $n \in \mathbb{Z}$. The group operation is given naturally by the addition of integers. $\mathbb{Z}$ is a discrete group but it is not a finite group.

(4) Additive group $\mathbb{Q}$ of rational numbers: it is a continuous group but not a Lie group.

(5) Additive group $\mathbb{R}$ of real numbers: it is a Lie group.

• *Ring*: a ring is a set R with a binary operation $\cdot$: $R \times R \to R$ and a binary operation $+$: $R \times R \to R$, such that

1. R is an abelian group under $+$, with a unique additive identity element 0. For all $a, b, c \in R$,

   (1) $(a + b) + c = a + (b + c)$

   (2) $a + b = b + a$

   (3) $a + 0 = a$

   (4) There exists a unique $(-a) \in R$ such that $a + (-a) = 0$.

2. Under multiplication $\cdot$, there exists a unique multiplicative identity element 1. For all $a, b, c \in R$,

   (1) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

   (2) $a \cdot 1 = 1 \cdot a = a$

3. The distribution law holds:

   (1) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

   (2) $(b + c) \cdot a = (b \cdot a) + (c \cdot a)$

Examples: (1) Ring of integers with the natural definitions of 0, 1, + and $\cdot$

(2) The ring of $n \times n$ matrices M_n, where the additive identity element is $0_{n\times n}$ and the multiplicative identity element is I. The addition is the matrix addition $+$ and the multiplication $\cdot$ is the matrix multiplication.

We also write $M_n(\mathbb{R})$ and $M_n(\mathbb{C})$ for $n \times n$ matrices with real and complex elements.

• *Field*: a ring F where the multiplication $\cdot$ also has the structure of an abelian group. For any $a \neq 0 \in F$, exists a unique element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.

Examples: (1) The field of real numbers $\mathbb{R}$

(2) The field of complex numbers $\mathbb{C}$

(3) The field of rational numbers $\mathbb{Q}$.

1.2 Matrix groups

In this course, we are going to study *continuous groups* G, such that G is a continuous topological space and the group operation $\cdot$ is a continuous map $G \times G \to G$. The precise definition of these will be introduced in section 2.

An extremely useful class of continuous groups in physics is the *matrix group* of $n \times n$ invertible matrices.

The (real) *dimension* of a matrix group G, $\dim(G)$, is defined as the total number of independent (continuous) real parameters in these matrices.

The group operation $\cdot : G \times G \to G$ is the matrix multiplication, while the identity element is the identity matrix I.

We can naturally define an action of a matrix group G of $n \times n$ matrices M on a n-dimensional vector space V using the multiplication $M \cdot v := Mv$. $v \in V$ is a n-dimensional column vector.

We will introduce the following notations of matrix groups that will be used through out the course, note that F is in general a field.

1. $GL(n, F)$: the *general linear group* of invertible $n \times n$ matrices M whose elements $\in F$. Note that $\det(M) \neq 0$. Examples:

   (1) $GL(n, \mathbb{R})$: the group of invertible $n \times n$ real matrices, in total n^2 independent real parameters, hence

   $$\dim(GL(n, \mathbb{R})) = n^2 \,. \tag{1.4}$$

   (2) $GL(n, \mathbb{C})$: the group of invertible $n \times n$ complex matrices, in total $2n^2$ independent real parameters, hence

   $$\dim(GL(n, \mathbb{C})) = 2n^2 \,. \tag{1.5}$$

   • Comments: all other matrix groups are subgroups of a general linear group.

   • The condition $\det(M) \neq 0$ does not reduce the total number of real parameters.

2. $SL(n, F)$: the *special linear group* of $n \times n$ matrices M whose elements $\in F$, with the additional condition that $\det(M) = 1$. Examples:

   (1) $SL(n, \mathbb{R}) \subset GL(n, \mathbb{R})$: the group of $n \times n$ real matrices with $\det(M) = 1$, in total $n^2 - 1$ independent parameters, hence

   $$\dim(SL(n, \mathbb{R})) = n^2 - 1 \,. \tag{1.6}$$

(2) $SL(n,\mathbb{C}) \subset GL(n,\mathbb{C})$: the group of $n \times n$ complex matrices with $\det(M) = 1$, in total $2n^2 - 2$ independent parameters, hence

$$\dim(SL(n,\mathbb{C})) = 2n^2 - 2\,. \tag{1.7}$$

• Comment: one can relax the condition that F is a field in this case. $GL(n,\mathbb{Z})$ and $SL(n,\mathbb{Z})$ also exists as a matrix group, despite being infinite discrete groups. They are 0-dimensional in the language of Lie groups. The case of $n = 2$ corresponds to the "modular group" of T^2.

3. $U(n) \subset GL(n,\mathbb{C})$: the *unitary group* of unitary $n \times n$ complex matrices M, $M^\dagger M = I$. $2n^2$ real parameters + n^2 constraints.

$$\dim(U(n)) = n^2\,. \tag{1.8}$$

Naively, there are $2n^2$ real parameters and $2n^2$ equations. However, the constraint $M^\dagger M - I = 0$ itself satisfy a condition:

$$(M^\dagger M - I)^\dagger = M^\dagger M - I\,. \tag{1.9}$$

Hence the number of independent constraints on real parameters is n^2.

• Example: $U(1)$ is one of the simplest matrix groups (Lie groups), which consists of elements $\{e^{i\theta}\}$ (or simply $e^{i\theta}$), ($\theta \in [0, 2\pi)$). $U(1)$ has the topology of a circle S^1.

• Note that $M \in U(n)$ satisfies $|\det(M)| = 1$, because $|\det(M)| \cdot |\det(M^\dagger)| = 1$, $|\det(M)| = |\det(M^\dagger)| > 0$.

4. $SU(n) \subset U(n)$: the *special unitary group* of unitary $n \times n$ complex matrices M with $M^\dagger M = I$ and $\det(M) = 1$.

$$\dim(SU(n)) = n^2 - 1\,. \tag{1.10}$$

Note that the dimension of $SU(n)$ is less than the dimension of $U(n)$ by one. This is because for a unitary matrix M, we already have

$$\det(M) = e^{i\theta}\ ,\ \theta \in \mathbb{R}\,, \tag{1.11}$$

the condition $\det(M) = 1$ only removed one free parameter.

• $U(n)$ and $SU(n)$ have a natural action on the vector space $\mathbb{C}^n$ of n-dimensional vectors $(x_1, x_2, \ldots, x_n)^T$, which preserves the inner product

$$\langle x, y\rangle = x^\dagger y = x_1^* y_1 + x_2^* y_2 + \cdots + x_n^* y_n\,. \tag{1.12}$$

As one can check, after an $U(n)$ or $SU(n)$ transformation $x' = Mx$, $y' = My$, the new inner product is

$$\langle x', y' \rangle = (x')^\dagger y' = x^\dagger M^\dagger M y = \langle x, y \rangle \,. \tag{1.13}$$

• Examples: (1) $SU(2)$ is a 3-dimensional group whose elements can be parametrized as

$$\begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix} \quad , \ (a^2+b^2+c^2+d^2=1), \ (a,b,c,d \in \mathbb{R}) \tag{1.14}$$

Hence $SU(2)$ has the geometry of a 3-dimensional sphere $S^3 \subset \mathbb{R}^4$.

One can also parametrize it using three parameters (α, β, γ) of S^3:

$$a = \cos\alpha \ , \ b = \sin\alpha\cos\beta \ , \ c = \sin\alpha\sin\beta\cos\gamma \ , \ d = \sin\alpha\sin\beta\sin\gamma \,. \tag{1.15}$$

The 2×2 matrices are

$$\begin{pmatrix} \cos\alpha + i\sin\alpha\cos\beta & \sin\alpha\sin\beta\cos\gamma + i\sin\alpha\sin\beta\sin\gamma \\ -\sin\alpha\sin\beta\cos\gamma + i\sin\alpha\sin\beta\sin\gamma & \cos\alpha - i\sin\alpha\cos\beta \end{pmatrix} \tag{1.16}$$

$SU(2)$ is the gauge group of weak interaction before electroweak symmetry breaking.

(2) $SU(3)$ is an 8-dimensional group, which is the gauge group of QCD.

5. $U(p,q)$ and $SU(p,q)$: the *indefinite unitary group* $U(p,q)$ is defined as the set of $(p+q) \times (p+q)$ matrices M which preserves the metric with signature (p,q)

$$g = \mathrm{diag}(\underbrace{1,\ldots,1}_{p},\underbrace{-1,\ldots,-1}_{q}) \,, \tag{1.17}$$

$$M^\dagger g M = g \,. \tag{1.18}$$

$SU(p,q)$ is the subgroup of $U(p,q)$ with $\det(M) = 1$.

6. $O(n) \subset GL(n,\mathbb{R})$: the *orthogonal group* of real $n \times n$ orthogonal matrices M such that $M^T M = I$. There are n^2 real parameters and $\frac{1}{2}n(n+1)$ constraint equations, resulting in $\frac{1}{2}n(n-1)$ independent parameters,

$$\dim(O(n)) = \frac{1}{2}n(n-1) \,. \tag{1.19}$$

Naively there are n^2 parameters and n^2 equations, however since the constraint equations satisfy

$$(M^T M - I)^T = M^T M - I\,, \tag{1.20}$$

only $\frac{1}{2}n(n+1)$ of the equations are independent.

• Comments: (1) $O(n)$ consists of matrices M with determinant $\det(M) = \pm 1$

(2) The natural multiplicative action of $O(n)$ on the n-dimensional Euclidean space $\mathbb{R}^n$ describes the linear transformation of an n-dimensional vector $x = (x_1, x_2, \ldots, x_n)^T$ which preserves the Euclidean inner product

$$\langle x, y\rangle = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n\,. \tag{1.21}$$

As one can check, after an $O(n)$ transformation $x' = Mx$, $y' = My$, the new inner product is

$$\langle x', y'\rangle = (x')^T y' = x^T M^T M y = \langle x, y\rangle\,. \tag{1.22}$$

(3) The matrices with $\det(M) = 1$ form a subgroup of $O(n)$, which is the *special orthogonal group* $SO(n)$.

(4) The matrices with $\det(M) = -1$ do not form a subgroup of $O(n)$. We call this subset $O(n)^- \subset O(n)$. They can always be written as

$$M = RN\,, \tag{1.23}$$

where R corresponds to the reflection

$$\begin{pmatrix} -1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix} \tag{1.24}$$

and N is an element of $O(n)$ with $\det(N) = 1$ (an element of $SO(n)$).

7. $SO(n)$: the *special orthogonal group* $SO(n)$, which is the subgroup of $O(n)$ consisting of matrices with $\det(M) = 1$.

$$\dim(SO(n)) = \frac{1}{2}n(n-1)\,. \tag{1.25}$$

• Comments: (1) the natural action of $SO(n)$ on the n-dimensional Euclidean space $\mathbb{R}^n$ describes the rotation of n-dimensional vectors

(2) $SO(2)$ can be parametrized as

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad , \ \theta \in [0, 2\pi) \tag{1.26}$$

One can see that $SO(2) \cong S^1 \cong U(1)$.

(3) There is a map (group homomorphism) from $SU(2)$ to $SO(3)$, which is defined as follows. For each (a, b, c, d) in (1.14) with

$$a^2 + b^2 + c^2 + d^2 = 1\,, \tag{1.27}$$

we define an $SO(3)$ group element

$$\begin{pmatrix} 1 - 2c^2 - 2d^2 & 2bc - 2ad & 2bd + 2ac \\ 2bc + 2ad & 1 - 2b^2 - 2d^2 & 2cd - 2ab \\ 2bd - 2ac & 2cd + 2ab & 1 - 2b^2 - 2c^2\,. \end{pmatrix} \tag{1.28}$$

Since (a, b, c, d) and $(-a, -b, -c, -d)$ defines the same group element in $SO(3)$. The map from $SU(2)$ to $SO(3)$ is a 2 to 1 homomorphism. In fact $SU(2)/\mathbb{Z}_2 \cong SO(3)$.

8. $O(p, q)$ and $SO(p, q)$: the *indefinite orthogonal group* or *pseudo-orthogonal group* $O(p, q)$ consists of real $n \times n$ matrices M that preserve the metric with signature (p, q)

$$g = \mathrm{diag}(\underbrace{1, \ldots, 1}_{p}, \underbrace{-1, \ldots, -1}_{q})\,, \tag{1.29}$$

$$M^T g M = g\,. \tag{1.30}$$

$SO(p, q)$ is the subgroup of $O(p, q)$ with $\det(M) = 1$.

• For an inner product with signature (p, q), it is preserved under $x' = Mx$, $y' = My$,

$$\langle x', y' \rangle = (x')^T g y' = x^T M^T g M y = x^T g y\,. \tag{1.31}$$

• Comment: $O(1, 3)$ is the (full) Lorentz group in four space-time dimensions.

9. Euclidean group $E(n)$ or $ISO(n)$: the matrix group whose elements are

$$\left(\begin{array}{c|c} R_{n\times n} & \mathbf{a}_{n\times 1} \\ \hline 0_{1\times n} & 1 \end{array}\right) \tag{1.32}$$

Here $R_{n\times n}$ is an element of the $O(n)$ group, and $\mathbf{a}_{n\times 1} = (a_1, a_2, \ldots, a_n)^T$ is a column vector in $\mathbb{R}^n$. Its action on $(x_1, x_2, \ldots, x_n, 1)^T$ exactly describes the space translation, rotation and reflection in the Euclidean space $\mathbb{R}^n$.

$$\dim(E(n)) = \frac{1}{2}n(n+1). \tag{1.33}$$

Note that $E(n)$ is a subgroup of $GL(n+1, \mathbb{R})$.

We can also write $O(n)$ as a quotient group:

$$O(n) = E(n)/T\,, \tag{1.34}$$

where $T = \mathbb{R}^n$ is the group of space translation. As one can check, T is indeed a normal subgroup of $E(n)$. For any

$$g = \left(\begin{array}{c|c} R & b \\ \hline 0 & 1 \end{array}\right) \in E(n)\ ,\ h = \left(\begin{array}{c|c} I & a \\ \hline 0 & 1 \end{array}\right) \in T\,, \tag{1.35}$$

$$g^{-1}hg \in T\,. \tag{1.36}$$

• Similarly, the Poincaré group $ISO(1, n-1)$ is the indefinite version of $ISO(n)$, where $R_{n\times n}$ is replaced by an element of the $O(1, n-1)$ group.

10. $Sp(2n, F)$: the *symplectic group* of $2n \times 2n$ matrices M with elements in F, such that

$$M^T \Omega M = \Omega\,, \tag{1.37}$$

$$\Omega = \begin{pmatrix} 0 & I_{n\times n} \\ -I_{n\times n} & 0 \end{pmatrix} \tag{1.38}$$

The dimension of $Sp(2n, \mathbb{R})$ and $Sp(2n, \mathbb{C})$ are

$$\dim(Sp(2n, \mathbb{R})) = n(2n+1)\ ,\ \dim(Sp(2n, \mathbb{C})) = 2n(2n+1)\,. \tag{1.39}$$

• $Sp(2n, F)$ preserves the inner product

$$x^T \Omega y = \sum_{j=1}^{n} (x_j y_{n+j} - x_{n+j} y_j)\,. \tag{1.40}$$

• Comment: the natural action of $Sp(2n, \mathbb{R})$ on $\mathbb{R}^{2n}$ describes the infinitesimal transformations on the $2n$ canonical coordinates $(q^1, \ldots, q^n, p_1, \ldots, p_n)$ in classical mechanics, such that the Poisson bracket

$$\{f, g\} = \frac{\partial f}{\partial q^m}\frac{\partial g}{\partial p_m} - \frac{\partial g}{\partial q^m}\frac{\partial f}{\partial p_m} \tag{1.41}$$

is preserved.

11. $Sp(n)$ or $USp(2n)$: the *unitary symplectic group*, which is defined as

$$Sp(n) = Sp(2n, \mathbb{C}) \cap SU(2n) \tag{1.42}$$

The dimension of $Sp(n)$ is

$$\dim(Sp(n)) = n(2n + 1)\,. \tag{1.43}$$

12. The groups of type GL, SL, U, SU, O, SO, Sp are all called "classical groups". There are also exceptional groups E_6, E_7, E_8, F_4, G_2, which can be written as matrix groups as well. But they are quite complicated and we do not discuss them in details here.

13. There exists Lie groups that are not matrix group, for example $\widetilde{SL}(2, \mathbb{R})$, the universal cover of $SL(2, \mathbb{R})$. (Will be explained in the topology section)

1.3 Subgroups

Here we present some examples of matrix subgroups. To prove a group H is a subgroup of G, we need to construct an injective homomorphism $\phi : H \to G$.

• As a reminder, let H and K to be subgroups of G, and let $f : H \times K \to G$ to be the multiplication map, defined as $f(h, k) = hk$. Then f is an isomorphism ($G = H \times K$) if and only if

(1) f is surjective.

(2) For any $h \in H$, $k \in K$, $hk = kh$ in G.

(3) $H \cap K$ is a trivial subgroup in G.

• As another remark, for two Lie groups G and H, $\dim(G \times H) = \dim(G) + \dim(H)$.

1. Matrix subgroups from definitions

   (1) $O(n) \subset U(n)$, since the condition $M^T M = I$ is a subcase of $M^\dagger M = I$ when M is a real matrix.

   (2) Similar to above, we have $SO(n) \subset SU(n)$, $O(p, q) \subset U(p, q)$, $SO(p, q) \subset SU(p, q)$

   (3) $Sp(2n, F) \subset SL(2n, F)$, since one can prove that the elements $M \in Sp(2n, F)$ satisfy $\det(M) = 1$ (see e.g. arXiv: 1505.04240).

   (4) $USp(2n) \subset Sp(2n, \mathbb{C})$, $USp(2n) \subset SU(2n)$.

2. $GL(m, F) \times GL(n - m, F) \subset GL(n, F)$.

   One takes the matrices in $GL(n, F)$ in the form of

$$\left(\begin{array}{c|c} M_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & I_{(n-m)\times(n-m)} \end{array}\right), \tag{1.44}$$

   which forms the subgroup $GL(m, F)$.

   Similarly, the matrices in the form of

$$\left(\begin{array}{c|c} I_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & N_{(n-m)\times(n-m)} \end{array}\right) \tag{1.45}$$

   form the subgroup $GL(n - m, F)$.

   Moreover, it is easy to check that the set of matrices in $GL(n, F)$ in form of

$$\left(\begin{array}{c|c} M_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & N_{(n-m)\times(n-m)} \end{array}\right) \tag{1.46}$$

   is a direct product $GL(m, F) \times GL(n - m, F)$.

The multiplication map $f : GL(m,F) \times GL(n-m,F) \to GL(m,F) \times GL(n-m,F)$ defined as

$$f(M_{m\times m}, N_{(n-m)\times(n-m)}) = \left(\begin{array}{c|c} M_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & N_{(n-m)\times(n-m)} \end{array}\right) \tag{1.47}$$

satisfies the criteria (1)(2)(3).

Thus it induces an injective homomorphism $f : GL(m,F) \times GL(n-m,F) \to GL(n,F)$.

• Similarly we have

$$\begin{aligned} &SL(m,F) \times SL(n-m,F) \subset SL(n,F) \\ &U(m) \times U(n-m) \subset U(n) \\ &SU(m) \times SU(n-m) \subset SU(n) \\ &O(m) \times O(n-m) \subset O(n) \\ &SO(m) \times SO(n-m) \subset SO(n) \\ &Sp(2m,F) \times Sp(2n-2m,F) \subset Sp(2n,F)\,. \end{aligned} \tag{1.48}$$

• The sub-cases also trivially hold:

$$\begin{aligned} &GL(m,F) \subset GL(n,F)\ (m<n) \\ &SL(m,F) \subset SL(n,F)\ (m<n) \\ &U(m) \subset U(n)\ (m<n) \\ &SU(m) \subset SU(n)\ (m<n) \\ &O(m) \subset O(n)\ (m<n) \\ &SO(m) \subset SO(n)\ (m<n) \\ &Sp(2m,F) \subset Sp(2n,F)\ (m<n)\,. \end{aligned} \tag{1.49}$$

3. $SL(n,\mathbb{R}) \subset GL(n,\mathbb{R})$

   This is trivial from the definitions. Nonetheless, there is another subgroup $GL(1,\mathbb{R}) \subset GL(n,\mathbb{R})$, whose elements are

$$\mathrm{diag}(a,a,\ldots,a)\ ,\ (a \neq 0 \in \mathbb{R}) \tag{1.50}$$

   The question is whether $SL(n,\mathbb{R}) \times GL(1,\mathbb{R}) \subseteq GL(n,\mathbb{R})$.

   Now for odd n, we claim the following result

$$SL(n,\mathbb{R}) \times GL(1,\mathbb{R}) \cong GL(n,\mathbb{R})\,, \tag{1.51}$$

after constructing the isomorphism $f : SL(n,\mathbb{R})\times GL(1,\mathbb{R}) \to GL(n,\mathbb{R})$: $f(M, aI_{n\times n}) = aM \in GL(n,\mathbb{R})$.

One can check that indeed

(1) the multiplication map $f : SL(n,\mathbb{R}) \times GL(1,\mathbb{R}) \to GL(n,\mathbb{R})$ is surjective, since every $G \in GL(n,\mathbb{R})$ can be written as $G = f(G \cdot (\det(G))^{-1/n}, (\det(G))^{1/n} I_{n\times n}) \in SL(n,\mathbb{R}) \times GL(1,\mathbb{R})$;

(2) M and $aI_{n\times n}$ always commute in $GL(n,\mathbb{R})$;

(3) $SL(n,\mathbb{R}) \cap GL(1,\mathbb{R}) = \{I_{n\times n}\}$ in $GL(n,\mathbb{R})$.

On the other hand, for even n, the criterion (3) is not satisfied, because the element $-I_{n\times n}$ is a non-trivial element in both $SL(n,\mathbb{R})$ and $GL(1,\mathbb{R})$. The criterion (1) is not satisfied either, since for a matrix $M' \in GL(n,\mathbb{R})$ with $\det(M') < 0$, it cannot be mapped to an element $(M, aI_{n\times n})$ in $SL(n,\mathbb{R}) \times GL(1,\mathbb{R})$.

Hence for even n

$$SL(n,\mathbb{R}) \times GL(1,\mathbb{R}) \neq GL(n,\mathbb{R})\,. \tag{1.52}$$

One can also check that there is no alternative way to embed $SL(n,\mathbb{R})\times GL(1,\mathbb{R})$ into $GL(n,\mathbb{R})$, satisfying (1)(2)(3).

4. $(SU(n) \times U(1))/\mathbb{Z}_n \cong U(n)$

   One asks whether $SU(n)\times U(1) \subseteq U(n)$, here $SU(n) \subset U(n)$ is trivial, and the elements of $U(1) \subset U(n)$ is

$$\mathrm{diag}(e^{i\theta}, e^{i\theta}, \ldots, e^{i\theta})\ ,\ (\theta \in [0, 2\pi)) \tag{1.53}$$

   After constructing the map $f : SU(n)\times U(1) \to U(n)$: $f(M, e^{i\theta} I_{n\times n}) = e^{i\theta} M \in U(n)$, the elements $e^{2\pi i/n} I_{n\times n}$ are in both of $SU(n)$ and $U(1)$. Thus $SU(n) \times U(1) \nsubseteq U(n)$.

   However, if we take the quotient group $(SU(n) \times U(1))/\mathbb{Z}_n$, where the $\mathbb{Z}_n$ generator acts on $SU(n) \times U(1)$ as

$$(e^{\frac{2\pi i}{n}}, e^{-\frac{2\pi i}{n}}) \cdot (M, e^{i\theta} I_{n\times n}) = (Me^{\frac{2\pi i}{n}}, e^{i(\theta-2\pi/n)} I_{n\times n})\,, \tag{1.54}$$

   the problem is solved.

   In other words, the elements of $(SU(n)\times U(1))/\mathbb{Z}_n$ are given by $(M, e^{i\theta} I_{n\times n}) \in SU(n) \times U(1)$, modding out the equivalence relation

$$(M, e^{i\theta} I_{n\times n}) \sim (Me^{\frac{2\pi i}{n}}, e^{i(\theta-2\pi/n)} I_{n\times n})\,, \tag{1.55}$$

   which one-to-one corresponds to $U(n)$ group elements.

5. $(SU(m) \times SU(n-m) \times U(1))/\mathbb{Z}_{lcm(m,n-m)} \subset SU(n)$

For $SU(n)$, note that apart from the subgroup $SU(m) \times SU(n-m)$ given by

$$\left(\begin{array}{c|c} M_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & N_{(n-m)\times(n-m)} \end{array}\right), \tag{1.56}$$

one can also write down the diagonal matrices

$$\mathrm{diag}(\underbrace{e^{\frac{2\pi ik}{m}},\ldots,e^{\frac{2\pi ik}{m}}}_{m},\underbrace{e^{\frac{-2\pi ik}{(n-m)}},\ldots,e^{\frac{-2\pi ik}{(n-m)}}}_{n-m}) \quad (k \in [0, lcm(m,n-m))\ ) \tag{1.57}$$

which form a $U(1)$ subgroup of $SU(n)$ as well (note that k has a different periodicity than 2π). One can also see that (1.57) commutes with all the $SU(m) \times SU(n-m)$ elements.

Now one may wonder whether $SU(m)\times SU(n-m)\times U(1)$ is a subgroup of $SU(n)$ (if $n > 2$). In other words, one need to check if the group homomorphism $f : SU(m) \times SU(n-m) \times U(1) \to SU(n)$ is injective,

$$f(M,N,e^{2\pi ik/lcm(m,n-m)}) = \left(\begin{array}{c|c} M & 0 \\ \hline 0 & N \end{array}\right)\cdot\mathrm{diag}(\underbrace{e^{\frac{2\pi ik}{m}},\ldots,e^{\frac{2\pi ik}{m}}}_{m},\underbrace{e^{\frac{-2\pi ik}{(n-m)}},\ldots,e^{\frac{-2\pi ik}{(n-m)}}}_{n-m}) \tag{1.58}$$

The answer is no, because for $k \in \mathbb{Z}$, the matrix $\mathrm{diag}(\underbrace{e^{\frac{2\pi ik}{m}},\ldots,e^{\frac{2\pi ik}{m}}}_{m},\underbrace{e^{\frac{-2\pi ik}{(n-m)}},\ldots,e^{\frac{-2\pi ik}{(n-m)}}}_{n-m})$ is in both $SU(m) \times SU(n-m)$ and $U(1)$.

The actual subgroup of $SU(n)$ is the quotient of $SU(m)\times SU(n-m)\times U(1)$ by $\mathbb{Z}_{lcm(m,n-m)}$. $\mathbb{Z}_{lcm(m,n-m)}$ is generated by the following element in $SU(m) \times SU(n-m) \times U(1)$

$$\left(\left(\begin{array}{c|c} e^{-\frac{2\pi i}{m}}I_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & I_{(n-m)\times(n-m)} \end{array}\right), \left(\begin{array}{c|c} I_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & e^{\frac{2\pi i}{n-m}}I_{(n-m)\times(n-m)} \end{array}\right), \left(\begin{array}{c|c} e^{\frac{2\pi i}{m}}I_{m\times m} & 0_{m\times(n-m)} \\ \hline 0_{(n-m)\times m} & e^{-\frac{2\pi i}{n-m}}I_{(n-m)\times(n-m)} \end{array}\right)\right) \tag{1.59}$$

• As a physical comment, in fact, the naive standard model gauge group $SU(3)\times SU(2)\times U(1)$ is not a subgroup of the GUT gauge group $SU(5)$! The one that can be embedded into $SU(5)$ is $SU(3)\times SU(2)\times U(1)/\mathbb{Z}_6$.

2 Generalities on Lie groups and Lie algebras

2.1 Matrix group as a topological space

We have already seen that there are many examples of matrix groups with a nice geometric structure. For example, $U(1) \cong S^1$ and $SU(2) \cong S^3$. Now we try to describe their geometric properties using the language of topology.

We first review some basic concepts in point-set topology, which will be relevant in the discussion of global structure of topological groups (Lie groups).

• A *topological space* is a set X with a collection of *open sets* $\mathcal{T} = \{U_i\}$, satisfies:

(1) $\varnothing, X \in \mathcal{T}$

(2) For any subcollection $\mathcal{S} = \{U_j | j \in J\} \subset \mathcal{T}$, $\bigcup_J U_j \in \mathcal{T}$

(3) For any finite subcollection $\mathcal{S} = \{U_j | j \in J\} \subset \mathcal{T}$, $\bigcap_J U_j \in \mathcal{T}$

• Example:

(1) the *usual topology* of $\mathbb{R}$ where the open sets are the open intervals (a, b) and their unions.

(2) *trivial topology* such that every subset of G is an open set.

In this course, we are always assuming that the topological space of interest has good properties, so that we can define a metric on it. The default topology of this space will be the metric topology.

• *Metric space*: a topological space with a metric function: $d : X \times X \to \mathbb{R}$, which satisfies

(1) $d(x, y) = d(y, x)$

(2) $d(x, y) \geq 0$ and the equality is satisfied if and only if $x = y$

(3) $d(x, y) + d(y, z) \geq d(x, z)$

• If X is a metric space, it has a *metric topology*, whose open subsets are

$$U_\epsilon(x) = \{y \in X | d(x, y) < \epsilon\} \tag{2.1}$$

and their unions.

• *Relative topology* of a subset $Y \subset X$: suppose that the collection of open sets of X is $\mathcal{T}$, the collection of open sets of Y is $\mathcal{S} = \{U \cap Y | \forall U \in \mathcal{T}\}$.

If a topological space Y is a subset of $\mathbb{R}^n$, its metric topology can be naturally constructed using the above concepts and the Euclidean metric on $\mathbb{R}^n$.

• *Product topology* of the product space $X \times Y$: denote the open sets of X as U_i^X and the open sets of Y as U_j^Y, the open sets of $X \times Y$ are unions of $U_i^X \times U_j^Y$.

• *Closed set*: a subset $Y \subset X$ is closed if and only if its complement $Y_c = X - Y$ in X is an open set.

One can also use the alternative equivalent definition of a closed set: a set $Y \subset X$ is closed if any sequence of elements $A_m \in Y$ converges to an element in Y.

Examples: (1) The empty set $\varnothing$ and X are both open and closed.

(2) on $\mathbb{R}$, the closed interval $[a, b]$ is the complement of the open set $(-\infty, a) \cup (b, \infty)$, and it is a closed set.

(3) The ball $\overline{U_\epsilon(x)} = \{y \in X | d(x, y) \leq \epsilon\}$ is closed.

(4) There exists subsets that are neither open or closed, e. g. $[a, b) \subset \mathbb{R}$

• *Continuous map*: a map $f : X \to Y$ is continuous if and only if the inverse image, $f^{-1}(U_Y)$, of any open set $U_Y \subset Y$ is an open set in X.

Example: for the piecewise function $f : \mathbb{R} \to \mathbb{R}$

$$f(x) = \begin{cases} x & (x \geq 0) \\ x - 1 & (x < 0) \end{cases} \tag{2.2}$$

The inverse image $f^{-1}((-0.1, 0.1)) = [0, 0.1)$, which is not an open set. Hence the notion of continuous map matches the intuition.

• *Homeomorphism* (equivalence relation at the level of topology): two topological spaces X and Y are homeomorphic if and only if there exists a continuous map $f : X \to Y$ such that its inverse map: $f^{-1} : Y \to X$ is also continuous.

With these definitions, we can introduce the notion of *topological groups*:

• A topological group G is a topological space with a group structure, such that the group operation $\cdot : \ G \times G \to G$ and the group inverse map ${}^{-1} : \ G \to G$ are continuous maps. Note that the topological space $G \times G$ has the product topology.

An isomorphism of topological groups $f : G_1 \to G_2$ is a group isomorphism that is also a homeomorphism of the underlying topological space.

From the definition, the inverse map ${}^{-1} : \ G \to G$ is a homeomorphism of G onto itself.

Note that the left product map $a\cdot : \ G \to G$: $g \to ag$ and right product map $\cdot a : \ G \to G$: $g \to ga$ are homeomorphisms of the topological space G onto itself, but they are not isomorphisms of topological groups.

• Note that strictly speaking, a discrete group G is also a topological group, with the trivial topology. Nonetheless, it is a *discrete space* in the topological sense, such that for any $x \in G$, there exists $s \in \mathbb{R}$, and $d(x, y) > s$ for any $y \in (G - \{x\})$.

• In many physics literature, a non-discrete group is called a continuous group, in contrast.

• Using the language of topology, we can define the matrix groups introduced in Section 1 as the *closed subgroups* of $GL(n, \mathbb{C})$. Equivalently we have:

Definition 1.4 (p.4 of B. Hall): a *matrix Lie group* G is a subgroup of $GL(n, \mathbb{C})$ with the following property: if A_m is any sequence of matrices in G, and A_m converges to some matrix A, then either A is in G or it is non-invertible.

• Note that the topology of the matrix Lie group G can be defined as the relative topology $G \subset M_n(\mathbb{C})$ from the metric topology of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

We also introduce the following notions in topology, which can be applied to topological groups.

• *Compactness*: if X is a subset of $\mathbb{R}^n$, then X is compact if and only if it is closed and bounded. "Bounded" means that there exists $s \in \mathbb{R}$, such that for any two points $x, y \in X$, $d(x, y) < s$. Note that the metric function is the Euclidean metric on $\mathbb{R}^n$.

Examples: (1) $S^1 \cong U(1) \cong SO(2)$ is compact

(2) $S^3 \cong SU(2)$ is compact

This can be proved by using the matrix form of elements in $SU(2)$:

$$\begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix} \quad , \ (a^2+b^2+c^2+d^2=1), \ (a,b,c,d \in \mathbb{R}) \tag{2.3}$$

Using the $\mathbb{R}^4$ Euclidean metric $d(x, y)$, one can choose for example $s = 3$ to show that $SU(2)$ is indeed bounded. It is also closed, because $S^3 = \mathbb{R}^4 - \{y|d(y, 0) < 1\} - \{y|d(y, 0) > 1\}$.

(3) The matrix groups $U(n)$, $SU(n)$, $O(n)$, $SO(n)$ and $Sp(n)(USp(2n))$ are all compact.

To show this, note that the group elements M ($n \times n$ matrix) all satisfy $M^\dagger M = I$, which implies that the columns of M are all unit vectors:

$$\sum_{j=1}^{n} |M_{jk}|^2 = 1 \, . \tag{2.4}$$

Hence every matrix element M_{jk} is bounded by $|M_{jk}| \leq 1$. Thus the group G is a closed and bounded subset of $M_n(\mathbb{C})$, which is compact.

(4) $GL(n, \mathbb{R})$, $GL(n, \mathbb{C})$, $SL(n, \mathbb{R})$, $SL(n, \mathbb{C})$, $Sp(2n, \mathbb{R})$, $Sp(2n, \mathbb{C})$ are non-compact. Check that they are unbounded!

For example, the group $SL(n, \mathbb{R})$ has elements of the form diag $(a, a^{-1}, 1, \ldots, 1)$, $a \in \mathbb{R}$, which are unbounded.

(5) The indefinite groups $O(p, q)$, $SO(p, q)$, $U(p, q)$, $SU(p, q)$ $(p, q \geq 1)$ are non-compact.

• Comment: importance of compactness of Lie groups in physics: consider a quantum gauge field theory with gauge group G, it is a unitary theory if and only if G is compact (related to the existence of unitary representation!).

• *Connectedness*: a topological space X is connected if it cannot be written as $X = X_1 \cup X_2$, where X_1 and X_2 are non-empty open subset of X, and $X_1 \cap X_2 = \varnothing$. Otherwise it is *disconnected*.

• In practice, one can also check a stronger condition *path-connectedness*. X is path-connected if for any $x, y \in X$, there exists a continuous map (path) $f : [0,1] \to X$ such that $f(0) = x$ and $f(1) = y$. In the cases of matrix groups, connectedness is equivalent to path-connectedness (p.17 of B. Hall).

Fix a point $x \in X$, the *connected component* containing x is the union of all connected open sets that contain x. If X has finitely many connected components, then each component is both open and closed.

Examples : (1) $GL(n, \mathbb{C})$, $SL(n, \mathbb{C})$, $U(n)$, $SU(n)$ and $SO(n)$ are all connected.

We prove for the example of $U(n)$: note that any unitary matrix M can be diagonalized by

$$M = UDU^{-1}\,, \tag{2.5}$$

where D is a diagonal matrix in the form of $D = \mathrm{diag}(e^{i\theta_1}, e^{i\theta_2}, \ldots, e^{i\theta_n})$, hence we can define a continuous path

$$M(t) = U \cdot \begin{pmatrix} e^{i(1-t)\theta_1} & 0 & \ldots & 0 \\ 0 & e^{i(1-t)\theta_2} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & e^{i(1-t)\theta_n} \end{pmatrix} \cdot U^{-1}\,, \tag{2.6}$$

which connects $M(0) = M$ and $M(1) = I$.

(2) $O(n)$ is disconnected as it can be written as $O(n) = SO(n) \cup O(n)^-$ and $SO(n) \cap O(n)^- = \varnothing$. $SO(n)$ and $O(n)^-$ are the two connected components of $O(n)$ (they are both open and closed).

(3) $O(p,q)$ and $SO(p,q)$ are disconnected, with four and two connected components, respectively.

• A *loop* in a topological space X is a continuous map $f : [0,1] \to X$ such that $f(0) = f(1)$. X is *simply-connected* if and only if every loop in X can be continuously shrunk to a point.

To be more quantitative, let us introduce the notion of fundamental group $\pi_1(X)$. We first introduce the notion of homotopy in algebraic topology, which has many applications in theoretical physics.

• Two maps $f_1 : X \to Y$ and $f_2 : X \to Y$ are called *homotopic*, $f_1 \sim f_2$, if they can be continuously deformed into other. In other words, there exists a continuous map $F : X \times [0,1] \to Y$, such that

$$F(x,0) = f_1(x)\ ,\ F(x,1) = f_2(x) \quad \text{for all } x \in X\,. \tag{2.7}$$

• Let us define the *fundamental group* with base point x_0: $\pi_1(X, x_0)$, or simply $\pi_1(X)$ when X is connected. A *based loop* is a map $l : [0,1] \to X$ such that

$$l(0) = l(1) = x_0 \in X\,. \tag{2.8}$$

The elements of $\pi_1(X, x_0)$ are the set of based loop mod out the homotopic equivalence $\sim$. In other words, each element of $\pi_1(X)$ is a *homotopy class* of based loop.

The identity element of $\pi_1(X)$ consists of the trivial homotopy class of the based loop that can be continuously shrunk to a point.

The group multiplication of two loops l_1 and l_2 is defined as follows:

$$l_1 \cdot l_2(t) = \begin{cases} l_1(2t) & 0 \le t \le \frac{1}{2} \\ l_2(2t-1) & \frac{1}{2} \le t \le 1 \end{cases} \tag{2.9}$$

And the inverse of l_1 is defined as

$$l_1^{-1}(t) = l_1(1-t)\,. \tag{2.10}$$

Namely, X is simply-connected if and only if the fundamental group $\pi_1(X)$ is trivial.

Examples:

(1) $\mathbb{R}^n$ is simply-connected.

(2) $U(1) \cong S^1$ is not simply-connected. $\pi_1(S^1) = \mathbb{Z}$, each element of $\pi_1(S^1)$ is labeled by an integer w, which is called the *winding number*. Choose the coordinate $x \sim x + 2\pi$ on S^1, the loop l_w with winding number w has the following property:

$$l_w(x+1) = l_w(x) + 2\pi w\,. \tag{2.11}$$

Moreover, $U(1) \times G$ are all non-simply-connected.

(3) $SU(2) \cong S^3$ is simply-connected. $SU(n)$ are all simply-connected.

(4) $SO(3) \cong SU(2)/\mathbb{Z}_2$ is homeomorphic to the *real projective space* RP^3, which is the space of vectors in $\mathbb{R}^4$. It is defined as the quotient space of $\mathbb{R}^4 - \{0\}$ by the equivalence relation $(x_1, x_2, x_3, x_4) \sim \lambda(x_1, x_2, x_3, x_4)$.

$SO(3)$ is not simply-connected, consider a loop starting from a point $(1,0,0,0)$ to its antipodal point $(-1,0,0,0)$. The loop cannot be shrunk to a point. Nonetheless, if we take twice of that loop, it is contractible to a trivial loop. Hence we have $\pi_1(SO(3)) = \mathbb{Z}_2$.

(5) Similarly, $SO(n)$ are all non-simply-connected. $\pi_1(SO(n)) = \mathbb{Z}_2$ (see B. Hall p. 377).

• The universal cover $\widetilde{G}$ of a connected Lie group G is a simply connected group, such that there exists a Lie group homomorphism $f : \widetilde{G} \to G$, which is a covering map in the topology language.

A covering map $f : \widetilde{G} \to G$ satisfies that for any $x \in G$, there exists an open neighborhood U_x of x and a discrete space D_x such that $f^{-1}(U_x) = \bigsqcup_{d \in D_x} V_d$ and $V_d \cong U_x$. In other words, a covering map describes the covering of G by a discrete set of copies.

• Example:

(1) $SU(2)$ is the universal cover of $SO(3) = SU(2)/\mathbb{Z}_2$.

(2) Similarly, $SU(N)$ is the universal cover of $PSU(N) = SU(N)/\mathbb{Z}_N$.

(3) $\mathbb{R}$ is the universal cover of $U(1) = \mathbb{R}/\mathbb{Z}$.

2.2 Lie group as a differential manifold

So far we have only studied the topological properties of matrix groups. In early 20th century, David Hilbert wondered what makes the set of "Lie groups", which include matrix groups, stand out of the rest of topological groups. This was also known as Hilbert's fifth problem. The answer is that an additional "smooth" structure needs to be imposed on the topological space. Later in the history, this additional structure is called differential structure, and the topological space is a differential manifold.

We define the first central mathematical object of this course: *Lie group* using the language of differential manifolds.

• A *Lie group* G is a differential manifold with a group structure, whose group operation $G \times G \to G$ and inverse map $G \to G$ are differentiable (smooth) maps.

• The *dimension* of the Lie group G is its dimension as a differential manifold. In this course we always assume that the dimension of Lie groups and differential manifolds are finite.

Now we introduce the concepts of differential manifold.

• A topological space M is an n-dimensional *differential manifold* (manifold) if

(1) M is provided with a collection of pairs $\{U_i, \varphi_i\}$

(2) $\{U_i\}$ is a collection of open sets which covers M, that is, $\bigcup_i U_i = M$.

(3) φ_i is a homeomorphism from U_i to an open subset $U_i' \subset \mathbb{R}^n$

(4) given U_i and U_j such that $U_i \cap U_j \neq \varnothing$, the map $\psi_{ij} = \varphi_i \circ \varphi_j^{-1}$: $\varphi_j(U_i \cap U_j) \to \varphi_i(U_i \cap U_j)$ is infinitely differentiable (C^∞ differentiable).

The pair $\{U_i, \varphi_i\}$ is called a *coordinate chart* or simply a chart, and the collection of coordinate charts is called an *atlas*. The map φ_i is the *coordinate* map or simply coordinate, which assigns local $\mathbb{R}^n$ coordinates to the subset $U_i \subset M$. The map ψ_{ij} is the *transition map* which relates the local coordinates of different patches.

The idea is to glue multiple pieces of (subset of) $\mathbb{R}^n$ into a more complicated geometric object, the differential manifold. The coordinate map φ_i allows one to talk about the analytic properties of functions on M locally. Now we introduce the notion of differentiable map.

• Let $f : M \to N$ be a map from an m-dimensional manifold M to an n-dimensional manifold N. A point $p \in M$ is mapped to $f(p) \in N$. Take a chart (U, φ) on M and (V, ψ) on N, such that $p \in U$ and $f(p) \in V$. f has the following coordinate presentation:

$$\psi \circ f \circ \varphi^{-1} : \mathbb{R}^m \to \mathbb{R}^n \,. \tag{2.12}$$

We write $\varphi(p) = \{x^\mu\}$ $(\mu = 1, \ldots, m)$ and $\psi(f(p)) = \{y^\alpha\}$ $(\alpha = 1, \ldots, n)$.

Hence

$$y^\alpha(x^\mu) \equiv f^\alpha(x^\mu) \tag{2.13}$$

is a map from an m-dimensional vector space to an n-dimensional vector space. Then if for each $(\alpha = 1, \ldots, n)$, $f^\alpha(x^\mu)$ is a C^∞ differentiable function respect to each x^μ, f is said to be *differentiable* or *smooth* at p.

• If f is differentiable (smooth) at every point $p \in M$, then f is called a *differentiable map* or *smooth* map from M to N.

• If $f : M \to N$ and its inverse map f^{-1} are differentiable maps, then f is called a *diffeomorphism* and the two manifolds M and N are *diffeomorphic*.

• *product manifold*: let M be an m-dimensional manifold with atlas (U_i, φ_i) and N be an n-dimensional manifold with atlas (V_j, ψ_j). Their product manifold $M \times N$ is an $(m+n)$-dimensional manifold whose points are $(p, q), p \in M, q \in N$. The atlas of $M \times N$ is $(U_i \times V_j, (\varphi_i, \psi_j))$. The coordinate map (φ_i, ψ_j) maps (p, q) to $(\varphi_i(p), \psi_j(q)) \in \mathbb{R}^{m+n}$.

• *embedded submanifold*: let M be an n-dimensional manifold, then a k-dimensional $(0 \leq k \leq n)$ embedded submanifold of M is a subset $S \subset M$ such that for every point $p \in S$ there exists a chart $\{U, \varphi\}$ containing p, and $\varphi(S \cap U)$ is the intersection of a k-dimensional plane (locally $(x_1, x_2, \ldots, x_k, 0, \ldots, 0) \in \mathbb{R}^n$) with U.

The charts on S are given by $(S \cap U, \varphi|_{S \cap U})$.

Examples: (1) $U(1) \cong S^1$: we write the circle as the equation

$$x^2 + y^2 = 1 \tag{2.14}$$

in the (x, y) plane. We need at least two coordinate patches to cover S^1, for example take

$$U_1 = S^1 - \{(1, 0)\} \ , \ \varphi_1(\cos\theta, \sin\theta) = \theta \ , \ (\theta \in (0, 2\pi)) \, . \tag{2.15}$$

The coordinate map φ_1 maps a point in the open subset U_1 to $U_1' = (0, 2\pi) \subset \mathbb{R}$.

$$U_2 = S^1 - \{(-1, 0)\} \ , \ \varphi_2(\cos\theta, \sin\theta) = \theta \ , \ (\theta \in (-\pi, \pi)) \tag{2.16}$$

The coordinate map φ_2 maps a point in U_2 to $U_2' = (-\pi, \pi) \subset \mathbb{R}$. Obviously φ_1 and φ_2 are homeomorphisms.

One can check that the transition maps $\psi_{12} = \varphi_1 \circ \varphi_2^{-1}$ and $\psi_{21} = \varphi_2 \circ \varphi_1^{-1}$ are indeed differentiable on $U_1 \cap U_2$.

Now let us consider the product manifold $S^1 \times S^1 \cong T^2$, whose atlas is

$$\{(U_1 \times U_1, (\varphi_1, \varphi_1)) \, , \, (U_1 \times U_2, (\varphi_1, \varphi_2)) \, , \, (U_2 \times U_1, (\varphi_2, \varphi_1)) \, , \, (U_2 \times U_2, (\varphi_2, \varphi_2)) \, . \tag{2.17}$$

The group operation of $U(1)\times U(1) \to U(1)$ maps $((\cos\theta_1, \sin\theta_1), (\cos\theta_2, \sin\theta_2))$ to $(\cos(\theta_1+\theta_2), \sin(\theta_1+\theta_2))$.

Now one can check that this map is differentiable for every values of θ_1 and θ_2, after choosing proper coordinate charts.

An easier thing to check is that the group inverse $U(1) \to U(1)$, which maps $(\cos\theta_1, \sin\theta_1)$ to $(\cos\theta_1, -\sin\theta_1)$, is differentiable.

(2) S^n: we write it as the equation

$$x_1^2 + x_2^2 + \cdots + x_{n+1}^2 = 1 \tag{2.18}$$

in $\mathbb{R}^{n+1}$. We can choose the stereographic coordinate patches (U_N, φ_N), (U_S, φ_S) to cover S^n:

$$\begin{aligned} U_N &= S^n - \{(0,0,\ldots,1)\}\ ,\ \varphi_N(x_1,\ldots,x_{n+1}) = \frac{(x_1,\ldots,x_n)}{1-x_{n+1}} \\ U_S &= S^n - \{(0,0,\ldots,-1)\}\ ,\ \varphi_S(x_1,\ldots,x_{n+1}) = \frac{(x_1,\ldots,x_n)}{1+x_{n+1}} \end{aligned} \tag{2.19}$$

One can check that $\psi_{NS} = \varphi_N \circ \varphi_S^{-1}$ and $\psi_{SN} = \varphi_S \circ \varphi_N^{-1}$ are differentiable maps, and S^n is an embedded submanifold of $\mathbb{R}^{n+1}$.

We can rephrase many group theory concepts using the language of differential manifolds.

(1) The product group $G \times H$ of two Lie groups G and H corresponds to the product manifold $G\times H$. Hence we have $\dim(G\times H) = \dim(G)+\dim(H)$.

• Example: the product group $U(1)^n$ corresponds to the product manifold $T_n = (S^1)^n$, which is the real n-torus.

(2) Cartan's closed subgroup theorem: any (topologically) closed subgroup $H \subset G$ of a Lie group G is an embedded submanifold of G. The group structure of H is induced from G.

• Example: $U(1)$ (S^1) is a Lie subgroup (submanifold) of $SU(2)$ (S^3).

(3) Lie group homomorphism: the same as the homomorphism of topological groups, which is a continuous group homomorphism. Note that in the cases of matrix Lie groups, this condition guarantees that the map is smooth in the differential manifold language (B. Hall p. 72 Corollary 3.50).

(4) The action of a Lie group G on a manifold M: a differentiable map $\cdot : G \times M \to M$ which satisfies

(i) For the identity element e of G, $e \cdot p = p$ for any $p \in M$;

(ii) For any $g_1, g_2 \in G$ and $p \in M$, $g_1 \cdot (g_2 \cdot p) = (g_1 g_2) \cdot p$.

• Example: the natural action of matrix multiplication on a vector space.

2.3 Basic notions of Lie algebra

Here we will introduce the second central concept of this course: Lie algebra. We first present the definition of Lie algebra, and then discuss its relations to Lie groups.

• A *finite dimensional Lie algebra* (Lie algebra) $\mathfrak{g}$ is a finite-dimensional vector space F^n ($F = \mathbb{R}$ or $\mathbb{C}$), together with a map (*Lie bracket*) $[.,.] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, such that:

(1) $[.,.]$ is bilinear ($[X,Y]+[X,Z] = [X,Y+Z]$, $[X,Z]+[Y,Z] = [X+Y,Z]$, $[aX,Y] = a[X,Y] = [X,aY]$ for all $X,Y,Z \in \mathfrak{g}$, $a \in F$)

(2) $[X,Y] = -[Y,X]$ for all $X,Y \in \mathfrak{g}$

(3) *Jacobi identity*:

$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0\,. \tag{2.20}$$

• The basis of the vector space are also called the *generators* of Lie algebra.

• The dimension of the real (complex) vector space is the *dimension* of real (complex) Lie algebra, it is also equal to the number of independent generators.

• If the vector space is real (complex), then $\mathfrak{g}$ is called a real (complex) Lie algebra.

• If one take a real Lie algebra $\mathfrak{g}$ and replace the real vector space $\mathbb{R}^n$ of $\mathfrak{g}$ with the complex vector space $\mathbb{C}^n$, the resulting complex Lie algebra $\mathfrak{g}_{\mathbb{C}}$ is called the *complexification* of $\mathfrak{g}$.

• Remark: a Lie algebra is not a ring in general, since the $[.,.]$ is not associative.

Now we further introduce some basic relevant concepts.

• Two elements X, Y of a Lie algebra *commute* if $[X,Y] = 0$. A Lie algebra $\mathfrak{g}$ is *abelian* (*commutative*) if $[X,Y] = 0$ for all $X,Y \in \mathfrak{g}$.

• A *Lie subalgebra* of a Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h}$ of $\mathfrak{g}$ such that $[H_1,H_2] \in \mathfrak{h}$ for all $H_1,H_2 \in \mathfrak{h}$.

• The *center* of a Lie algebra $\mathfrak{g}$ is set of all elements $X \in \mathfrak{g}$, such that $[X,Y] = 0$ for all $Y \in \mathfrak{g}$.

• If $\mathfrak{g}$ and $\mathfrak{h}$ are Lie algebras, then a linear map $\phi : \mathfrak{g} \to \mathfrak{h}$ is called a *Lie algebra homomorphism* if $\phi([X,Y]) = [\phi(X),\phi(Y)]$ for all $X,Y \in \mathfrak{g}$. If ϕ is a one-to-one map, then ϕ is called a *Lie algebra isomorphism*.

• The *direct sum* $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ of two Lie algebras $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is the vector space direct sum of $\mathfrak{g}_1$ and $\mathfrak{g}_2$, with the bracket

$$[(X_1,X_2),(Y_1,Y_2)] = ([X_1,Y_1],[X_2,Y_2])\,. \tag{2.21}$$

Note that $X_1,Y_1 \in \mathfrak{g}_1$ and $X_2,Y_2 \in \mathfrak{g}_2$.

The dimension of the direct sum Lie algebra satisfies $\dim(\mathfrak{g}_1 \oplus \mathfrak{g}_2) = \dim(\mathfrak{g}_1) + \dim(\mathfrak{g}_2)$.

• Let $X_1, \ldots, X_N$ be a basis of Lie algebra $\mathfrak{g}$, the *structure constants* c_{ijk} of $\mathfrak{g}$ are defined as[1]

$$[X_i, X_j] = \sum_{k=1}^{N} c_{ijk} X_k \,. \tag{2.22}$$

Then the conditions (2) and (3) in the definition of Lie algebra are translated to

$$c_{ijk} = -c_{jik} \tag{2.23}$$

and

$$\sum_{n=1}^{N} (c_{jkn} c_{nlm} + c_{kln} c_{njm} + c_{ljn} c_{nkm}) = 0 \,. \tag{2.24}$$

Examples: (1) $\mathfrak{u}(1)$ Lie algebra with a single generator X, and the Lie bracket $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$:

$$[X, X] = 0 \,. \tag{2.25}$$

$\mathfrak{u}(1)$ is an abelian Lie algebra. The direct sum of $\mathfrak{u}(1)$, $\mathfrak{u}(1)^{\oplus n}$ is also an abelian Lie algebra.

(2) the 3d vector space $\mathbb{R}^3$ with the 3d cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ as the Lie bracket.

One can easily check the bilinear and anticommuting conditions. For the Jacobi identity, this is equivalent to

$$X \times (Y \times Z) + Y \times (Z \times X) + Z \times (X \times Y) = 0 \tag{2.26}$$

for any 3d vectors X, Y and Z.

Suppose that the basis of this "cross product" Lie algebra are X_1, X_2, X_3, the Lie bracket is

$$[X_i, X_j] = \epsilon_{ijk} X_k \tag{2.27}$$

and the structure constant is

$$c_{ijk} = \epsilon_{ijk} \,. \tag{2.28}$$

(3) Matrix Lie algebra $\mathfrak{gl}(n, \mathbb{R})$ ($\mathfrak{gl}(n, \mathbb{C})$): the space of real (complex) $n \times n$ matrices $M_n(\mathbb{R})$ ($M_n(\mathbb{C})$), with the Lie bracket

$$[X, Y] = XY - YX \,. \tag{2.29}$$

The three conditions of Lie algebra can be easily checked.

[1] In some physics context, the definition is up to an i factor: $[X_i, X_j] = \sum_{k=1}^{N} i c_{ijk} X_k$.

• All other matrix Lie algebras are subalgebras of some $\mathfrak{gl}(n, \mathbb{C})$.

(4) $\mathfrak{sl}(n, \mathbb{R})$ ($\mathfrak{sl}(n, \mathbb{C})$): the Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$ ($\mathfrak{gl}(n, \mathbb{C})$) which consists of traceless matrices. The set is closed under Lie bracket, because for any $n \times n$ matrices X and Y,

$$tr(XY - YX) = tr(XY) - tr(YX) = 0\,. \tag{2.30}$$

Obviously it also holds for traceless matrices X and Y.

2.4 From Lie group to Lie algebra

Why study Lie algebra? What's the relation between Lie group and Lie algebra?

Roughly speaking, Lie algebra consists of the infinitesimal generators of the Lie group. Physically, it consists of infinitesimal transformation of a continuous transformation group. In other words, it encodes the "local" information of a Lie group, which is easier to study using linear algebra. More precisely, the Lie algebra $\mathfrak{g}$ associated to a Lie group G is isomorphic to the *tangent space* $T_e(G)$ at the identity point $e \in G$ (see e. g. Nakahara for more details).

• Hence the real dimension of $\mathfrak{g}$ ($\mathfrak{g}$ as a real vector space) is equal to the dimension of G

$$\dim_{\mathbb{R}}(\mathfrak{g}) = \dim(G)\,. \tag{2.31}$$

On the other hand, the full Lie group has many global properties, that can only be studied with topology or differential manifolds. In many physics applications, one only needs to use the Lie algebra and its representations, and does not need to worry about the full Lie group structure.

The classification of Lie algebras (up to Lie algebra isomorphism) is also much easier than the classification of Lie groups (up to diffeomorphism). For example, the different Lie groups $SU(2)$ and $SO(3)$ have the same Lie algebra, which is the "cross product" Lie algebra mentioned before.

• Most generally, we have the (modern form of) *Lie's third theorem*: for any finite-dimensional real Lie algebra $\mathfrak{g}$, there exists a connected Lie subgroup of $GL(n, \mathbb{C})$ whose Lie algebra is isomorphic to $\mathfrak{g}$.

Now the question is how to construct the Lie algebra $\mathfrak{g}$ associated to a Lie group G. For the cases of matrix Lie group G, we have the following "simple" definition:

• The (associated) Lie algebra $\mathfrak{g}$ of a matrix Lie group G is the set of all matrices X, such that $e^{tX} \in G$ for all real number t. The Lie bracket is the matrix commutator $[X, Y] = XY - YX$.

Here e^{tX} is the *matrix exponential*, which is defined as a Taylor expansion

$$e^{tX} = \sum_{m=0}^{\infty} \frac{t^m}{m!} X^m\,. \tag{2.32}$$

The leading order term $X^0 \equiv I$.

The Taylor expansion converges absolutely for any $n \times n$ complex matrix X. (Proof: B. Hall p. 31-32)

• Note that in physics literature, e^{tX} in the definition of matrix Lie algebra is typically replaced with e^{itX}, and the Lie algebra elements differ by a factor of i.

One can prove the following properties for matrix exponential, where X and Y are two $n \times n$ matrices:

(1) $e^0 = I$.

(2) $(e^X)^\dagger = e^{X^\dagger}$.

(3) e^X is invertible and $(e^X)^{-1} = e^{-X}$.

(4) $e^{(\alpha+\beta)X} = e^{\alpha X} e^{\beta X}$ for all $\alpha, \beta \in \mathbb{C}$.

(5) If $XY = YX$, then $e^{X+Y} = e^X e^Y = e^Y e^X$.

(6) For $C \in GL(n, \mathbb{C})$, $e^{CXC^{-1}} = Ce^X C^{-1}$.

(7) $\frac{d}{dt} e^{tX} = Xe^{tX} = e^{tX} X$.

Note that for two generic matrices X and Y, $e^{X+Y} = e^X e^Y$ does not hold, since

$$e^{X+Y} = \sum_{m=0}^{\infty} \frac{(X+Y)^m}{m!} \tag{2.33}$$

and

$$\begin{aligned} e^X e^Y &= \sum_{m=0}^{\infty} \sum_{k=0}^{m} \frac{X^k}{k!} \frac{Y^{m-k}}{(m-k)!} \\ &= \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{k=0}^{m} \frac{m!}{k!(m-k)!} X^k Y^{m-k} \,. \end{aligned} \tag{2.34}$$

Only when X and Y commutes, we have

$$(X+Y)^m = \sum_{k=0}^{m} \frac{m!}{k!(m-k)!} X^k Y^{m-k} \,, \tag{2.35}$$

and $e^{X+Y} = e^X e^Y$.

Despite of the fact above, we have the *Lie product formula* (Proof: B. Hall p. 40):

$$e^{X+Y} = \lim_{m \to \infty} (e^{\frac{X}{m}} e^{\frac{Y}{m}})^m \,. \tag{2.36}$$

• The property (7) above is also quite important, since after assigning $t = 0$, we have

$$\left(\frac{d}{dt} e^{tX} \right) |_{t=0} = X \,. \tag{2.37}$$

This is a way of mapping the matrix group element $e^{tX} \in G$ to the corresponding matrix Lie algebra element X.

Now we can check the following properties of the Lie algebra $\mathfrak{g}$ associated to the Lie group G. Here X and Y are two elements of $\mathfrak{g}$.

1. For any $C \in G$, we have $CXC^{-1} \in \mathfrak{g}$.

This is because of the property (6) of matrix exponential, we have

$$e^{CXC^{-1}} = Ce^X C^{-1} \,, \tag{2.38}$$

which is an element of G.

2. $sX \in \mathfrak{g}$ for all real numbers s.

This is simply because $e^{t(sX)} = e^{(ts)X}$, which by definition is an element of G.

3. $X + Y \in \mathfrak{g}$.

This is a non-trivial fact from the Lie product formula

$$e^{X+Y} = \lim_{m\to\infty}(e^{\frac{X}{m}}e^{\frac{Y}{m}})^m \,. \tag{2.39}$$

The limit is also an element of G because G is closed.

4. $XY - YX \in \mathfrak{g}$.

Because of the property (7) of matrix exponential,

$$\frac{d}{dt}(e^{tX}Ye^{-tX})|_{t=0} = XY - YX \,. \tag{2.40}$$

From the point 1, we have $e^{tX}Ye^{-tX} \in \mathfrak{g}$. Since we can write

$$XY - YX = \lim_{t\to 0}\frac{e^{tX}Ye^{-tX} - Y}{t} \,, \tag{2.41}$$

along with the point 3 and $\mathfrak{g}$ is topologically closed (as a vector space), we can conclude that $XY - YX \in \mathfrak{g}$.

• From the properties 2, 3, 4, we indeed see that $\mathfrak{g}$ is closed under the Lie algebra operations.

• Examples:

(1) For the abelian Lie group $G = U(1)$, whose elements are $e^{i\theta}$ ($\theta \in [0, 2\pi)$, the associated (real) Lie algebra is $\mathfrak{g} = \mathfrak{u}(1)$, whose elements are $i\theta$ ($\theta \in \mathbb{R}$). The generator of $\mathfrak{u}(1)$ is i, which commutes with itself

$$[i, i] = 0 \tag{2.42}$$

Moreover, for $G = U(1)^N$, the associated Lie algebra is $\mathfrak{g} = \mathfrak{u}(1)^{\oplus N}$, whose generators are

$$(i, 0, \ldots, 0) \ , \ (0, i, \ldots, 0), \ldots, (0, 0, \ldots, i) \,. \tag{2.43}$$

In fact, we have the following theorem (Proof: B. Hall p. 57): for any abelian Lie group G, the associated Lie algebra $\mathfrak{g}$ is also abelian.

(2) For $G = GL(n, \mathbb{R})$, the associated Lie algebra is the matrix Lie algebra of $n \times n$ real matrices, $\mathfrak{gl}(n, \mathbb{R})$. For $G = GL(n, \mathbb{C})$, the associated (complex) Lie algebra is $\mathfrak{gl}(n, \mathbb{C})$. From the definition one can see that $\mathfrak{gl}(n, \mathbb{C})$ is the complexification of $\mathfrak{gl}(n, \mathbb{R})$.

The generators of $\mathfrak{gl}(n)$ can be taken as $E_{k,l}(k,l \in 1,\dots n)$, whose matrix elements are

$$(E_{k,l})_{i,j} = \delta_{ik}\delta_{jl}\,. \tag{2.44}$$

To derive the commutation relation between these generators, we compute

$$\begin{aligned}(E_{i,j}E_{k,l} - E_{k,l}E_{i,j})_{m,n} &= (E_{i,j})_{m,p}(E_{k,l})_{p,n} - (E_{k,l})_{m,p}(E_{i,j})_{p,n} \\ &= \delta_{im}\delta_{jp}\delta_{kp}\delta_{ln} - \delta_{km}\delta_{lp}\delta_{ip}\delta_{jn} \\ &= \delta_{im}\delta_{jk}\delta_{nl} - \delta_{km}\delta_{il}\delta_{jn} \\ &= \delta_{jk}(E_{i,l})_{m,n} - \delta_{il}(E_{k,j})_{m,n}\end{aligned} \tag{2.45}$$

Hence the commutation relation is

$$[E_{i,j}, E_{k,l}] = \delta_{jk}E_{i,l} - \delta_{il}E_{k,j}\,. \tag{2.46}$$

• Remark: if one views $\mathfrak{gl}(n,\mathbb{C})$ as a real Lie algebra, it is $2n^2$-dimensional ($\mathbb{C}^{n^2} = \mathbb{R}^{2n^2}$), and the generators are $E_{k,l}, iE_{k,l}(k,l \in 1,\dots n)$

(3) For $G = SL(n,\mathbb{R})$, the associated Lie algebra is the matrix Lie algebra of $n \times n$ traceless real matrices, $\mathfrak{sl}(n,\mathbb{R})$. For $G = SL(n,\mathbb{C})$, the associated (complex) Lie algebra is $\mathfrak{sl}(n,\mathbb{C})$. One can see that $\mathfrak{sl}(n,\mathbb{C})$ is the complexification of $\mathfrak{sl}(n,\mathbb{R})$.

The associated Lie algebra of $SL(n,\mathbb{R})$ (or $SL(n,\mathbb{C})$) should consist of matrices X satisfying

$$\det(e^{tX}) = 1 \tag{2.47}$$

for any $t \in \mathbb{R}$. It implies that

$$e^{tr(X)} = \det(e^X) = 1\,, \tag{2.48}$$

and hence

$$tr(X) = 0\,. \tag{2.49}$$

The generators of $\mathfrak{sl}(n)$ can be taken as $E_{k,l}$ $(k,l = 1,\dots,n\ , k \neq l)$ and $E_{k,k} - E_{k+1,k+1}$ $(k = 1,\dots,n-1)$.

(4) For $G = U(n)$, the associated Lie algebra is $\mathfrak{u}(n)$, which consists of all $n \times n$ complex matrices with $X^\dagger = -X$. For $G = SU(n)$, the associated Lie algebra is $\mathfrak{su}(n)$, which consists of all $n \times n$ complex traceless matrices with $X^\dagger = -X$.

Since a matrix U is unitary if and only if $U^\dagger = U^{-1}$, e^{tX} is unitary if and only if

$$(e^{tX})^\dagger = (e^{tX})^{-1}\,. \tag{2.50}$$

Using the property (2) of the matrix exponential, the l.h.s. equals to $e^{tX^\dagger}$. Using the property (3) of the matrix exponential, the r.h.s. equals to e^{-tX}. Hence the above equation is equivalent to

$$X^\dagger = -X\,. \tag{2.51}$$

For the case of $SU(n)$, one combines the arguments above with the traceless argument in the case (3).

The generators of $\mathfrak{u}(n)$ can be taken as $E_{k,l} - E_{l,k}$, $iE_{k,l} + iE_{l,k}$ $(k, l = 1, \ldots, n,\ k < l)$ and $iE_{k,k}$ $(k = 1, \ldots, n)$.

The generators of $\mathfrak{su}(n)$ can be taken as $E_{k,l} - E_{l,k}$, $iE_{k,l} + iE_{l,k}$ $(k, l = 1, \ldots, n,\ k < l)$ and $iE_{k,k} - iE_{k+1,k+1}$ $(k = 1, \ldots, n-1)$.

• Note that $\mathfrak{u}(n) = \mathfrak{su}(n) \oplus \mathfrak{u}(1)$. The $\mathfrak{u}(1)$ factor is generated by the diagonal element $iI_{n\times n}$.

For example, the generators of $\mathfrak{su}(2)$ can be chosen as

$$J_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}\ ,\ J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\ ,\ J_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}\,. \tag{2.52}$$

Their commutation relation is

$$[J_a, J_b] = -2\epsilon_{abc} J_c\,. \tag{2.53}$$

One can also choose the rescaled generators $X_i = -\frac{1}{2}J_i$, which gives the cross product algebra:

$$[X_a, X_b] = \epsilon_{abc} X_c\,, \tag{2.54}$$

or the Pauli matrices $\sigma_a = -iJ_a$,

$$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c\,. \tag{2.55}$$

One can also check that (from the counting of independent parameters) the complexification of $\mathfrak{u}(n)$ is $\mathfrak{gl}(n, \mathbb{C})$, and the complexification of $\mathfrak{su}(n)$ is $\mathfrak{sl}(n, \mathbb{C})$. For instance, the complexification of $\mathfrak{u}(n)$ has complex dimension n^2, which is also a subalgebra of $\mathfrak{gl}(n, \mathbb{C})$. However, since the complex dimension of $\mathfrak{gl}(n, \mathbb{C})$ is equal to n^2, $\mathfrak{u}(n)_\mathbb{C} \cong \mathfrak{gl}(n, \mathbb{C})$ has to hold, because a Lie algebra is a vector space.

(5) For $G = O(n)$ and $SO(n)$, the associated real Lie algebra consists of all $n \times n$ real matrices with $X^T = -X$. It is denoted as $\mathfrak{so}(n)$.

One can simply apply the arguments in the cases of $G = U(n)$ and $SU(n)$, with the restriction that the matrices are all real. Hence $\mathfrak{so}(n)$ is a subalgebra of $\mathfrak{su}(n)$.

The generators of $\mathfrak{so}(n)$ can be taken as $E_{k,l} - E_{l,k}$ $(k, l = 1, \ldots, n,\ k < l)$.

• For the generators of $\mathfrak{so}(3)$, we can take the following ones

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \ , \ L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \ , \ L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} . \tag{2.56}$$

They satisfy

$$[L_a, L_b] = \epsilon_{abc} L_c \, , \tag{2.57}$$

which is isomorphic to $\mathfrak{su}(2)$.

(6) For $G = O(p,q)$ and $SO(p,q)$ $(p+q=n)$, the associated real Lie algebra $\mathfrak{so}(p,q)$ are generally different from $\mathfrak{so}(n)$.

The Lie algebra $\mathfrak{so}(p,q)$ consists of $(p+q)\times(p+q)$ matrices X satisfying

$$gX^T g = -X \, , \tag{2.58}$$

$$g = \mathrm{diag}(\underbrace{1,\ldots,1}_{p},\underbrace{-1,\ldots,-1}_{q}) \, . \tag{2.59}$$

To derive the generators of $\mathfrak{so}(p,q)$, we write X in block form:

$$X = \begin{pmatrix} A_{p\times p} & B_{p\times q} \\ C_{q\times p} & D_{q\times q} \end{pmatrix} . \tag{2.60}$$

The equation (2.58) can be written as

$$\begin{pmatrix} A^T & -C^T \\ -B^T & D^T \end{pmatrix} = -\begin{pmatrix} A & B \\ C & D \end{pmatrix} . \tag{2.61}$$

The generators are hence $E_{k,l} - E_{l,k}$ $(k,l = 1,\ldots,p,\ k<l)$, $E_{k,l} - E_{l,k}$ $(k,l = p+1,\ldots,p+q,\ k<l)$, $E_{k,l} + E_{l,k}$, $(k = 1,\ldots,p,\ l = p+1,\ldots,p+q)$.

Let us consider the example of $\mathfrak{so}(2,1)$, whose generators are

$$L'_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \ , \ L'_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \ , \ L'_3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} . \tag{2.62}$$

Their commutation relations are

$$[L'_1, L'_2] = -L'_3 \ , \ [L'_2, L'_3] = L'_1 \ , \ [L'_3, L'_1] = L'_2 \tag{2.63}$$

It is impossible to perform a real linear $\mathfrak{gl}(\mathfrak{g},\mathbb{R})$ transformation M_{ij} on the generators L'_i, such that the new generators $L_i = M_{ij}L'_j$ satisfy the commutation relation of $\mathfrak{so}(3)$. Hence $\mathfrak{so}(3)$ and $\mathfrak{so}(2,1)$ are non-isomorphic as real Lie algebras.

Nonetheless, if one consider the complexification $\mathfrak{so}(3)_{\mathbb{C}}$ and $\mathfrak{so}(2,1)_{\mathbb{C}}$, they are equivalent. One can take for example $L_1 = iL'_1$, $L_2 = iL'_2$, $L_3 = L'_3$ and we have

$$[L_i, L_j] = \epsilon_{ijk} L_k \tag{2.64}$$

(7) For $G = Sp(2n, \mathbb{R})$, the associated Lie algebra $\mathfrak{sp}(2n, \mathbb{R})$ consists of real matrices X with

$$\Omega X^T \Omega = X \,, \tag{2.65}$$

where Ω is the one in the definition of symplectic groups (1.38). For $G = Sp(2n, \mathbb{C})$, the Lie algebra $\mathfrak{sp}(2n, \mathbb{C})$ consists of complex matrices obeying the same condition. In block diagonal form, X has the form of

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{2.66}$$

which satisfies

$$A = -D^T \ , \ B = B^T \ , \ C = C^T \,. \tag{2.67}$$

The generators can be taken as $E_{k,l} - E_{l+n,k+n}$ $(k, l = 1, \ldots, n)$, $E_{k,l+n} + E_{l,k+n}$ $(k, l = 1, \ldots, n, \ k < l)$, $E_{k,k+n}$ $(k = 1, \ldots, n)$, $E_{k+n,l} + E_{l+n,k}$ $(k, l = 1, \ldots, n, \ k < l)$, $E_{k+n,k}$ $(k = 1, \ldots, n)$. The number of generators is indeed

$$\dim(\mathfrak{sp}(2n, \mathbb{R})) = n(2n+1) \,. \tag{2.68}$$

Obviously the complexification of $\mathfrak{sp}(2n, \mathbb{R})$ is $\mathfrak{sp}(2n, \mathbb{C})$.

• Remark: the relation between the classifications of Lie groups, real and complex Lie algebras.

(1) If Lie groups $G_1 \cong G_2$, then their corresponding real Lie algebras satisfy $\mathfrak{g}_1 \cong \mathfrak{g}_2$, but not vice versa.

Example: $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong \mathfrak{o}(3)$, but $SU(2) \ncong SO(3) \ncong O(3)$;

(2) Each Lie algebra $\mathfrak{g}$ corresponds to a unique (connected) simply-connected Lie group G (the universal cover).

Example: $\mathfrak{su}(2)$ corresponds to $SU(2)$. $\mathfrak{sl}(2, \mathbb{C})$ corresponds to $SL(2, \mathbb{C})$.

(3) If real Lie algebras $\mathfrak{g}_1 \cong \mathfrak{g}_2$, then their complexifications satisfy $\mathfrak{g}_{1,\mathbb{C}} \cong \mathfrak{g}_{2,\mathbb{C}}$, but not vice versa.

Example: $\mathfrak{so}(3)_{\mathbb{C}} \cong \mathfrak{so}(2,1)_{\mathbb{C}}$, but $\mathfrak{so}(3) \ncong \mathfrak{so}(2,1)$.

2.5 General concepts of representations

2.5.1 Representations of Lie group and Lie algebra

In this section we introduce the representations of Lie groups G and Lie algebras $\mathfrak{g}$, which are ways to "write down" the elements in G and $\mathfrak{g}$. Here we are only interested in the "linear" representations which has the form of matrices. Formally we define:

• A *finite-dimensional complex representation* of a Lie group G is a Lie group homomorphism $\Pi : G \to GL(V) \cong GL(n, \mathbb{C})$, where $V = \mathbb{C}^n$ is a finite dimensional complex vector space. The dimension of the complex representation $\dim(\Pi) = \dim(V) = n$.

Recall the definition of Lie group homomorphism, which means for all $g_1, g_2, g \in G$,

$$\Pi(g_1 \cdot g_2) = \Pi(g_1) \cdot \Pi(g_2) \ , \ \Pi(e) = I \ , \ \Pi(g^{-1}) = \Pi(g)^{-1} \, . \tag{2.69}$$

• A *finite-dimensional complex representation* of a Lie algebra $\mathfrak{g}$ is a Lie algebra homomorphism $\pi : \mathfrak{g} \to \mathfrak{gl}(V) \cong \mathfrak{gl}(n, \mathbb{C})$, where $V = \mathbb{C}^n$ is a finite dimensional complex vector space. The dimension of the complex representation $\dim(\pi) = \dim(V) = n$.

Recall the definition of Lie algebra homomorphism, which means

$$\pi([X, Y]) = [\pi(X), \pi(Y)] \ , \ \forall X, Y \in \mathfrak{g} \tag{2.70}$$

• If the homomorphism Π or π is injective, then Π or π is called a *faithful representation.*

When we talk about a representation later, it means a complex representation (even if the Lie algebra $\mathfrak{g}$ is real) by default. Nonetheless one can still define the notion of real representation of G or $\mathfrak{g}$ by letting V to be a real vector space. The dimension is still defined as $\dim(V)$.

Note that when we talk about the action of a group G or Lie algebra $\mathfrak{g}$ on a vector space V, we often use the notation

$$\Pi(g)v := g \cdot v \quad (g \in G, v \in V) \, , \tag{2.71}$$

which is nothing but a matrix multiplied on a column vector.

Now using the relation between matrix Lie group G and its associated matrix Lie algebra $\mathfrak{g}$, one can naturally construct an (associated) representation π of $\mathfrak{g}$ from a representation Π of G which satisfies

$$\Pi(e^X) = e^{\pi(X)} \tag{2.72}$$

for all $X \in \mathfrak{g}$. One can also compute $\pi(X)$ with

$$\pi(X) = \frac{d}{dt}\Pi(e^{tX})|_{t=0}\,. \tag{2.73}$$

• Note that the representation Π of G and its associated representation π of $\mathfrak{g}$ act on the same vector space $\mathbb{C}^n$ (finite and infinitesimal transformation act on the same physical object)!

• A representation Π of G is *unitary* if $\Pi(g)$ is a unitary operator ($\Pi(g)^\dagger\Pi(g) = I$) for every element $g \in G$. In other words, an n-dimensional unitary representation is a group homomorphism $\Pi:\ G \to U(n)$.

• Similarly, the associated representation π of $\mathfrak{g}$ is unitary if $\pi^\dagger(X) = -\pi(X)$ for all $X \in \mathfrak{g}$. See the relation between the unitary group $U(n)$ and its Lie algebra $\mathfrak{u}(n)$.

• Theorem: a Lie group G has a faithful finite-dimensional unitary representation if and only if G is compact.

As an example, consider the 1-dimensional Lie groups $GL(1,\mathbb{R})$ and $U(1)$. Obviously the former does not a have a faithful finite-dimensional unitary representation. While the $U(1)$ group has faithful 1-dimensional unitary representations $\Pi_a:\ e^{i\theta} \to e^{ia\theta}$ $(a = \pm 1)$.

For a Lie group or Lie algebra, the classification of representations is an important question. In physics, this corresponds to classifying all the possible matter fields charged under some gauge group or global symmetry group. To talk about classification of representations, we first need to define the homomorphism and isomorphism between representations.

• A *homomorphism* (*intertwining map*) between two representations $\Pi : G \to GL(V)$ and $\Sigma : G \to GL(W)$ of G, $\phi : V \to W$ can be defined as

$$\phi(\Pi(g)v) = \Sigma(g)\phi(v)\,, \tag{2.74}$$

for any $g \in G$ and $v \in V$.

• Two representations are *isomorphic* if ϕ is invertible.

The notions above can be analogously introduced for a Lie algebra $\mathfrak{g}$, replacing $g \in G$ with $X \in \mathfrak{g}$ and Π with π.

The classification of all the representations of G or $\mathfrak{g}$ is too complicated and tedious. A simpler thing to do is classifying the building blocks of representations, which are the irreducible representations.

• Now we define the notion of irreducible representations (irreps). For a representation Π of a Lie group G acting on vector space V, a subspace $W \subset V$ is called *invariant* if $\Pi(g)w \in W$ for all $w \in W$ and $g \in G$. If the only invariant subspaces of V are $\{0\}$ and V, then V is called *irreducible*.

The classification of irreps for a Lie algebra $\mathfrak{g}$ up to isomorphism is a central subject of this course.

We present the Lie group version of Schur's lemma:

(1) Let V and W be irreducible real or complex representations of a group or Lie algebra and let $\phi : V \to W$ be an intertwining map. Then either $\phi = 0$ or ϕ is an isomorphism.

(2) Let V be an irreducible complex representation of a group or Lie algebra and let $\phi : V \to V$ be an intertwining map. Then $\phi = \lambda I$ for some $\lambda \in \mathbb{C}$.

(3) Let V and W be irreducible complex representations of a group or Lie algebra and let $\phi_1, \phi_2 : V \to W$ be intertwining maps. Then $\phi_1 = \lambda\phi_2$ for some $\lambda \in \mathbb{C}$.

As an important consequence, if Π is an irreducible representation of G, and A is in the center of a matrix Lie group G, then $\Pi(A) = \lambda I$ for some $\lambda \in \mathbb{C}$. Similar statement holds for π of the Lie algebra $\mathfrak{g}$.

Proof for the group version: if A is in the center of G, then for any $B \in G$,

$$\Pi(A)\Pi(B) = \Pi(AB) = \Pi(BA) = \Pi(B)\Pi(A)\,. \tag{2.75}$$

For any $v \in V$, we have

$$\Pi(A)\Pi(B)v = \Pi(B)\Pi(A)v\,. \tag{2.76}$$

Hence $\Pi(A)$ is exactly an intertwining map between V and itself. Due to Schur's lemma (2), $\Pi(A) = \lambda I$ for some $\lambda \in \mathbb{C}$.

We have the following propositions relating a matrix Lie group G and its associated (real) Lie algebra $\mathfrak{g}$, assuming G is connected (proof see B. Hall p. 80):

(1) π of $\mathfrak{g}$ is irreducible if and only if Π of G is irreducible.

(2) If π_1, π_2 of $\mathfrak{g}$ are isomorphic if and only if Π_1, Π_2 of G are isomorphic.

Assuming G is simply-connected, we have additionally

(3) every representation π of $\mathfrak{g}$ can be constructed from a representation Π of G.

• Examples of Lie group and Lie algebra representations:

(1) The representations of $U(1)$ and $\mathfrak{u}(1)$

From Schur's lemma, the irreducible representation of a commutative group of Lie algebra is always one-dimensional. Take an abelian group G for example, since the center of G is itself, from the Schur's lemma, one can see that $\Pi(A) = \lambda I$ for all elements $A \in G$. Hence any subspace of Π is an invariant space, and Π cannot be irreducible if $\dim(\Pi) > 1$.

Take $G = U(1)$, the representations $\Pi_a :\ e^{i\theta} \to e^{ia\theta}$ $(a \in \mathbb{Z})$ are the only irreducible representations of $U(1)$. The corresponding representation π_a of $\mathfrak{u}(1)$ is $\pi_a : i\theta \to ia\theta$ $(a \in \mathbb{Z})$.

Nonetheless, the full set of irreducible representions of $\mathfrak{u}(1)$ is $\pi_a : i\theta \to ia\theta$ $(a \in \mathbb{C})$ $(a \in \mathbb{R}$ if we want unitary representations). This is because $U(1)$ is not simply-connected, and there is not a one-to-one correspondence between irreps of $U(1)$ and $\mathfrak{u}(1)$.

The number a in the representation Π_a or π_a is usually called the "charge" of the $U(1)$ or $\mathfrak{u}(1)$ representation. This is just the electric charge of a $U(1)$ gauge theory.

• Remark: The representations of $GL(1,\mathbb{R})$ and $\mathfrak{u}(1) = \mathfrak{gl}(1,\mathbb{R})$ are one-to-one correspond to each other. For a non-compact $GL(1,\mathbb{R})$ gauge theory, the electric charge is not quantized.

(2) The *standard representation* Π_s of a matrix group G: the inclusion map from G to $GL(n,\mathbb{C})$. By definition these are all faithful representations.

a) For $G = GL(n,\mathbb{R})$, $GL(n,\mathbb{C})$, $SL(n,\mathbb{R})$, $SL(n,\mathbb{C})$, $\dim(\Pi_s) = n$, non-unitary

b) For $G = U(n)$, $SU(n)$, $\dim(\Pi_s) = n$, unitary

c) For $G = SO(n)$, $\dim(\Pi_s) = n$, unitary

d) For $G = SO(p,q)$, $(p,q > 0)$, $\dim(\Pi_s) = p + q$, non-unitary

e) For $G = Sp(2n,\mathbb{R})$, $Sp(2n,\mathbb{C})$, $\dim(\Pi_s) = 2n$, non-unitary

f) For $G = USp(2n) = Sp(n)$, $\dim(\Pi_s) = 2n$, unitary

• Similarly the corresponding representation π_s of the associated Lie algebra $\mathfrak{g}$ is also called the standard representation. The standard representations are typically called the "fundamental representation" in physics literature. For example, the n-dimensional fundamental representation of $\mathfrak{su}(n)$ is typically called $\mathbf{n}$.

• Note that the above examples are mostly irreducible representations. But for abelian groups such as $SO(2)$, $SO(1,1)$, the 2-dimensional standard representation is reducible as complex representations.

(3) The trivial representation: $\Pi : G \to GL(1,\mathbb{C})$ satisfies $\Pi(g) = 1$ for all $g \in G$. Similarly for a Lie algebra $\mathfrak{g}$, the trivial representation $\pi : \mathfrak{g} \to \mathfrak{gl}(1,\mathbb{C})$ satisfies $\pi(g) = 0$ for all $g \in \mathfrak{g}$. The trivial representation is one-dimensional, and not faithful.

(4) The adjoint representation of G and $\mathfrak{g}$.

For each $A \in G$, define a linear map (the *adjoint map*) $\mathrm{Ad}_A : \mathfrak{g} \to \mathfrak{g}$ by

$$\mathrm{Ad}_A(X) = AXA^{-1}\,. \tag{2.77}$$

Let $GL(\mathfrak{g})$ by the group of all linear transformations of $\mathfrak{g}$. Then the map $A \to \mathrm{Ad}_A$ is a homomorphism from G to $GL(\mathfrak{g})$, as one can easily check $Ad_{AB}(X) = ABXB^{-1}A^{-1} = Ad_A Ad_B(X)$.

The map Ad defines the *adjoint representation* of G, $adj(G)$. Its dimension

$$\dim(adj(G)) = \dim(\mathfrak{g}) = \dim(G)\,. \tag{2.78}$$

For an element $X \in \mathfrak{g}$, we define the linear map $\mathrm{ad}_X : \mathfrak{g} \to \mathfrak{g}$ by

$$\mathrm{ad}_X Y = [X, Y]\,. \tag{2.79}$$

The map from X to ad_X is a Lie algebra homomorphism from $\mathfrak{g}$ to $\mathfrak{gl}(\mathfrak{g})$. One can check that

$$\mathrm{ad}_{[X,Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y]\,. \tag{2.80}$$

This map defines the *adjoint representation* of $\mathfrak{g}$, $adj(\mathfrak{g})$. Its dimension is also

$$\dim(adj(\mathfrak{g})) = \dim(\mathfrak{g})\,. \tag{2.81}$$

As an example, consider the adjoint representation of $\mathfrak{su}(2)$. Let us take the basis of $\mathfrak{su}(2)$ to be X_1, X_2, X_3, with commutation relation

$$[X_i, X_j] = \epsilon_{ijk} X_k\,. \tag{2.82}$$

Now from the definition of the adjoint representation, we have

$$\mathrm{ad}_{X_i} X_j = \epsilon_{ijk} X_k\,. \tag{2.83}$$

Consider a generic vector $a_1X_1 + a_2X_2 + a_3X_3$ in the 3d vector space of the representation ad, we have

$$\begin{aligned}
\mathrm{ad}_{X_1}(a_1X_1 + a_2X_2 + a_3X_3) &= a_2X_3 - a_3X_2 \\
\mathrm{ad}_{X_2}(a_1X_1 + a_2X_2 + a_3X_3) &= a_3X_1 - a_1X_3 \\
\mathrm{ad}_{X_3}(a_1X_1 + a_2X_2 + a_3X_3) &= a_1X_2 - a_2X_1
\end{aligned} \tag{2.84}$$

The 3×3 matrices forms of the representation ad_{X_i} acting on $(a_1, a_2, a_3)^T$ are then

$$\mathrm{ad}_{X_1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}\,,\ \mathrm{ad}_{X_2} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}\,,\ \mathrm{ad}_{X_3} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\,. \tag{2.85}$$

Note that the matrix elements of $(\mathrm{ad}_{X_i})_{jk}$ are equal to the structure constant c_{ikj} (this holds in general).

• An important example of adjoint representation is the gauge boson A_μ in gauge theory, e. g. gluon in QCD transforms under the adjoint representation **8** of the $SU(3)$ gauge group.

(5) Direct sum of representations: let $\Pi_1, \Pi_2, \ldots, \Pi_k$ be representations of G, acting on $V_1, V_2, \ldots, V_k$ respectively. The *direct sum* representation $\Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k$ is defined as

$$[\Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k(g)](v_1, v_2, \ldots, v_k) = (\Pi_1(g)v_1, \Pi_2(g)v_2, \ldots, \Pi_k(g)v_k)\,. \tag{2.86}$$

In matrix form,

$$\Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k(g) = \begin{pmatrix} \Pi_1(g) & 0 & \ldots & 0 \\ 0 & \Pi_2(g) & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \Pi_k(g) \end{pmatrix}. \tag{2.87}$$

One can define the direct sum $\pi_1 \oplus \pi_2 \oplus \cdots \oplus \pi_k$ of Lie algebra representations in the same way. The dimension of the direct sum representation is

$$\dim(\Pi_1 \oplus \Pi_2 \oplus \cdots \oplus \Pi_k) = \dim(\Pi_1) + \dim(\Pi_2) + \cdots + \dim(\Pi_k)\,, \tag{2.88}$$

$$\dim(\pi_1 \oplus \pi_2 \oplus \cdots \oplus \pi_k) = \dim(\pi_1) + \dim(\pi_2) + \cdots + \dim(\pi_k)\,. \tag{2.89}$$

An important question is, given a Lie group G or a Lie algebra $\mathfrak{g}$, whether an arbitrary representation can be decomposed into a direct sum of irreps? We introduce the notion of *complete reducibility*:

• A finite-dimensional representation of G or $\mathfrak{g}$ is *completely reducible* if it is isomorphic to a direct sum of irreps.

• G or $\mathfrak{g}$ is said to have complete reducibility if any representation is completely reducible.

• It turns out that many Lie groups and Lie algebras and their representations do not have this property. But the following important classes do (B. Hall p. 92, 275):

1. If G is a matrix Lie group and Π is a unitary representation, then Π is completely reducible. Similarly, if $\mathfrak{g}$ is a real Lie algebra and π is a unitary representation, then π is completely reducible.

2. If G is a compact matrix Lie group, then it has complete reducibility.

3. If a complex Lie algebra $\mathfrak{g}$ is a "semisimple" Lie algebra, then it has complete reducibility. (the notion of semisimple will be introduced later)

(6) Tensor product of representations (version 1): let Π_1 be a representation of G_1, acting on space V_1 and Π_2 be a representation of G_2, acting on space V_2. The *tensor product* representation of $G_1 \times G_2$ acting on $V_1 \otimes V_2$ is defined by

$$(\Pi_1 \otimes \Pi_2)(A, B) = \Pi_1(A) \otimes \Pi_2(B)\,, \tag{2.90}$$

for any $A \in G_1$, $B \in G_2$.

In matrix form, suppose that $\Pi_1(A)_{ij} = a_{ij}$ $(i, j = 1, \ldots, m)$, $\Pi_2(B)_{ij} = b_{ij}$ $(i, j = 1, \ldots, n)$, $(\Pi_1 \otimes \Pi_2)(A, B) =$

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & \ldots & a_{11}b_{1n} & a_{12}b_{11} & \ldots & a_{12}b_{1n} & a_{1m}b_{11} & \ldots & a_{1m}b_{1n} \\ a_{11}b_{21} & a_{11}b_{22} & \ldots & a_{11}b_{2n} & a_{12}b_{21} & \ldots & a_{12}b_{2n} & a_{1m}b_{21} & \ldots & a_{1m}b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{11}b_{n1} & a_{11}b_{n2} & \ddots & a_{11}b_{nn} & a_{12}b_{n1} & \ldots & a_{12}b_{nn} & a_{1m}b_{n1} & \ldots & a_{1m}b_{nn} \\ a_{21}b_{11} & a_{21}b_{12} & \ldots & a_{21}b_{1n} & a_{22}b_{11} & \ldots & a_{22}b_{1n} & a_{2m}b_{11} & \ldots & a_{2m}b_{1n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{21}b_{n1} & a_{21}b_{n2} & \ldots & a_{21}b_{nn} & a_{22}b_{n1} & \ldots & a_{22}b_{nn} & a_{2m}b_{n1} & \ldots & a_{2m}b_{nn} \\ a_{m1}b_{11} & a_{m1}b_{12} & \ldots & a_{m1}b_{1n} & a_{n2}b_{11} & \ldots & a_{m2}b_{1n} & a_{mm}b_{11} & \ldots & a_{mm}b_{1n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{m1}b_{n1} & a_{m1}b_{n2} & \ldots & a_{m1}b_{nn} & a_{n2}b_{n1} & \ldots & a_{m2}b_{nn} & a_{mm}b_{n1} & \ldots & a_{mm}b_{nn} \end{pmatrix}. \tag{2.91}$$

Let π_1 be a representation of $\mathfrak{g}_1$, acting on V_1 and π_2 be a representation of $\mathfrak{g}_2$, acting on V_2, the *tensor product* representation of $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ acting on $V_1 \otimes V_2$ is defined by

$$(\pi_1 \otimes \pi_2)(X, Y) = \pi_1(X) \otimes I + I \otimes \pi_2(Y)\,, \tag{2.92}$$

for any $X \in \mathfrak{g}_1$, $Y \in \mathfrak{g}_2$.

To derive (2.92), we have

$$\begin{aligned} (\pi_1 \otimes \pi_2)(X, Y) &= \left[\frac{d}{dt}(\Pi_1 \otimes \Pi_2)(e^{tX}, e^{tY})\right]|_{t=0} \\ &= \left[\frac{d}{dt}(\Pi_1(e^{tX}) \otimes \Pi_2(e^{tY}))\right]|_{t=0} \\ &= \left[\frac{d}{dt}\Pi_1(e^{tX}) \otimes \Pi_2(e^{tY}) + \Pi_1(e^{tX}) \otimes \frac{d}{dt}\Pi_2(e^{tY})\right]|_{t=0} \\ &= \pi_1(X) \otimes I + I \otimes \pi_2(Y)\,. \end{aligned} \tag{2.93}$$

Here we used

$$\left[\frac{d}{dt}\Pi_i(e^{tX})\right]|_{t=0} = \pi_i(X) \quad (i = 1, 2)\,. \tag{2.94}$$

The dimension of the tensor product representation is

$$\dim(\Pi_1 \otimes \Pi_2) = \dim(\Pi_1) \cdot \dim(\Pi_2)\,, \tag{2.95}$$

$$\dim(\pi_1 \otimes \pi_2) = \dim(\pi_1) \cdot \dim(\pi_2)\,. \tag{2.96}$$

In physics, consider two gauge groups $G_1 = SU(M)$ and $G_2 = SU(N)$, the "bifundamental matter field" is in the representation $(\mathbf{M}, \mathbf{N})$ of $SU(M) \times SU(N)$. As an example, the standard model left-handed quarks are in the fundamental representation $\mathbf{3}$ of the QCD gauge group $SU(3)$ and the fundamental representation $\mathbf{2}$ of the weak interaction gauge group $SU(2)$, hence it can be written as $(\mathbf{3}, \mathbf{2})$.

(7) Tensor product of representations (version 2): let Π_1 and Π_2 be representations of G, acting on V_1 and V_2, the *tensor product* representation of G acting on $V_1 \otimes V_2$ is defined by

$$(\Pi_1 \otimes \Pi_2)(g) = \Pi_1(g) \otimes \Pi_2(g)\,. \tag{2.97}$$

Let π_1 and π_2 be representations of $\mathfrak{g}$, acting on V_1 and V_2, the *tensor product* representation of $\mathfrak{g}$ acting on $V_1 \otimes V_2$ is defined by

$$(\pi_1 \otimes \pi_2)(g) = \pi_1(g) \otimes I + I \otimes \pi_2(g)\,. \tag{2.98}$$

(Proof by the matrix exponential formula)

Similar to the case above, we have the same dimension formula

$$\dim(\Pi_1 \otimes \Pi_2) = \dim(\Pi_1) \cdot \dim(\Pi_2)\,, \tag{2.99}$$

$$\dim(\pi_1 \otimes \pi_2) = \dim(\pi_1) \cdot \dim(\pi_2)\,. \tag{2.100}$$

• An important question is how to write the tensor product $\Pi_1 \otimes \Pi_2$ (or $\pi_1 \otimes \pi_2$) as a direct sum of irreducible representations of G (or $\mathfrak{g}$). This is the "tensor product decomposition" or "Clebsch-Gordan" theory, which will be discussed later.

• Note that the direct sum and tensor product of representations satisfy the usual rules in linear algebra, e. g.

$$\begin{aligned}
\Pi_1 \oplus \Pi_2 &= \Pi_2 \oplus \Pi_1 \\
(\Pi_1 \oplus \Pi_2) \oplus \Pi_3 &= \Pi_1 \oplus (\Pi_2 \oplus \Pi_3) \\
\Pi_1 \otimes \Pi_2 &= \Pi_2 \otimes \Pi_1 \\
(\Pi_1 \otimes \Pi_2) \otimes \Pi_3 &= \Pi_1 \otimes (\Pi_2 \otimes \Pi_3) \\
(\Pi_1 \oplus \Pi_2) \otimes \Pi_3 &= (\Pi_1 \otimes \Pi_3) \oplus (\Pi_2 \otimes \Pi_3)\,.
\end{aligned} \tag{2.101}$$

(8) Dual representation: let Π be a representation of G acting on V, its *dual representation* Π^* is the representation acting on the dual vector space V^*, given by

$$\Pi^*(g) = [\Pi(g^{-1})]^T \,. \tag{2.102}$$

One can check that it satisfies the group homomorphism condition

$$\begin{aligned} \Pi^*(g_1)\Pi^*(g_2) &= [\Pi(g_1^{-1})]^T[\Pi(g_2^{-1})]^T = (\Pi(g_2^{-1})\Pi(g_1^{-1}))^T \\ &= \Pi(g_2^{-1}g_1^{-1})^T = \Pi((g_1g_2)^{-1})^T \,. \end{aligned} \tag{2.103}$$

Let π be a representation of $\mathfrak{g}$ acting on V, its dual representation π^* is the representation acting on V^*, given by

$$\pi^*(X) = -\pi(X)^T \,. \tag{2.104}$$

Obviously $\dim(\Pi^*) = \dim(\Pi)$, $\dim(\pi^*) = \dim(\pi)$.

• Note that for a unitary representation Π, the dual representation Π^* coincides with the *conjugate representation* defined as

$$\overline{\Pi}(g) = \overline{\Pi(g)} \,, \tag{2.105}$$

$$\overline{\pi}(X) = \overline{\pi(X)} \,. \tag{2.106}$$

• For the fundamental representation $\mathbf{n}$ of $SU(n)$, its conjugate representation is the anti-fundamental representation $\bar{\mathbf{n}}$.

• A *real representation* is the one where all $\Pi(g)$ or $\pi(X)$ are real. A real representation is always the conjugate of itself. But the converse it not true, a self-conjugate representation that is not real is called a *pseudo-real representation*.

• Anti-quarks are in the anti-fundamental representation $\bar{\mathbf{3}}$ of the QCD gauge group $SU(3)$.

(9) In the future sections, we typically consider a complex Lie algebra. Nonetheless, the representations for a real Lie algebra $\mathfrak{g}$ and its complexification $\mathfrak{g}_{\mathbb{C}}$ are closely related. We have the following proposition (B. Hall p. 80):

for every finite-dimensional complex representation π of $\mathfrak{g}$, there is a unique extension of π into a complex representation of $\mathfrak{g}_{\mathbb{C}}$: $\pi(X + iY) = \pi(X) + i\pi(Y)$ for all $X, Y \in \mathfrak{g}$. Moreover, π of $\mathfrak{g}_{\mathbb{C}}$ is irreducible if and only if π of $\mathfrak{g}$ is irreducible.

2.5.2 Representations of $SU(2)$, $\mathfrak{su}(2)$ and $SO(3)$

We discuss the classification irreducible representations of matrix Lie groups $SU(2)$, $SO(3)$ and Lie algebra $\mathfrak{su}(2)$. Let's first talk about $G = SU(2)$.

• Let $V_m \cong \mathbb{C}^{m+1}$ be the space of homogeneous polynomials of degree m in two complex variables z_1 and z_2. An element of V_m takes the form of

$$f(z_1, z_2) = a_0 z_1^m + a_1 z_1^{m-1} z_2 + \cdots + a_m z_2^m \,, \tag{2.107}$$

each a_i is an arbitrary complex number.

Now for each $U \in SU(2)$, which acts on $z = (z_1, z_2)^T$ as

$$Uz = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \,, \tag{2.108}$$

define a linear transformation $\Pi_m(U)$ on the space V_m as

$$[\Pi_m(U) f](z) = f(U^{-1} z) \,. \tag{2.109}$$

Expand it out explicitly, we can write down

$$[\Pi_m(U) f](z_1, z_2) = \sum_{k=0}^{m} a_k (U_{11}^{-1} z_1 + U_{12}^{-1} z_2)^{m-k} (U_{21}^{-1} z_1 + U_{22}^{-1} z_2)^k \,. \tag{2.110}$$

To find the explicit form of $\Pi_m(U)$, one just needs expand out the r. h. s. and read off the new coefficients

$$a'_k = (\Pi_m(U))_{kj} a_j \,. \tag{2.111}$$

of each monomial $z_1^{m-k} z_2^k$.

The map Π_m is thus a representation of $SU(2)$ of dimension $\dim(\Pi_m) = m + 1$, as one can check that

$$\begin{aligned} \Pi_m(U_1)[\Pi_m(U_2) f](z) &= [\Pi_m(U_2) f](U_1^{-1} z) \\ &= f(U_2^{-1} U_1^{-1} z) \\ &= \Pi_m(U_1 U_2) f(z) \,. \end{aligned} \tag{2.112}$$

• Note that this definition has the similar form as a dual representation. For example, for the case of $m = 1$, Π_1 defines the dual representation of the standard representation Π_s of $SU(2)$. Plug in Π_1 into (2.110):

$$[\Pi_1(U) f](z_1, z_2) = a_0 (U_{11}^{-1} z_1 + U_{12}^{-1} z_2) + a_1 (U_{21}^{-1} z_1 + U_{22}^{-1} z_2) \,. \tag{2.113}$$

The l.h.s. is equal to

$$(a_0', a_1') \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = (\Pi_1(U) \begin{pmatrix} a_0 \\ a_1 \end{pmatrix})^T \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \tag{2.114}$$

while the r.h.s. is equal to

$$(a_0, a_1) U^{-1} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} , \tag{2.115}$$

hence $\Pi_1(U)^T = U^{-1}$, and

$$\Pi_1(U) = (U^{-1})^T = \overline{U} . \tag{2.116}$$

(Note that for a unitary representation, the dual representation is isomorphic to the conjugate representation)

• As a comment, the 2-dimensional standard representation **2** of $SU(2)$ is actually self-dual. Since it is not a real representation, **2** is pseudo-real.

Let us write out

$$U = \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix} . \tag{2.117}$$

Thus $\Pi_1(U)$ is given by

$$\Pi_1(U) = \overline{U} = \begin{pmatrix} a - bi & c - di \\ -c - di & a + bi \end{pmatrix} . \tag{2.118}$$

The conjugate rep. of Π_1 is

$$\overline{\Pi}_1(U) = U = \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix} . \tag{2.119}$$

Now we define the isomorphism between the two representations:

$$\phi : \Pi_1(U) \to \overline{\Pi}_1(U) , \quad \phi = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} , \tag{2.120}$$

which is clearly invertible. One can check this is indeed a homomorphism between Lie algebra representations:

$$\phi(\Pi_1(U)v) = \overline{\Pi}_1 \phi(v) \tag{2.121}$$

by

$$\phi \cdot \bar{U} = U \cdot \phi . \tag{2.122}$$

• Now we work out the corresponding representation π_m of $\mathfrak{su}(2)$, using (2.73).

$$(\pi_m(X)f)(z) = \frac{d}{dt} f(e^{-tX}z)|_{t=0} \tag{2.123}$$

Denote $z(t) = e^{-tX}z = (z_1(t), z_2(t))$, using

$$\pi_m(X)f = \frac{\partial f}{\partial z_1}\frac{dz_1}{dt}|_{t=0} + \frac{\partial f}{\partial z_2}\frac{dz_2}{dt}|_{t=0}\,, \tag{2.124}$$

since $dz/dt|_{t=0} = -Xz$, we have

$$\pi_m(X)f = -\frac{\partial f}{\partial z_1}(X_{11}z_1 + X_{12}z_2) - \frac{\partial f}{\partial z_2}(X_{21}z_1 + X_{22}z_2)\,. \tag{2.125}$$

We can then choose a set of generators for the Lie algebra $\mathfrak{su}(2)$, such as (2.52):

$$J_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}\ ,\ J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\ ,\ J_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}\ , \tag{2.126}$$

The representation of π_m in terms of these generators are

$$\begin{aligned} \pi_m(J_1) &= -iz_1\frac{\partial}{\partial z_2} - iz_2\frac{\partial}{\partial z_1} \\ \pi_m(J_2) &= z_1\frac{\partial}{\partial z_2} - z_2\frac{\partial}{\partial z_1} \\ \pi_m(J_3) &= -iz_1\frac{\partial}{\partial z_1} + iz_2\frac{\partial}{\partial z_2}\,. \end{aligned} \tag{2.127}$$

It seems that the forms of these $\pi_m(J_i)$ are the same for every m, but they actually act on polynomials in different V_m.

Now we check the relation between π_1 and the standard representation **2** of $\mathfrak{su}(2)$. For $m = 1$, the action of (2.127) on an element $a_0z_1 + a_1z_2$ of the 2d vector space V_1 is

$$\begin{aligned} \pi_1(J_1)(a_0z_1 + a_1z_2) &= -ia_0z_2 - ia_1z_1 \\ \pi_1(J_2)(a_0z_1 + a_1z_2) &= a_1z_1 - a_0z_2 \\ \pi_1(J_3)(a_0z_1 + a_1z_2) &= -ia_0z_1 + ia_1z_2\,. \end{aligned} \tag{2.128}$$

Hence we have for $(i = 1, 2, 3)$,

$$(\pi_1(J_i)\begin{pmatrix} a_0 \\ a_1 \end{pmatrix})^T \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = (a_0, a_1)(-J_i)\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}\ . \tag{2.129}$$

Hence $\pi_1(J_i) = -J_i^T$, which is exactly the definition of the dual representation of $\mathbf{2}$.

Nonetheless, the representation $\mathbf{2}$ of $\mathfrak{su}(2)$ is self-dual as well, and it is pseudo-real.

Another convenient choice is to first consider the complexification $\mathfrak{su}(2)_{\mathbb{C}}$, and use the Lie algebra isomorphism $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C})$. Then we use the $\mathfrak{sl}(2,\mathbb{C})$ generators

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \ , \ X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \ , \ Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \ , \tag{2.130}$$

which satisfies the commutation relations

$$[H, X] = 2X \ , \ [H, Y] = -2Y \ , \ [X, Y] = H \, . \tag{2.131}$$

Using these generators, we can write (2.125) in terms of the Lie algebra generators

$$\begin{aligned} \pi_m(H) &= -z_1 \frac{\partial}{\partial z_1} + z_2 \frac{\partial}{\partial z_2} \\ \pi_m(X) &= -z_2 \frac{\partial}{\partial z_1} \\ \pi_m(Y) &= -z_1 \frac{\partial}{\partial z_2} \, . \end{aligned} \tag{2.132}$$

Applying to the element $z_1^{m-k} z_2^k$ of V_m, we get

$$\begin{aligned} \pi_m(H)(z_1^{m-k} z_2^k) &= (2k - m) z_1^{m-k} z_2^k \\ \pi_m(X)(z_1^{m-k} z_2^k) &= (k - m) z_1^{m-k-1} z_2^{k+1} \\ \pi_m(Y)(z_1^{m-k} z_2^k) &= -k z_1^{m-k+1} z_2^{k-1} \end{aligned} \tag{2.133}$$

Hence the element (state) $z_1^{m-k} z_2^k$ has eigenvalue $(2k-m)$ under the operator $\pi_m(H)$. The eigenvalue is called *weight* of the state. The operator $\pi_m(X)$ and $\pi_m(Y)$ has the effect of shifting the eigenvalue(weight) by $+2$ and -2 respectively, which can be thought as creation and annihilation operators in quantum mechanics (after a scaling factor 2).

More explicitly, consider an eigenvector u of $\pi_m(H)$, with eigenvalue α:

$$\pi_m(H) u = \alpha u \, . \tag{2.134}$$

Because

$$[\pi_m(H), \pi_m(X)] = \pi_m([H, X]) = 2\pi_m(X) \, , \tag{2.135}$$

we have

$$\begin{aligned} \pi_m(H)\pi_m(X)u &= \pi_m(X)\pi_m(H)u + 2\pi_m(X)u \\ &= (\alpha + 2)\pi_m(X)u \, . \end{aligned} \tag{2.136}$$

Similarly one can prove that

$$\pi_m(H)\pi_m(Y)u = (\alpha - 2)\pi_m(Y)u\,. \tag{2.137}$$

One can conclude that (1) either $\pi_m(X)u = 0$ or $\pi_m(X)u$ is an eigenvector of $\pi_m(H)$ with eigenvalue $\alpha + 2$; (2) either $\pi_m(Y)u = 0$ or $\pi_m(Y)u$ is an eigenvector of $\pi_m(H)$ with eigenvalue $\alpha - 2$.

Hence the representation π_m of $\mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{su}(2)_{\mathbb{C}}$ has $(m+1)$ states $z_1^m, z_1^{m-1}z_2, \ldots, z_2^m$ with weights $-m, -(m-2), \ldots, (m-2), m$. In particular, the state z_2^m with weight m is called the *highest weight state*. The representation π_m is also called the representation of the highest weight m.

In quantum mechanics, the representation π_m of $\mathfrak{su}(2)$ describes a spin $s = \frac{1}{2}m$ system. The $2s+1$ states in the system has $s_z = s, s-1, \ldots, -(s-1), -s$, which is the eigenvalue under $\frac{1}{2}\pi_m(H)$.

• Proposition: each representation π_m is irreducible.

One needs to show that every non-zero invariant subspace of V_m is equal to V_m. Recall the definition of invariant space $W \subset V_m$: $\pi_m(g)w \in W$ for all $g \in \mathfrak{sl}(2,\mathbb{C})$, $w \in W$.

Now suppose that W is such a space and

$$w = a_0 z_1^m + a_1 z_1^{m-1} z_2 + \cdots + a_m z_2^m \tag{2.138}$$

is a non-zero element of W. There's at least one $a_k \neq 0$. Let k_0 be the smallest k such that $a_k \neq 0$, and consider

$$\pi_m(X)^{m-k_0} w\,. \tag{2.139}$$

Since each application of $\pi_m(X)$ raises the power of z_2 by 1 in the homogeneous polynomial, and kills the term when the power of z_2 exceeds m, the resulting $\pi_m(X)^{m-k_0}w$ is just a non-zero multiple of z_2^m. Since W is invariant, one concludes that the basis z_2^m has to be contained in W.

Then one can apply $\pi_m(Y)$ k times on z_2^m, which generates every monomial $z_1^k z_2^{m-k}$. Hence $W = V_m$ and we have proved that V_m is irreducible.

• Moreover, following a longer proof of theorem 4.32 on (B. Hall p.96), one can show that every irreducible complex representation of $\mathfrak{sl}(2,\mathbb{C})$ is isomorphic to one of the π_m ($m \in \mathbb{N}$). Hence if one know that π of $\mathfrak{sl}(2,\mathbb{C})$ is irreducible and its dimension $\dim(\pi) = m+1$, it is always isomorphic to π_m, the irrep with highest weight m.

• Furthermore, since there is a one-to-one correspondence between irreducible complex representations of a real Lie algebra $\mathfrak{g}$ and that of its complexification $\mathfrak{g}_{\mathbb{C}}$, we can conclude that the irreps of $\mathfrak{su}(2)$ also falls into the same classification.

Then since the corresponding Lie group $SU(2)$ is simply-connected, we confirmed the one-to-one correspondence between the representations of $\mathfrak{su}(2)$ and $SU(2)$. (see theorem 5.6 on p. 119 on B. Hall for a more rigorous statement)

• Now we talk about the representations of $G = SO(3)$, which has the same Lie algebra $\mathfrak{g} = \mathfrak{so}(3) \cong \mathfrak{su}(2)$ as $SU(2)$.

Nonetheless, the only irreducible representation of $SO(3)$ are realized as the space V_l of homogeneous sphere harmonic polynomials on $\mathbb{R}^3$ of degree l. The way $SO(3)$ acts on V_l is

$$(\Pi_l(R)f)(\vec{x}) = f(R^{-1}\vec{x})\,, \tag{2.140}$$

where $R \in SO(3)$ is a 3d rotation matrix in $\mathbb{R}^3$, and $\vec{x} = (x, y, z)^T$.

For example, the vector space V_l for $l = 0$ corresponds to the constant function 1; $l = 1$ corresponds to the spherical harmonic polynomials x, y and z; $l = 2$ corresponds to the spherical harmonic polynomials xy, yz, xz, $x^2 - y^2$, $y^2 - z^2$. For general l, the basis of the space V_l is Y_{lm}, $m = -l, -(l-1), \ldots, (l-1), l$.

In quantum mechanics, the representation Π_l of $SO(3)$ describes a system with angular momentum $J = l$. The eigenvalue of a state under $\frac{1}{2}\pi_{2l}(H)$ is the J_z.

• Now we show that $SO(3)$ does not have even-dimensional irreps, by showing that the Lie algebra representation π_m of $\mathfrak{su}(2)$ cannot be uplifted to a representation Σ_m of $SO(3)$ when m is odd.

We take the following generators X_i for $\mathfrak{su}(2)$ and L_i for $\mathfrak{so}(3)$, which satisfy the same commutation relations:

$$X_1 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}\,,\ X_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\,,\ X_3 = \frac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}\,. \tag{2.141}$$

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}\,,\ L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}\,,\ L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\,. \tag{2.142}$$

Which $[X_a, X_b] = \epsilon_{abc}X_c$, $[L_a, L_b] = \epsilon_{abc}L_c$.

Now suppose that m is odd. Since $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, the representation σ_m of $\mathfrak{so}(3)$ amd π_m of $\mathfrak{su}(2)$ are isomorphic, we have

$$\sigma_m(L_3) = \pi_m(X_3)\,. \tag{2.143}$$

Note that $X_3 = -iH/2$, where

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\,. \tag{2.144}$$

Since we know that there is a basis u_0, u_1, $\ldots$, u_m for V_m, such that u_j is an eigenvector for $\pi_m(H)$ with eigenvalue $m-2j$. It means that u_j is an eigenvector for $\sigma_m(L_3)$, with eigenvalue $-i(m-2j)/2$. Hence in the basis of u_j, we can write

$$\sigma_m(L_3) = \mathrm{diag}(\frac{i}{2}(-m), \frac{i}{2}(-m+2), \ldots, \frac{i}{2}m)\,. \tag{2.145}$$

For the corresponding Lie group representation Σ_m of $SO(3)$, we should have

$$\Sigma_m(e^{2\pi L_3}) = e^{2\pi\sigma_m(L_3)}\,. \tag{2.146}$$

But one can evaluate the l.h.s. from

$$e^{2\pi L_3} = \begin{pmatrix} \cos 2\pi & -\sin 2\pi & 0 \\ \sin 2\pi & \cos 2\pi & 0 \\ 0 & 0 & 1 \end{pmatrix} = I\,, \tag{2.147}$$

while on the r.h.s.,

$$e^{2\pi\sigma_m(L_3)} = -I \tag{2.148}$$

when m is odd.

Hence it leads to contradiction, and the representation Σ_m for $SO(3)$ does not exist for odd m.

The irreducible representations of $SO(3)$ are all with dimension $2l+1$, which corresponds to the Lie algebra representation π_{2l} of $\mathfrak{so}(3) \cong \mathfrak{su}(2)$. This is an example where not every representation of the associated Lie algebra $\mathfrak{g}$ comes from the Lie group G, due to that G is not simply-connected.

• Comment: (1) all representations of $SO(3)$ are real (2) the odd-dimensional irreducible representations of $SU(2)$ and $\mathfrak{su}(2)$ are real (in the sense that it's isomorphic to a real representation), while the even-dimensional irreps of $SU(2)$ and $\mathfrak{su}(2)$ are pseudo-real.

• Comment: the even dimensional irreps of $\mathfrak{su}(2)$ correspond to the *projective representations* of $SO(3)$, but not actual representations.

2.6 Applications in gauge theory

For a continuous global symmetry, there is a non-trivial action of matrix Lie group element $g \in G$ on the physical fields. A field in physics means a map $\phi : M \to V$ from the spacetime manifold M to a vector space V where the field takes value.

For a field ϕ in the representation (Π, V), the group action is

$$\phi \to \Pi(g)\phi \equiv g \cdot \phi \,. \tag{2.149}$$

Here g is independent of the space-time coordinate. For example, for a free N-component complex scalar field $\phi \in \mathbb{C}^N$,

$$L = \partial_\mu \phi^\dagger \partial^\mu \phi \,, \tag{2.150}$$

the Lagrangian is invariant under a $G = U(N)$ global symmetry $\phi \to g\phi \equiv \Pi(g)\phi$, $g \in U(N)$ is a space-time independent parameter. In this case, Π is the standard representation $\mathbf{N}$ of G.

If we want G to become a gauge group, and G becomes local symmetry (also known as gauging the global symmetry), the transformation parameter g becomes dependent on the space-time coordinates.

Now the partial derivative $\partial_\mu \phi$ is no longer gauge covariant, because in general

$$\partial_\mu \phi' = \partial_\mu(\Pi(g(x))\phi) \neq \Pi(g(x))\partial_\mu \phi \,. \tag{2.151}$$

Hence the kinetic term

$$\partial_\mu \phi^\dagger \partial^\mu \phi \tag{2.152}$$

in the Lagrangian is not gauge invariant.

To restore the gauge invariance, we define a *covariant derivative*

$$D_\mu = \partial_\mu + \pi(A_\mu) \,, \tag{2.153}$$

where $A_\mu : M \to \mathfrak{g}$ is the *gauge field*, and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ is the corresponding representation of Π.

and require that the *gauge field* A_μ transforms as

$$A'_\mu = g A_\mu g^{-1} - (\partial_\mu g) g^{-1} \tag{2.154}$$

under the gauge transformation.

Note that in terms of the representation π and Π, we have

$$\pi(A'_\mu) = \Pi(g)\pi(A_\mu)\Pi(g)^{-1} - (\partial_\mu \Pi(g))\Pi(g)^{-1} \,. \tag{2.155}$$

Now we can check that

$$\begin{aligned}(D_\mu\phi)' &= (\partial_\mu + \pi(A_\mu)')(\Pi(g)\phi) \\ &= (\partial_\mu\Pi(g))\phi + \Pi(g)\partial_\mu\phi + \Pi(g)\pi(A_\mu)\phi - (\partial_\mu\Pi(g))\phi \\ &= \Pi(g)(D_\mu\phi)\,,\end{aligned} \tag{2.156}$$

which is gauge covariant.

After the gauge transformation $\phi' = \Pi(g)\phi$, the new kinetic term is

$$\begin{aligned}(D_\mu\phi')^\dagger(D^\mu\phi') &= (\Pi(g)D_\mu\phi)^\dagger\Pi(g)D^\mu\phi \\ &= (D_\mu\phi)^\dagger\Pi(g)^\dagger\Pi(g)D^\mu\phi \\ &= (D_\mu\phi)^\dagger(D^\mu\phi)\,.\end{aligned} \tag{2.157}$$

The kinetic term for the matter field ϕ is then gauge invariant. Note that the above derivation applies to any unitary representation Π, and the symmetry group G is in general a subgroup of $U(N)$.

Remark: for any compact matrix Lie group G, there exists a unitary group $U(N)$ such that G is a subgroup of $U(N)$, because G has a faithful finite-dimensional unitary representation.

• Now let's discuss more about the gauge field itself.

The gauge field A_μ is an element of the Lie algebra $\mathfrak{g}$ associated to the matrix Lie group G. In other words, A_μ is in the adjoint representation of G (acted by the gauge group G).

We can decompose A_μ in terms of the Lie algebra generators X_a:

$$A_\mu = A^a_\mu X_a\,. \tag{2.158}$$

Now we construct a gauge covariant field strength for A_μ:

$$\begin{aligned}F_{\mu\nu} &= [D_\mu, D_\nu] \\ &= \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]\,.\end{aligned} \tag{2.159}$$

After plugging in (2.158), the component form is

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + c_{bca}A^b_\mu A^c_\nu\,. \tag{2.160}$$

One can show that after the gauge transformation (2.154), using $D'_\mu = gD_\mu g^{-1}$,

$$F'_{\mu\nu} = gF_{\mu\nu}g^{-1}\,, \tag{2.161}$$

and the kinetic term

$$tr(F_{\mu\nu}F^{\mu\nu}) \tag{2.162}$$

is gauge invariant.

For a general matter field ϕ_i in the representation (Π, V) of G, $(i = 1, \ldots, \dim(V))$, the action of A_μ on ϕ can be written as $A_\mu \phi \equiv \pi(A_\mu)\phi$,

$$(A_\mu \phi)_i = A_\mu^a \pi(X_a)_{ij} \phi_j \,, \tag{2.163}$$

where $\pi(X_a)_{ij}$ are the matrix elements of X_a under the representation (π, V) of $\mathfrak{g}$.

• More detailed examples:

(1) $G = U(1)$, with charged matter fields

Let us consider a $U(1)$ gauge field A_μ coupled to complex scalar ϕ with charge $a \in \mathbb{Z}$. In the Lie group language, ϕ is in the representation Π_a of $U(1)$, which maps

$$\Pi_a(e^{i\theta}) = e^{ia\theta} \,. \tag{2.164}$$

The associated representation π_a of $\mathfrak{u}(1)$ is

$$\pi_a(i\theta) = ia\theta \,. \tag{2.165}$$

The gauge transformations of ϕ and A_μ are

$$\begin{aligned} \phi' &= \Pi_a(e^{i\theta})\phi = e^{ia\theta}\phi \\ A'_\mu &= A_\mu - i\partial_\mu \theta \,. \end{aligned} \tag{2.166}$$

$\theta(x) \in [0, 2\pi)$ is the periodic parameter of $U(1)$.

The covariant derivative is

$$D_\mu = \partial_\mu + \pi(A_\mu) = \partial_\mu + aA_\mu \,. \tag{2.167}$$

The gauge invariant field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \,. \tag{2.168}$$

The gauge invariant Lagrangian is

$$L = \frac{1}{2}(D_\mu \phi)^\dagger (D^\mu \phi) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \,. \tag{2.169}$$

• Now let us discuss an important extended operator in the $U(1)$ gauge theory, which is the Wilson loop

$$W_q(\mathcal{C}) = \exp\left(q \oint_{\mathcal{C}} A_\mu dx^\mu \right) \quad (q \in \mathbb{Z}) \,. \tag{2.170}$$

Why is q quantized? Let us consider the gauge transformation

$$A_\mu \to A_\mu - i\partial_\mu \theta \,, \tag{2.171}$$

$$\oint_{\mathcal{C}} A_\mu dx^\mu \to \oint_{\mathcal{C}} A_\mu dx^\mu - i \oint_{\mathcal{C}} \partial_\mu \theta dx^\mu \,. \tag{2.172}$$

Now the key point: there exists *large gauge transformation* $\theta_m : \mathcal{C} \to U(1)$, which corresponds to the non-trivial homotopy class $m \in \pi_1(U(1)) = \mathbb{Z}$. Quantatively, $\oint_{\mathcal{C}} \partial_\mu \theta_m dx^\mu$ is exactly 2π times the winding number m!:

$$\oint_{\mathcal{C}} \partial_\mu \theta_m dx^\mu = 2\pi m \,. \tag{2.173}$$

Hence to construct a gauge invariant operator out of $\oint_{\mathcal{C}} A_\mu dx^\mu$, that is also invariant under the large gauge transformations, we need to use (2.170).

• Remark: the quantization of Wilson loop is directly related to the compactness of $U(1)$ gauge group, and the quantization of charged particles in the $U(1)$ gauge theory. If the gauge group is $G = \mathbb{R}$, q is unquantized.

(2) Non-abelian gauge theory

Let us consider a gauge theory with non-abelian gauge group G (associated Lie algebra $\mathfrak{g}$). The gauge invariant kinetic term is

$$\frac{1}{2} tr F_{\mu\nu} F^{\mu\nu} \,. \tag{2.174}$$

We expand $F_{\mu\nu} = F^a_{\mu\nu} X_a$, and take the normalization

$$tr(X_a X_b) = \frac{1}{2}\delta_{ab} \,. \tag{2.175}$$

If so, the kinetic term can be written as

$$\frac{1}{4} F^a_{\mu\nu} F^{\mu\nu,a} \,. \tag{2.176}$$

For the matter fields ϕ_i in the representation (Π, V) of G, they are coupled to the gauge field A_μ via the covariant derivative as we presented before:

$$D_\mu \phi_i = \partial_\mu \phi_i - A^a_\mu \pi(X_a)_{ij} \phi_j \,. \tag{2.177}$$

In particular, if the matter is in the adjoint representation of G, it can be written as

$$\begin{aligned} D_\mu \phi_a &= \partial_\mu \phi_a - A^c_\mu \pi(X_c)_{ab} \phi_b \\ &= \partial_\mu \phi_a - A^c_\mu \phi_b c_{cba} \\ &= \partial_\mu \phi_a - [A_\mu, \phi]_a \,. \end{aligned} \tag{2.178}$$

Or equivalently, $D_\mu \phi = \partial_\mu \phi - [A_\mu, \phi]$ when we omit the indices.

3 Semisimple Lie algebra

3.1 Levi's decomposition

We have seen that the classification of Lie algebras is much simpler than the classification of Lie groups, since we do not need to care about the global structures of the latter.

Now we want to study the classifications of finite-dimensional Lie algebras. A particularly important subset of Lie algebras is the *semisimple Lie algebras*, which is defined as:

• a Lie algebra $\mathfrak{g}$ is *semisimple* if and only if it has no non-zero abelian ideals.

Here an *ideal* is a Lie subalgebra $\mathfrak{i} \subset \mathfrak{g}$ satisfying: for any $g \in \mathfrak{g}$, $h \in \mathfrak{i}$, $[g, h] \in \mathfrak{i}$.

For example, the Heisenberg algebra $[A, B] = C$, $[A, C] = [B, C] = 0$ has a non-zero ideal $\{C\}$, hence it is not semisimple. Abelian Lie algebras such as $\mathfrak{u}(1)^{\oplus N}$ are not semisimple either.

On the other hand, one can prove that the $\mathfrak{su}(2)$ Lie algebra with the Lie bracket $[X_a, X_b] = \epsilon_{abc} X_c$ is semisimple.

• Furthermore, a Lie algebra is semisimple if and only if it can be decomposed into a direct sum of *simple Lie algebras*.

• A simple Lie algebra $\mathfrak{g}$ is a non-abelian Lie algebra whose only ideals are $\{0\}$ and $\mathfrak{g}$.

• In contrast, a Lie algebra is called *solvable* if the sequence of Lie algebras

$$\mathfrak{g} \supset [\mathfrak{g}, \mathfrak{g}] \supset [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \supset \dots \tag{3.1}$$

terminates at the zero Lie subalgebra. Here $[\mathfrak{g}, \mathfrak{g}]$ means the Lie subalgebra of $\mathfrak{g}$ whose elements are $[g, h]$ for all $g, h \in \mathfrak{g}$.

A semisimple Lie algebra is never solvable, and vice versa. For example, the abelian algebras $\mathfrak{u}(1)$ and $\mathfrak{u}(1)^{\oplus N}$ are solvable, the Heisenberg algebra is also solvable.

For an arbitrary finite dimensional Lie algebra, we have the following *Levi's decomposition theorem*:

• Any finite dimensional Lie algebra $\mathfrak{g}$ can be decomposed as the semi-direct sum of a solvable Lie algebra $\mathfrak{s}$ and a semisimple Lie algebra $\mathfrak{l}$:

$$\mathfrak{g} = \mathfrak{s} \oplus_S \mathfrak{l}\,. \tag{3.2}$$

The *semi-direct sum* $\mathfrak{g} = \mathfrak{s} \oplus_S \mathfrak{l}$ of two Lie algebras $\mathfrak{s}$ and $\mathfrak{l}$ is a Lie algebra $\mathfrak{g} = \mathfrak{s} \cup \mathfrak{l}$, where $\mathfrak{s} \cap \mathfrak{l} = \{0\}$, $[\mathfrak{s}, \mathfrak{s}] \subset \mathfrak{s}$, $[\mathfrak{l}, \mathfrak{l}] \subset \mathfrak{l}$, $[\mathfrak{s}, \mathfrak{l}] \subset \mathfrak{s}$.

• Examples:

(1) Solvable Lie algebra: $\mathfrak{u}(1)$
(2) Simple Lie algebra $\mathfrak{su}(2)$
(3) Semisimple Lie algebra: $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\mathfrak{su}(2)$
(4) An example that can be decomposed into the direct sum of a semisimple Lie algebra and a solvable Lie algebra: $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$

Note that although the results above hold for both the case of real and complex Lie algebra. For our main purpose of the classification, we are going to focus on the complex Lie algebras. Since a real Lie algebra and its complexification share the same complex representations, in physics it is often enough to study the complex version. The classification of simple *complex* Lie algebras into four infinite series $A_n \cong \mathfrak{sl}(n+1, \mathbb{C})$, $B_n \cong \mathfrak{so}(2n+1)_{\mathbb{C}}$, $C_n \cong \mathfrak{sp}(2n, \mathbb{C})$, $D_n \cong \mathfrak{so}(2n)_{\mathbb{C}}$ and five *exceptional Lie algebras* E_6, E_7, E_8, F_4, G_2 will be the central subject of the next sections.

3.2 Root systems

In this section, we further explore the structures of a semisimple complex Lie algebra. We first introduce the inner product of two Lie algebra elements. Note that we take a complex Lie algebra $\mathfrak{g}$ as the complexification of the real Lie algebra $\mathfrak{r} \subset \mathfrak{g}$. Equivalently we can write $\mathfrak{g} = \mathfrak{r} \oplus i\mathfrak{r}$. For a real matrix Lie algebra $\mathfrak{r}$ with real coefficients, the elements of $\mathfrak{g}$ are the complex linear combinations of the matrices in $\mathfrak{r}$.

• We define the complex *inner product* of Lie algebra elements $\langle .,. \rangle : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$, which takes real values when restricted to $\mathfrak{r}$, such that

$$\langle \mathrm{ad}_X(Y), Z \rangle = -\langle Y, \mathrm{ad}_X(Z) \rangle \,. \tag{3.3}$$

for all $X \in \mathfrak{r}$, $Y, Z \in \mathfrak{g}$.

This implies that

$$\langle \mathrm{ad}_X(Y), Z \rangle = \langle Y, \mathrm{ad}_{X^*}(Z) \rangle \,, \tag{3.4}$$

if we define the matrix adjoint X^* as

$$(X_1 + iX_2)^* = -X_1 + iX_2 \,, \tag{3.5}$$

for $X_1, X_2 \in \mathfrak{r}$. Note that if $\mathfrak{r}$ is a subalgebra of $\mathfrak{u}(n)$, we have

$$(X_1 + iX_2)^* \equiv (X_1 + iX_2)^\dagger \,, \tag{3.6}$$

because $X_1^\dagger = -X_1$, $X_2^\dagger = -X_2$. Hence $\langle .,. \rangle$ has the meaning of a "unitary" inner product.

• Of course, the following conditions for a complex inner product hold ($\forall X, Y, Z \in \mathfrak{g}, k \in \mathbb{C}$):

(1) $\langle X, Y \rangle = \overline{\langle Y, X \rangle}$
(2) $\langle X + Y, Z \rangle = \langle X, Z \rangle + \langle Y, Z \rangle$
(3) $\langle kX, Y \rangle = \bar{k} \langle X, Y \rangle$
(4) $\langle X, kY \rangle = k \langle X, Y \rangle$
(5) $\langle X, X \rangle \geq 0$ and the equality holds only when $X = 0$.

• For example, if $\mathfrak{g}$ is a complex Lie subalgebra of $\mathfrak{gl}(n, \mathbb{C})$ and assume that for all $X \in \mathfrak{g}$, we have $X^\dagger \in \mathfrak{g}$. Then $\mathfrak{r} = \mathfrak{g} \cap \mathfrak{u}(n)$ is the real part of $\mathfrak{g}$, and the following inner product satisfies all the conditions:

$$\langle X, Y \rangle = tr(X^\dagger Y) \,. \tag{3.7}$$

One can check that $\langle X, Y \rangle = tr(X^\dagger Y)$ satisfies the condition $\langle \mathrm{ad}_X(Y), Z \rangle = -\langle Y, \mathrm{ad}_X(Z) \rangle$ for $X \in \mathfrak{r} \subset \mathfrak{u}(n)$, because

$$tr([X, Y]^\dagger Z) = tr(Y^\dagger X^\dagger Z - X^\dagger Y^\dagger Z) = tr(-Y^\dagger X Z + X Y^\dagger Z) \,, \tag{3.8}$$

while

$$-tr(Y^\dagger[X,Z]) = -tr(Y^\dagger XZ - Y^\dagger ZX)\,, \tag{3.9}$$

which equals to (3.8). We used $X^\dagger = -X$.

• On the other hand, for an arbitrary complex or real Lie algebra $\mathfrak{g}$, we can define a different notion of inner product called the *Killing form*:

$$K(X,Y) = tr(\mathrm{ad}_X \mathrm{ad}_Y)\,. \tag{3.10}$$

It is a symmetric bilinear inner product (not a complex inner product). Given a basis X_i of $\mathfrak{g}$, it can also be written in the metric form

$$g_{ij} = tr(\mathrm{ad}_{X_i}\mathrm{ad}_{X_j})\,. \tag{3.11}$$

Since we have

$$\mathrm{ad}_{X_i}\mathrm{ad}_{X_j}X_k = [X_i,[X_j,X_k]] = [X_i, c_{jkl}X_l] = c_{ilm}c_{jkl}X_m\,, \tag{3.12}$$

Hence we can evaluate

$$g_{ij} = tr(\mathrm{ad}_{X_i}\mathrm{ad}_{X_j}) = c_{ikl}c_{jlk}\,. \tag{3.13}$$

For example, for $\mathfrak{g} = \mathfrak{su}(2)$, with $c_{ijk} = \epsilon_{ijk}$, we can compute

$$g_{ij} = -2\delta_{ij}\,. \tag{3.14}$$

• Remark: the Killing form g_{ij} can be used as a metric to raise and lower indices. For example in many physics literature, the structure constant c_{ijk} is written as $c_{ij}^{\ k}$

$$X_i = g_{ij}X^j\ ,\ [X_i,X_j] = c_{ij}^{\ k}X_k\,. \tag{3.15}$$

• Remark: for a real Lie algebra $\mathfrak{g}$, its Killing form is negative-definite if and only if $\mathfrak{g}$ is associated to a compact Lie group G.

• Remark: $\mathfrak{g}$ is semisimple if and only if its Killing form is non-degenerate (with maximal rank).

We separate the vector space of the Lie algebra $\mathfrak{g}$ into distinct parts. First we define

• the *Cartan subalgebra* $\mathfrak{h}$ of $\mathfrak{g}$ to be the maximal abelian subalgebra of $\mathfrak{g}$. It satisfies

(1) for all $H_1, H_2 \in \mathfrak{h}$, $[H_1,H_2] = 0$;

(2) if an $X \in \mathfrak{g}$ satisfies $[H,X] = 0$ for all $H \in \mathfrak{h}$, then $X \in \mathfrak{h}$;

(3) For all $H \in \mathfrak{h}$, ad_H are simultaneously diagonalizable.

• Using this definition, we can now define the *rank* of a Lie algebra $\mathrm{rk}(\mathfrak{g})$ to be the dimension of its Cartan subalgebra.

Note that we can also decompose $\mathfrak{h}$ into the real part and imaginary part by $\mathfrak{h} = \mathfrak{t} \oplus i\mathfrak{t}$. The definition of $\mathrm{rk}(\mathfrak{g})$ for a complex Lie algebra can be naturally extended to its real part as $\mathrm{rk}(\mathfrak{r}) = \mathrm{rk}(\mathfrak{r}_{\mathbb{C}}) = \mathrm{rk}(\mathfrak{g})$ (which is also equal to the rank of the corresponding matrix Lie group G).

• Examples:

(1) $\mathrm{rk}(\mathfrak{u}(1)_{\mathbb{C}}^{\oplus N}) = N$

(2) $\mathrm{rk}(\mathfrak{su}(2)_{\mathbb{C}}) = 1$, more generally $\mathrm{rk}(\mathfrak{su}(N)_{\mathbb{C}}) = N - 1$.

(3) $\mathrm{rk}(\mathfrak{g}_1 \oplus \mathfrak{g}_2) = \mathrm{rk}(\mathfrak{g}_1) + \mathrm{rk}(\mathfrak{g}_2)$.

Now let us introduce the notion of roots and the root system R of a semisimple Lie algebra $\mathfrak{g}$.

• A non-zero element $\alpha \in \mathfrak{h}$ is called a *root* if there exists a non-zero $X \in \mathfrak{g}$ such that

$$[H, X] = \langle \alpha, H \rangle X \tag{3.16}$$

for all $H \in \mathfrak{h}$.

• A non-obvious fact is that if we choose H to be an element of the real part of Cartan subalgebra $\mathfrak{t} \subset \mathfrak{h}$, $\langle \alpha, H \rangle$ is actually pure imaginary. The reason is that because of

$$\langle \mathrm{ad}_H(Y), Z \rangle = -\langle Y, \mathrm{ad}_H(Z) \rangle \quad (\forall Y, Z \in \mathfrak{g}), \tag{3.17}$$

ad_H $(H \in \mathfrak{t})$ is anti-Hermitian and has pure imaginary eigenvalues.

As a simple proof in linear algebra, we derive

$$\begin{aligned} \langle \mathrm{ad}_H X, X \rangle &= -\langle X, \mathrm{ad}_H X \rangle \\ \langle \langle \alpha, H \rangle X, X \rangle &= -\langle X, \langle \alpha, H \rangle X \rangle \\ \overline{\langle \alpha, H \rangle} &= -\langle \alpha, H \rangle \end{aligned} \tag{3.18}$$

Thus if α is a root, $\langle \alpha, H \rangle$ must be pure imaginary. Because for any $X, H \in \mathfrak{r}$, $\langle X, H \rangle \in \mathbb{R}$, we conclude that the root $\alpha \in i\mathfrak{t}$.

• Since for any X satisfying (3.16), its multiple aX also satisfies (3.16). The set of these Lie algebra elements form a vector space $\mathfrak{g}_\alpha$, which is called the *root space* of the root α. A non-zero element of $\mathfrak{g}_\alpha$ is called a *root vector*.

• Using these notations, the vector space of $\mathfrak{g}$ can be decomposed as a direct sum of vector spaces

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha} \mathfrak{g}_\alpha \,. \tag{3.19}$$

This is not a direct sum of Lie algebras. In fact, for a semisimple Lie algebra, each root space $\mathfrak{g}_\alpha$ ($\alpha \neq 0$) is one-dimensional (B. Hall Theorem 7.23, p. 180). On the other hand, the Cartan subalgebra $\mathfrak{h}$ can also be thought as $\mathfrak{g}_0$.

• Proposition: for any $\alpha, \beta \in \mathfrak{h}$, we have

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta} \tag{3.20}$$

if $\alpha + \beta$ is a root. In particular, if $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_{-\alpha}$, then $[X, Y] \in \mathfrak{h}$. Furthermore, if $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, and $\alpha + \beta$ is neither zero nor a root, then $[X, Y] = 0$.

Proof: for the main part we use the Jacobi identity

$$[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]]\,. \tag{3.21}$$

If $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, r.h.s. is

$$\begin{aligned} [[H, X], Y] + [X, [H, Y]] &= [\langle \alpha, H\rangle X, Y] + [X, \langle \beta, H\rangle Y] \\ &= \langle \alpha + \beta, H\rangle [X, Y]\,. \end{aligned} \tag{3.22}$$

Identify this with $[H, [X, Y]]$, we conclude that $[X, Y] \in \mathfrak{g}_{\alpha+\beta}$ if $\alpha + \beta$ is a root.

Specifically, when $\beta = -\alpha$, we have $[H, [X, Y]] = 0$ and we can conclude $[X, Y] \in \mathfrak{h}$.

The final sentence is trivial from the definition of roots.

• Proposition: if α is a root, so is $-\alpha$. If $X \in \mathfrak{g}_\alpha$, then $X^* \in \mathfrak{g}_{-\alpha}$.

Proof: we decompose $X = X_1 + iX_2$, where $X_1, X_2 \in \mathfrak{r}$, we write

$$X^* = -X_1 + iX_2. \tag{3.23}$$

If $H \in \mathfrak{t} \subset \mathfrak{r}$, we have

$$([H, X])^* = -[H, X_1] + i[H, X_2] = [H, X^*]\,. \tag{3.24}$$

In particular, if X is a root vector with root $\alpha \in i\mathfrak{t}$, then for all $H \in \mathfrak{t}$, we have

$$[H, X^*] = ([H, X])^* = (\langle \alpha, H\rangle X)^* = -\langle \alpha, H\rangle X^* \tag{3.25}$$

Because $\langle \alpha, H\rangle$ is a pure imaginary number for $H \in \mathfrak{t}$. Finally, one can extend the statement to $H \in \mathfrak{h}$ and conclude that X^* is a vector of the root $-\alpha$.

• In fact, for each root α, we can find linearly independent elements $X_\alpha \in \mathfrak{g}_\alpha$, $Y_\alpha \in \mathfrak{g}_{-\alpha}$, $H_\alpha \in \mathfrak{h}$, such that they satisfy the commutation relation of $\mathfrak{sl}(2, \mathbb{C})$:

$$[H_\alpha, X_\alpha] \quad = 2X_\alpha$$

$$\begin{aligned} [H_\alpha, Y_\alpha] &= -2Y_\alpha \\ [X_\alpha, Y_\alpha] &= H_\alpha \end{aligned} \tag{3.26}$$

One can choose $Y_\alpha = X_\alpha^*$.

Note that from the definition $[H_\alpha, X_\alpha] = \langle \alpha, H_\alpha \rangle X_\alpha$, we derive

$$\langle \alpha, H_\alpha \rangle = 2\,, \tag{3.27}$$

and

$$H_\alpha = \frac{2\alpha}{\langle \alpha, \alpha \rangle}\,. \tag{3.28}$$

• The H_α defined in the above equation is called the *coroot* of the root α.

• Now we formally define a *root system* R as a finite set of non-zero elements of a real inner product space E $(= i\mathfrak{t} \subset \mathfrak{h})$, with the following additional properties

(1) R spans E

(2) If $\alpha \in R$, then $-\alpha \in R$, and the only multiples of α in R are $\pm\alpha$

(3) If $\alpha, \beta \in R$, then $s_\alpha \cdot \beta \in R$. Here $s_\alpha : E \to E$ is a linear map defined by

$$s_\alpha \cdot \beta = \beta - 2\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}\alpha\,, \quad \beta \in E\,. \tag{3.29}$$

All such s_α with $\alpha \in R$ generates the *Weyl group* of R, which is a finite group.

(4) For all $\alpha, \beta \in R$, the quantity

$$\langle \beta, H_\alpha \rangle = 2\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \tag{3.30}$$

is an integer.

Proof of these statement for the root system of a semisimple Lie algebra $\mathfrak{g}$ with Cartan subalgebra $\mathfrak{h}$:

(1) If R does not span $E = i\mathfrak{t}$, then there exists a non-zero $H \in i\mathfrak{t}$ such that $\langle \alpha, H \rangle = 0$ for all $\alpha \in R$. Then we have $[H, H'] = 0$ for all $H' \in \mathfrak{h}$ and

$$[H, X] = \langle \alpha, H \rangle = 0 \tag{3.31}$$

for all $X \in \mathfrak{g}_\alpha$. Thus H commutes with every element in $\mathfrak{g}$, and this contradicts the definition of a semisimple Lie algebra.

(2) The first part was already proven, for the second part, see B. Hall p. 181.

(3) See B. Hall p. 182-183

(4) See B. Hall p. 184

• Now we introduce the notion of simple roots.

• If (E, R) is a root system, a subset $\Delta \subset R$ is called the set of *simple roots* if

(1) Δ is a basis of the vector space E (thus the number of simple roots also equals to $\mathrm{rk}(\mathfrak{g})$)

(2) Each root $\alpha \in R$ can be written as a linear combination of elements of Δ with integer coefficients, and these coefficients are either all non-negative or non-positive.

• Using the simple roots as a basis of E, the roots with non-negative coefficients are called *positive roots* while the others are *negative roots*.

• Existence theorem of simple roots (B. Hall p. 207).

• Using the simple roots α_i, one can define a basis for $\mathfrak{g}$ which is called the *Chevalley basis*. The Cartan subalgebra basis are H_{α_i} $(i = 1, \ldots, r)$ and the root vectors are E_α and $E_{-\alpha}$ for all the positive roots α. The commutation relations are

$$\begin{aligned} &[H_{\alpha_i}, H_{\alpha_j}] = 0 \\ &[H_{\alpha_i}, E_{\alpha_j}] = \frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_i, \alpha_i \rangle} E_{\alpha_j} \\ &[E_{\alpha_i}, E_{-\alpha_i}] = H_{\alpha_i} \\ &[E_\beta, E_\gamma] = N_{\beta\gamma} E_{\beta+\gamma} \end{aligned} \tag{3.32}$$

$N_{\beta\gamma}$ is an integer.

• Examples:

(1) $\mathfrak{sl}(n+1, \mathbb{C}) = A_n$: the Cartan subalgebra elements are diagonal matrices

$$H = \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_{n+1} \end{pmatrix} \tag{3.33}$$

with

$$\lambda_1 + \lambda_2 + \cdots + \lambda_{n+1} = 0\,. \tag{3.34}$$

Then for the matrix $E_{j,k}$ $(j, k = 1, \ldots, n+1)$, we can compute

$$[H, E_{j,k}] = (\lambda_j - \lambda_k) E_{j,k}\,. \tag{3.35}$$

Hence when $j \neq k$, $E_{j,k}$ is a root vector for the root called α_{jk}, which satisfy

$$\langle \alpha_{jk}, H \rangle = \lambda_j - \lambda_k\,. \tag{3.36}$$

To find a basis for the roots, let us take the orthonormal basis $e_1, e_2, \ldots, e_{n+1}$ of the vector space $\mathbb{C}^{n+1}$ such that $e_i = E_{i,i}$ and the bilinear form is just

$$\langle X, Y \rangle = tr(X^\dagger Y)\,. \tag{3.37}$$

Hence we can write

$$\alpha_{jk} = e_j - e_k \tag{3.38}$$

with

$$\langle \alpha_{jk}, \alpha_{jk} \rangle = 2\,. \tag{3.39}$$

The *length* of each root α_{jk} is $\sqrt{2}$ in this basis.

The simple roots can be taken as

$$\Delta = \{e_1 - e_2, e_2 - e_3, \ldots, e_n - e_{n+1}\}\,, \tag{3.40}$$

and the other positive roots can be written as a non-negative linear combination of the simple roots, while the negative roots can be written as a non-positive linear combination of the simple roots.

This root system is also noted as A_n, associated to the A_n Lie algebra.

The Weyl group element

$$s_{\alpha_{jk}} = 1 - \langle \alpha_{jk},\ \rangle \alpha_{jk} \tag{3.41}$$

interchanges the j-th and k-th component of a vector in $\mathbb{C}^{n+1}$. Hence the Weyl group of A_n Lie algebra is isomorphic to the permutation group S_{n+1}.

The Chevalley basis is

$$H_i \equiv H_{\alpha_i} = E_{i,i} - E_{i+1,i+1}\,,\ E_{\alpha_i} = E_{i,i+1}\,,\ E_{-\alpha_i} = E^\dagger_{\alpha_i} = E_{i+1,i} \quad (i = 1, \ldots, n)\,, \tag{3.42}$$

where α_i means the simple roots. The other E_α can be generated by the fourth line of (3.32)

(2) $\mathfrak{so}(2n)_\mathbb{C} = D_n$

The Cartan subalgebra of $\mathfrak{so}(2n)_\mathbb{C}$ consists of block diagonal matrices

$$H = \left(\begin{array}{cc|cc|c|cc} 0 & a_1 & 0 & 0 & \ldots & 0 & 0 \\ -a_1 & 0 & 0 & 0 & \ldots & 0 & 0 \\ \hline 0 & 0 & 0 & a_2 & \ldots & 0 & 0 \\ 0 & 0 & -a_2 & 0 & \ldots & 0 & 0 \\ \hline \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \hline 0 & 0 & 0 & 0 & \ldots & 0 & a_n \\ 0 & 0 & 0 & 0 & \ldots & -a_n & 0 \end{array}\right) \tag{3.43}$$

where $a_i \in \mathbb{C}$.

The root vectors are $2n \times 2n$ matrices with a 2×2 matrix C in the (j,k) block $(j < k)$, and $-C^T$ in the (k,j)-block, where C is one of the follows:

$$C_1 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix} \ , \ C_2 = \begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix} \ , \ C_3 = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \ , \ C_4 = \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} . \tag{3.44}$$

One can check that these $2n \times 2n$ matrices are indeed root vectors and the corresponding roots are

$$\begin{aligned} \langle \alpha^{(1)}_{jk}, H \rangle &= i(a_j + a_k) \ , \ \langle \alpha^{(2)}_{jk}, H \rangle = -i(a_j + a_k) \ , \\ \langle \alpha^{(3)}_{jk}, H \rangle &= i(a_j - a_k) \ , \ \langle \alpha^{(4)}_{jk}, H \rangle = -i(a_j - a_k) \, . \end{aligned} \tag{3.45}$$

We use the inner product $\langle X, Y \rangle = tr(X^\dagger Y)/2$, and then when we identify H with an element of $\mathbb{C}^n$:

$$H \to i(a_1, a_2, \ldots, a_n) \, , \tag{3.46}$$

the inner product on $\mathfrak{h}$ is exactly the standard inner product on $\mathbb{C}^n$.

An orthonormal basis for the roots is then chosen as $e_k = iE_{2k-1,2k} - iE_{2k,2k-1}$ $(k = 1, \ldots, n)$. And the roots are all the $\pm e_j \pm e_k$ $(j, k = 1, \ldots, n \ , \ j < k)$.

A set of simple roots can be chosen as

$$\Delta = \{e_1 - e_2, e_2 - e_3, \ldots, e_{n-1} - e_n, e_{n-1} + e_n\} \, , \tag{3.47}$$

and all the other roots are either non-positive or non-negative linear combinations of them.

(3) $\mathfrak{so}(2n+1)$ and $\mathfrak{sp}(2n)$: see section 7.7.3, 7.7.4 of B. Hall.

3.3 Classification of rank-2 Lie algebras

• Now the problem is to classify all the root systems that give rise to different semisimple Lie algebras. In this section we solve this problem in rank-2. We first specify the notion of isomorphic root systems.

• Two root systems (E, R) and (F, S) are called *isomorphic* if there exists an invertible linear transformation $\phi : E \to F$ such that ϕ maps R onto S and such that for all $\alpha \in R$, $\beta \in E$, we have

$$\phi(s_\alpha \cdot \beta) = s_{\phi\alpha} \cdot (\phi\beta)\,. \tag{3.48}$$

Thus an isomorphism of root systems preserves the reflection (the Weyl group structure).

For each root system R, there exists a semisimple Lie algebra $\mathfrak{g}$ whose root system is isomorphic to R. Also, if two root systems R_1 and R_2 are isomorphic, the corresponding semisimple Lie algebra $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are also isomorphic. (See B. Hall Theorem 8.20, p. 237)

• Furthermore, a root system (E, R) is called *reducible* if there exists an orthogonal decomposition $E = E_1 \oplus E_2$ with $\dim E_1 > 0$, $\dim E_2 > 0$ such that every element of R is either in E_1 or in E_2. If no such decomposition exists, (E, R) is called *irreducible*. Irreducible root systems correspond to simple Lie algebras. (proof: B. Hall p. 186-188)

• Now we prove the following propositions for root systems:

(1) Suppose α, β are roots in R, α is not a multiple of β, and $\langle\alpha, \alpha\rangle \geq \langle\beta, \beta\rangle$, then one of the following holds:

1. $\langle\alpha, \beta\rangle = 0$.
2. $\langle\alpha, \alpha\rangle = \langle\beta, \beta\rangle$ and the angle between α and β is $\pi/3$ or $2\pi/3$.
3. $\langle\alpha, \alpha\rangle = 2\langle\beta, \beta\rangle$ and the angle between α and β is $\pi/4$ or $3\pi/4$.
4. $\langle\alpha, \alpha\rangle = 3\langle\beta, \beta\rangle$ and the angle between α and β is $\pi/6$ or $5\pi/6$.

Proof: let $m_1 = 2\langle\alpha, \beta\rangle/\langle\alpha, \alpha\rangle$ and $m_2 = 2\langle\alpha, \beta\rangle/\langle\beta, \beta\rangle$. From the property (4) of a root system, m_1 and m_2 are integers. We have

$$m_1 m_2 = \frac{4\langle\alpha, \beta\rangle^2}{\langle\alpha, \alpha\rangle\langle\beta, \beta\rangle} = 4\cos^2\theta \tag{3.49}$$

where θ is the angle between α and β. As α is not a multiple of β, the only possible values for $m_1 m_2$ are 0, 1, 2, 3.

It is easy to check these four cases exactly correspond to the four cases in the proposition.

(2) For two simple roots $\alpha, \beta \in \Delta$, $\langle \alpha, \beta \rangle \leq 0$.

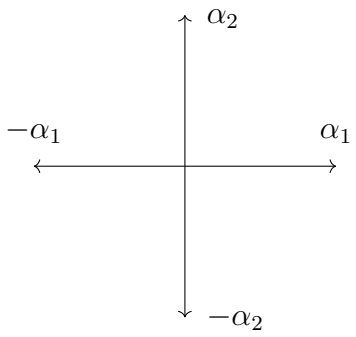
Proof: if $\langle \alpha, \beta \rangle > 0$, the root

$$s_\alpha \cdot \beta = \beta - 2\frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle}\alpha \tag{3.50}$$

does not have non-negative nor non-positive coefficients in α and β. This contradicts the definition of simple roots.

• Now for the case of rank-2 semisimple Lie algebra, there are only four possible configurations of the two simple roots α_1 and α_2, which leads to four non-isomorphic root systems.

(1) $\langle \alpha_1, \alpha_2 \rangle = 0$. The root system is



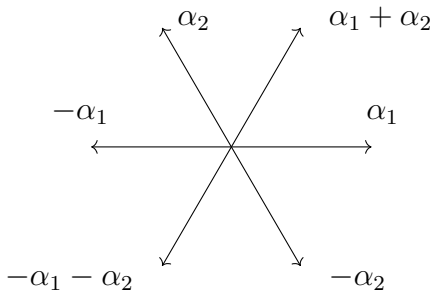
(3.51)

This is the root system of $A_1 \oplus A_1$, which corresponds to the Lie algebra $\mathfrak{su}(2)_\mathbb{C} \oplus \mathfrak{su}(2)_\mathbb{C}$. There are four roots and two generators in $\mathfrak{h}$, hence

$$\dim(A_1 \oplus A_1) = 6\,. \tag{3.52}$$

(2) $\langle \alpha_1, \alpha_1 \rangle = \langle \alpha_2, \alpha_2 \rangle$ and the angle between α_1 and α_2 is $2\pi/3$.

The root system is

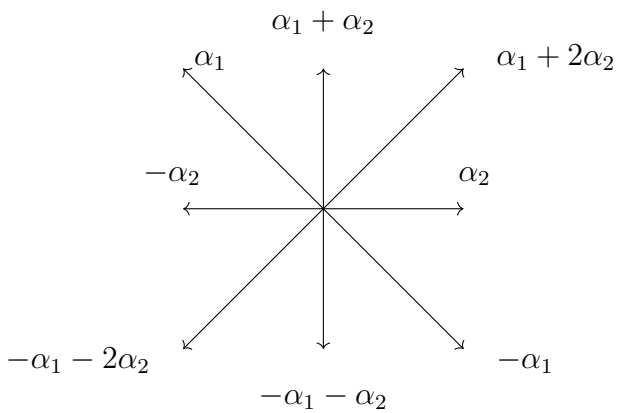


(3.53)

This is the root system of A_2, which corresponds to the Lie algebra $\mathfrak{su}(3)_\mathbb{C} = \mathfrak{sl}(3, \mathbb{C})$. There are six roots and two generators in $\mathfrak{h}$, hence

$$\dim(A_2) = 8\,. \tag{3.54}$$

(3) $\langle \alpha_1, \alpha_1 \rangle = 2\langle \alpha_2, \alpha_2 \rangle$ and the angle between α_1 and α_2 is $3\pi/4$
The root system is

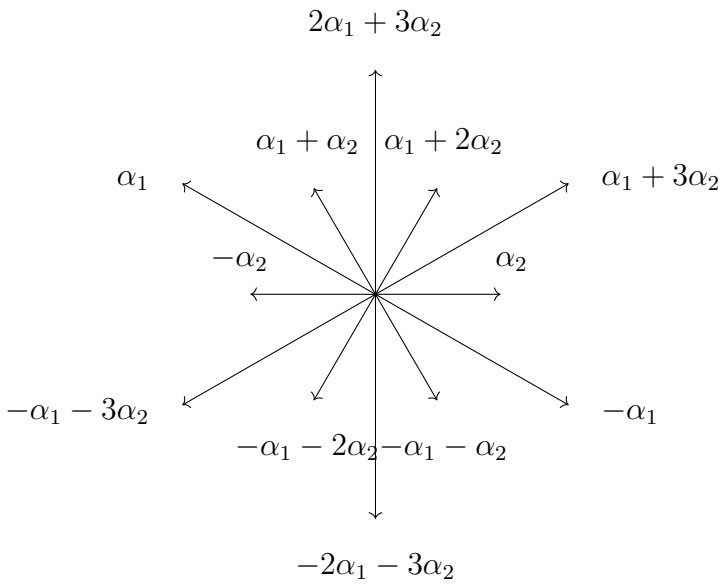


(3.55)

This is the root system of B_2, which corresponds to the Lie algebra $\mathfrak{so}(5)_{\mathbb{C}} = \mathfrak{sp}(4, \mathbb{C})$. There are eight roots and two generators in $\mathfrak{h}$, hence

$$\dim(B_2) = 10\,. \tag{3.56}$$

(4) $\langle \alpha_1, \alpha_1 \rangle = 3\langle \alpha_2, \alpha_2 \rangle$ and the angle between α_1 and α_2 is $5\pi/6$
The root system is



(3.57)

This is the root system of G_2, which corresponds to the (complex) exceptional Lie algebra $\mathfrak{g}_2$. There are 12 roots and two generators in $\mathfrak{h}$, hence

$$\dim(G_2) = 14\,. \tag{3.58}$$

This is the first exceptional Lie algebra we have encountered in this course.

• Finally, note that A_2, B_2 and G_2 root systems are all irreducible, which corresponds to simple Lie algebras. While $A_1 \oplus A_1$ is reducible.

3.4 Classification of simple Lie algebras

• We want to classify the root systems which one-to-one correspond to a simple Lie algebra, which corresponds to irreducible root systems.

• Inspired by the rank-two classification, we introduce the notion of *Dynkin diagram*, which is a graph defines as follows:

1. The nodes v_i one-to-one correspond to the simple roots α_i.

2. Between two distinct vertices v_i and v_j we draw zero, one, two or three lines if the angle between α_i and α_j are $\pi/2$, $2\pi/3$, $3\pi/4$ or $5\pi/6$, respectively.

3. In addition, when the length of roots α_i and α_j are unequal, we draw an arrow from the *long root* to the *short root*.

We draw the Dynkin diagrams for the rank-two semisimple Lie algebras:

$$A_1 \oplus A_1 : \quad \bullet \quad \bullet \tag{3.59}$$

$$A_2 : \quad \bullet\!\!-\!\!\bullet \tag{3.60}$$

$$B_2 : \quad \bullet\!\!\Rightarrow\!\!\bullet \tag{3.61}$$

$$G_2 : \quad \bullet\!\!\Rrightarrow\!\!\bullet \tag{3.62}$$

As one can see, the Dynkin diagram for $A_1 \oplus A_1$ is disconnected.

We have the following propositions (proof: B. Hall p. 217-218):

(1) A root system is irreducible if and only if its Dynkin diagram is connected (thus the corresponding Lie algebra is simple)

(2) Two root systems are isomorphic if and only if their Dynkin diagrams are isomorphic (as graphs).

Hence from (2), we can classify complex Lie algebras by classifying inequivalent Dynkin diagrams.

• We also introduce the Cartan matrix $\mathcal{C}_{ij}$ of a root system, which one-to-one corresponds to a Dynkin diagram:

$$\begin{aligned} \mathcal{C}_{ij} &= \frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle} \\ &= \langle \alpha_i, H_{\alpha_j} \rangle \,. \end{aligned} \tag{3.63}$$

Each element of $\mathcal{C}_{ij}$ is an integer, due to the property (4) of a root system. The diagonal terms all equal to two, and the off-diagonal terms are all non-positive ($\langle \alpha_i, \alpha_j \rangle \leq 0$).

The Cartan matrix for the rank-two semisimple Lie algebras are

$$A_1 \oplus A_1 : \quad \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \tag{3.64}$$

$$A_2 : \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{3.65}$$

$$B_2 : \quad \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \tag{3.66}$$

$$G_2 : \quad \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} \tag{3.67}$$

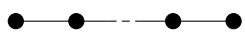
• Proposition: the Cartan matrix for a semisimple Lie algebra is positive-definite.

Proof: multiplying each row of $\mathcal{C}_{ij}$ by $\langle \alpha_j, \alpha_j \rangle$, the resulting matrix is just $M_{ij} = 2\langle \alpha_i, \alpha_j \rangle$ which is a positive definite symmetric inner product on $\mathbb{R}^{\mathrm{rk}(\mathfrak{g})}$.

• Corollary: $\det(\mathcal{C}_{ij}) > 0$.

• Classification of simple Lie algebras using irreducible root systems (connected Dynkin diagrams):

(1) A_n series, corresponding to $\mathfrak{sl}(n+1, \mathbb{C})$ or $\mathfrak{su}(n+1)_{\mathbb{C}}$ Lie algebra:



(3.68)

The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 \\ 0 & 0 & 0 & \dots & -1 & 2 \end{pmatrix} \tag{3.69}$$

The simple roots of A_n can be written with the standard basis of $\mathbb{R}^{n+1}$:

$$\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ \dots, \ \alpha_n = e_n - e_{n+1} \, . \tag{3.70}$$

(2) B_n series, corresponding to $\mathfrak{so}(2n+1)_{\mathbb{C}}$ Lie algebra:

(3.71)

The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -2 \\ 0 & 0 & 0 & \dots & -1 & 2 \end{pmatrix} \tag{3.72}$$

The simple roots of B_n can be written with the standard basis of $\mathbb{R}^n$:

$$\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ \dots, \ \alpha_{n-1} = e_{n-1} - e_n \ , \ \alpha_n = e_n \, . \tag{3.73}$$

(3) C_n series, corresponding to $\mathfrak{sp}(2n, \mathbb{C})$ Lie algebra:

(3.74)

The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 \\ 0 & 0 & 0 & \dots & -2 & 2 \end{pmatrix} \tag{3.75}$$

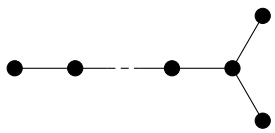
The simple roots of C_n can be written with the standard basis of $\mathbb{R}^n$:

$$\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ \dots, \ \alpha_{n-1} = e_{n-1} - e_n \ , \ \alpha_n = 2e_n \, . \tag{3.76}$$

• Note that the Dynkin diagram for B_n and C_n are quite similar. The difference is that most of the simple roots of B_n are long roots, while most of the simple roots of C_n are short roots.

• From the isomorphism of Dynkin diagrams, we can easily see $B_2 \cong C_2$.

(4) D_n series, corresponding to $\mathfrak{so}(2n)_{\mathbb{C}}$ Lie algebra



(3.77)

The Cartan matrix is

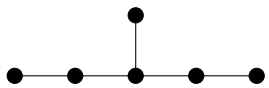
$$\begin{pmatrix} 2 & -1 & 0 & \dots & 0 & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 & -1 \\ 0 & 0 & 0 & \dots & -1 & 2 & 0 \\ 0 & 0 & 0 & \dots & -1 & 0 & 2 \end{pmatrix} \tag{3.78}$$

The simple roots of D_n can be written with the standard basis of $\mathbb{R}^n$:

$$\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ \dots, \ \alpha_{n-1} = e_{n-1} - e_n \ , \ \alpha_n = e_{n-1} + e_n \, . \tag{3.79}$$

• From the isomorphism of Dynkin diagrams, one can see $D_3 = \mathfrak{so}(6)_{\mathbb{C}} \cong A_3 = \mathfrak{su}(4)_{\mathbb{C}}$, $D_2 = \mathfrak{so}(4)_{\mathbb{C}} \cong A_1 \oplus A_1 = \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}$.

(5) The exceptional E_6 Lie algebra



(3.80)

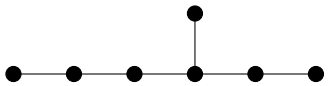
The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix} . \tag{3.81}$$

The simple roots of E_6 can be written with the standard basis of $\mathbb{R}^6$:

$$\begin{aligned} &\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ , \ \alpha_3 = e_3 - e_4 \ , \ \alpha_4 = e_4 + e_5 \ , \\ &\alpha_5 = -\frac{1}{2}(e_1 + e_2 + e_3 + e_4 + e_5) + \frac{\sqrt{3}}{2} e_6 \ , \ \alpha_6 = e_4 - e_5 \, . \end{aligned} \tag{3.82}$$

(6) The exceptional E_7 Lie algebra



(3.83)

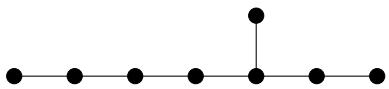
The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix} . \tag{3.84}$$

The simple roots of E_7 can be written with the standard basis of $\mathbb{R}^7$:

$$\begin{aligned} &\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ , \ \alpha_3 = e_3 - e_4 \ , \ \alpha_4 = e_4 - e_5 \ , \ \alpha_5 = e_5 + a_6 \\ &\alpha_6 = -\frac{1}{2}(e_1 + e_2 + e_3 + e_4 + e_5 + e_6) + \frac{\sqrt{2}}{2} e_7 \ , \ \alpha_7 = e_5 - e_6 \, . \end{aligned} \tag{3.85}$$

(7) The exceptional E_8 Lie algebra



(3.86)

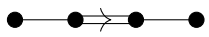
The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix} . \tag{3.87}$$

The simple roots of E_8 can be written with the standard basis of $\mathbb{R}^8$:

$$\begin{aligned} &\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3, \ , \ \alpha_3 = e_3 - e_4 \ , \ \alpha_4 = e_4 - e_5 \ , \ \alpha_5 = e_5 - a_6 \\ &\alpha_6 = e_6 - e_7 \ , \ \alpha_7 = -\frac{1}{2}(e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7 + e_8) \ , \ \alpha_8 = e_6 - e_7 \, . \end{aligned} \tag{3.88}$$

(8) The exceptional F_4 Lie algebra



(3.89)

The Cartan matrix is

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \tag{3.90}$$

The simple roots of F_4 can be written with the standard basis of $\mathbb{R}^4$:

$$\alpha_1 = e_2 - e_3 \ , \ \alpha_2 = e_3 - e_4, \ , \ \alpha_3 = e_4 \ , \ \alpha_4 = \frac{1}{2}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 - \frac{1}{2}e_4 \, . \tag{3.91}$$

(9) The exceptional G_2 Lie algebra

(3.92)

$$\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} \tag{3.93}$$

• If one of the following equivalent conditions is satisfied for a (semi)simple Lie algebra $\mathfrak{g}$:

(1) the simple roots of $\mathfrak{g}$ all have the same length;

(2) there is no arrow in the Dynkin diagram of $\mathfrak{g}$;

(3) the Cartan matrix of $\mathfrak{g}$ is symmetric,

then $\mathfrak{g}$ is called a *simply-laced Lie algebra*. Otherwise $\mathfrak{g}$ is a *non-simply-laced Lie algebra*.

• The A_n, D_n and E_n type Lie algebras are simply-laced, while B_n, C_n, F_4 and G_2 are non-simply-laced.

• The structure of simply-laced Lie algebras often appear in different branches mathematics with the name "ADE classification". For example, the finite subgroups of $SO(3)$ ($SU(2)$) are classified in this way:

$$\begin{aligned} A_n &\leftrightarrow \text{ cyclic groups } \mathbb{Z}_n \\ D_n &\leftrightarrow \text{ dihedral groups} \\ E_6 &\leftrightarrow \text{ tetrahedral group} \\ E_7 &\leftrightarrow \text{ octahedral group} \\ E_8 &\leftrightarrow \text{ icosahedral group} \end{aligned} \tag{3.94}$$

• The deep mathematics behind it is called "McKay correspondence".

• We summarize the dimension of the simple Lie algebras in the following table:

A_n	$n(n+2)$
B_n	$n(2n+1)$
C_n	$n(2n+1)$
D_n	$n(2n-1)$
E_6	78
E_7	133
E_8	248
F_4	52
G_2	14

(3.95)

3.5 Subalgebras

• In physics, it is important to study the subalgebras $\mathfrak{s}$ of a given semisimple Lie algebra $\mathfrak{g}$. For example, in the Higgs mechanism, one breaks a gauge group (algebra) to a smaller gauge group (algebra). Note that here $\mathfrak{s}$ can be a direct sum of a semisimple algebra with a number of $\mathfrak{u}(1)$ factors.

Since a semisimple Lie algebra can be written as a direct sum of simple Lie algebras: $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_k$. It is sufficient to study the subalgebras $\mathfrak{s}_i$ of each simple Lie algebras $\mathfrak{g}_i$ and then

$$\mathfrak{s}_1 \oplus \mathfrak{s}_2 \oplus \cdots \oplus \mathfrak{s}_k \subset \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_k \,. \tag{3.96}$$

Now let us present some sufficient conditions for $\mathfrak{s} \subset \mathfrak{g}$:

1. From the relation between the matrix Lie group G and its associated Lie algebra $\mathfrak{g}$, it is easy to see that:

   • Sufficient condition: if the matrix group Lie S is a subgroup of the matrix Lie group G, then the associated Lie algebra $\mathfrak{s}$ of S is a Lie subalgebra of $\mathfrak{g}$. Furthermore, $\mathfrak{s}_{\mathbb{C}} \subset \mathfrak{g}_{\mathbb{C}}$.

   • For example:

   (1) $(SU(k) \times SU(n-k) \times U(1))/\mathbb{Z}_{lcm(k,n-k)} \subset SU(n)$, $(0 < k < n)$. Hence $A_{k-1} \oplus A_{n-k-1} \oplus \mathfrak{u}(1)_{\mathbb{C}} \subset A_{n-1}$.

   (2) $SO(2n-1) \subset SO(2n) \subset SO(2n+1)$, which implies $B_{n-1} \subset D_n \subset B_n$.

   (3) $USp(2n) \cong Sp(n) \subset SU(2n)$, which implies $C_n \subset A_{2n-1}$.

   • Of course, one can also easily check the above relations using matrix Lie subalgebra arguments.

2. The subalgebra criterion in terms of root systems (R-type subalgebra): if the semisimple Lie algebra $\mathfrak{g}$ has the Cartan subalgebra $\mathfrak{h}$ and root system (E, R), and the vector space of $\mathfrak{s}$ can be decomposed as

   $$\mathfrak{s} = \mathfrak{t} \oplus \bigoplus_{\alpha \in S} \mathfrak{g}_\alpha \,, \tag{3.97}$$

   where $\mathfrak{t}$ is a subspace of $\mathfrak{h}$ and S is a sub-root system of R, then $\mathfrak{s}$ is an *R-type subalgebra* of $\mathfrak{g}$.

   • Example:

   (1) A_1 is a sub-root system of the root system of any semisimple Lie algebra $\mathfrak{g}$, hence $A_1 \subset \mathfrak{g}$.

(2) From the root system of rank-2 Lie algebras, we can see that $A_1 \oplus A_1 \subset B_2$ and $A_1 \oplus A_1 \subset G_2$, but $A_1 \oplus A_1 \nsubseteq A_2$. We also have $A_2 \subset G_2$, $A_2 \nsubseteq B_2$, $B_2 \nsubseteq G_2$.

• From the structure of simple roots α_i, encoded in the Dynkin diagram of $\mathfrak{g}$, we can also pick a set of sub-diagrams that form the Dynkin diagram of a semisimple Lie subalgebra $\mathfrak{s} \subset \mathfrak{g}$. Furthermore, if there are nodes $\alpha_{i_1}, \ldots, \alpha_{i_l}$ in the Dynkin diagram of $\mathfrak{g}$ which are not included into $\mathfrak{s}$, where $l = \mathrm{rk}(\mathfrak{g}) - \mathrm{rk}(\mathfrak{s})$, $\mathfrak{s} \oplus \mathfrak{u}(1)_{\mathbb{C}}^{\oplus l}$ is also a Lie subalgebra of $\mathfrak{g}$.

The first sentence is trivial. For the second sentence, we orthogonally decompose the Cartan subalgebra $\mathfrak{h}_{\mathfrak{g}}$ of $\mathfrak{g}$ into

$$\mathfrak{h}_{\mathfrak{g}} = \mathfrak{h}_{\mathfrak{s}} \oplus \mathfrak{h}_{\mathfrak{s}}^{\perp}\,, \tag{3.98}$$

where $\mathfrak{h}_{\mathfrak{s}}$ is the Cartan subalgebra of $\mathfrak{s}$. Hence $\mathfrak{h}_{\mathfrak{s}}^{\perp} = \mathfrak{u}(1)_{\mathbb{C}}^{\oplus l}$. We can prove that its elements commute with any root vectors in $\mathfrak{s}$, as follows: for any root α of $\mathfrak{s}$ with root vector X_α, they satisfy

$$[H, X_\alpha] = \langle \alpha, H \rangle X_\alpha \tag{3.99}$$

for any $H \in \mathfrak{h}$. Hence if $H \in \mathfrak{h}_{\mathfrak{s}}^{\perp}$, since $\alpha \in \mathfrak{h}_{\mathfrak{s}}$, we have $\langle \alpha, H \rangle = 0$.

• For example:

(1) $A_{n-1} \oplus \mathfrak{u}(1) \subset B_n$ $(\mathfrak{su}(n)_{\mathbb{C}} \oplus \mathfrak{u}(1)_{\mathbb{C}} \subset \mathfrak{so}(2n+1)_{\mathbb{C}})$

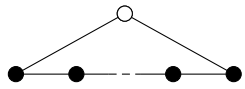
(2) $A_{n-1} \oplus \mathfrak{u}(1) \subset C_n$ $(\mathfrak{su}(n)_{\mathbb{C}} \oplus \mathfrak{u}(1)_{\mathbb{C}} \subset \mathfrak{sp}(n)_{\mathbb{C}})$

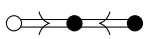
(3) $A_{n-1} \oplus \mathfrak{u}(1) \subset D_n$ $(\mathfrak{su}(n)_{\mathbb{C}} \oplus \mathfrak{u}(1)_{\mathbb{C}} \subset \mathfrak{so}(2n)_{\mathbb{C}})$

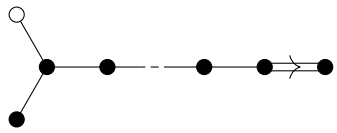
• Now we present a more powerful condition of an R-type subalgebra of a simple Lie algebra.

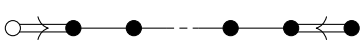
We first introduce the *extended Dynkin diagram* (*affine Dynkin diagram*) of a simple Lie algebra $\mathfrak{g}$, denoted as $\hat{\mathfrak{g}}$:

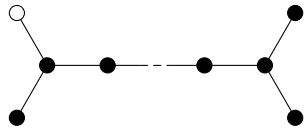
$\widehat{A}_1$: (3.100)

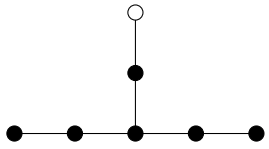
$\widehat{A}_n$:  (3.101)

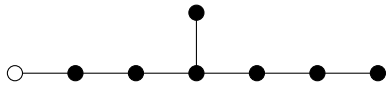
$\widehat{B}_2 \cong \widehat{C}_2$:  (3.102)

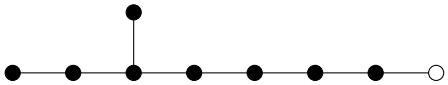
$\widehat{B}_n$:  (3.103)

$\widehat{C}_n$:  (3.104)

$\widehat{D}_n$:  (3.105)

$\widehat{E}_6$:  (3.106)

$\widehat{E}_7$:  (3.107)

$\widehat{E}_8$:  (3.108)

$\widehat{F}_4$: (3.109)

$\widehat{G}_2$: (3.110)

Note that the extended Dynkin diagram of $\mathfrak{g}$ always has one more node than the Dynkin diagram of $\mathfrak{g}$. In fact, it is the Dynkin diagram of the *affine Lie algebra* $\widehat{\mathfrak{g}}$, which is an infinite dimensional Lie algebra that contains $\mathfrak{g}$ as a subalgebra. As another comment, the Cartan matrix associated to the extended Dynkin diagram $\widehat{\mathfrak{g}}$ has $\det(\mathcal{C}) = 0$.

• Then starting from $\mathfrak{g}$, we can construct the following R-type subalgebras:

(1) Deleting one node from the extended Dynkin diagram of $\mathfrak{g}$, and the resulting Dynkin diagram is that of a non-abelian Lie algebra $\mathfrak{s}$, then $\mathfrak{s}$ is an R-type subalgebra of $\mathfrak{g}$.

(2) Deleting k-nodes from the extended Dynkin diagram of $\mathfrak{g}$, and the resulting Dynkin diagram is that of a non-abelian Lie algebra $\mathfrak{s}$, then $\mathfrak{s} \oplus \mathfrak{u}(1)_{\mathbb{C}}^{k-1}$ is an R-type subalgebra of $\mathfrak{g}$.

(3) To get all the R-type subalgebras, one needs to apply this to all the subalgebras $\mathfrak{g}' \subset \mathfrak{g}$ recursively.

• More examples:

(1) $A_5 \oplus A_1 \subset E_6$ (or $\mathfrak{su}(6)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}} \subset E_6$)

(2) $A_7 \subset E_7$ (or $\mathfrak{su}(8)_{\mathbb{C}} \subset E_7$)

(3) $A_8 \subset E_8$ (or $\mathfrak{su}(9)_{\mathbb{C}} \subset E_8$)

(4) $E_6 \oplus A_2 \subset E_8$

(5) $A_2 \oplus A_2 \oplus A_2 \subset E_6$. Note that the fact $A_2 \oplus A_2 \oplus A_2 \oplus A_2 \subset E_8$ cannot be deduced from deleting one node of $\widehat{E}_8$ directly.

(6) $E_7 \oplus A_1 \subset E_8$

(7) $B_4 \subset F_4$ (or $\mathfrak{so}(9)_{\mathbb{C}} \subset F_4$)

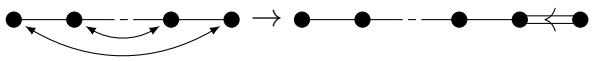
• Note that a non-simply-laced algebra cannot be an R-type subalgebra of a simply-laced one.

3. • If a Lie subalgebra is not an R-type subalgebra, then it is an *S-type subalgebra*.

The classification of S-type subalgebras is a very complicated problem. One can refer to Section 2.2 of (Yamatsu 15' arXiv: 1511.08771) for more details. One thing for sure is that the rank of an S-type subalgebra $\mathfrak{s}$ satisfies $\mathrm{rk}(\mathfrak{s}) < \mathrm{rk}(\mathfrak{g})$. A notable subclass of S-type subalgebra comes from the "folding", which is a way of embedding non-simply-laced Lie algebra into simply-laced ones.

• For example:

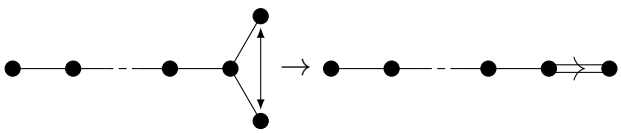
(1) $C_n \subset A_{2n-1}$



(3.111)

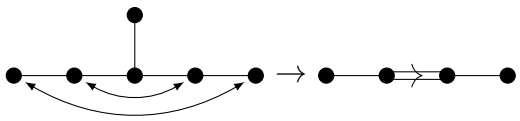
Since all roots of A_{2n-1} have the same length while C_n do not, the roots of C_n cannot form a subsystem of the A_{2n-1} root system, and C_n is not an R-type subalgebra of A_{2n-1}.

(2) $B_n \subset D_{n+1}$



(3.112)

(3) $F_4 \subset E_6$



(3.113)

(3) $G_2 \subset B_3$

(3.114)

The sufficient and necessary condition for $\mathfrak{s} \subset \mathfrak{g}$ is quite non-trivial. Here we only present a few necessary condition for $\mathfrak{s} \subset \mathfrak{g}$, which can be easily proved:

1. $\dim(\mathfrak{s}) \leq \dim(\mathfrak{g})$.
2. $\mathrm{rk}(\mathfrak{s}) \leq \mathrm{rk}(\mathfrak{g})$.
3. If $\mathrm{rk}(\mathfrak{s}) = \mathrm{rk}(\mathfrak{g})$, then $\mathfrak{s}$ has to be an R-type subalgebra of $\mathfrak{g}$, which means that the Dynkin diagram of (the semisimple part) of $\mathfrak{s}$ can be embedded in the extended Dynkin diagram $\hat{\mathfrak{g}}'$ of $\mathfrak{g}' \subset \mathfrak{g}$.

4 Representation of semisimple Lie algebras

4.1 Highest weight representations

In this section, we are going to systematically study the representations of semisimple Lie algebras. We first classify and study the irreducible representations of a semisimple Lie algebra $\mathfrak{g}$, and a general representation of $\mathfrak{g}$ can be written as a direct sum of irreducible representations. The applicability of this method is based on the following theorem:

• For a semisimple Lie algebra $\mathfrak{g}$, it is said to have the "complete reducibility", such that any finite dimensional representation of $\mathfrak{g}$ can be written as a direct sum of irreducible representations (p. 275, theorem 10.9 of B. Hall).

Now, for each irreducible representation of $\mathfrak{g}$, it one-to-one corresponds to a "highest weight representation" that is characterized by a unique "highest weight". To define this notion, we first introduce the concept of weights, which is a generalization of the weights in the case of $\mathfrak{su}(2)$.

• Let (π, V) be a representation of $\mathfrak{g}$, an element $\lambda \in \mathfrak{h}$ is a *weight* of π if there exists a non-zero vector $v \in V$ such that

$$\pi(H)v = \langle \lambda, H \rangle v \tag{4.1}$$

for all $H \in \mathfrak{h}$. The *weight space* corresponding to λ is the set of all *weight vectors* $v \in V$ satisfying (4.1) and the *multiplicity* of λ is the dimension of the corresponding weight space. The set of weights λ is called the *weight system* of (π, V).

Using a basis $H_{\alpha_1}, \ldots, H_{\alpha_r}$ for $\mathfrak{h}$, a weight λ can also be written as

$$\lambda = (\langle \lambda, H_{\alpha_1} \rangle, \langle \lambda, H_{\alpha_2} \rangle, \ldots, \langle \lambda, H_{\alpha_r} \rangle) \,. \tag{4.2}$$

α_i are the simple roots of $\mathfrak{g}$.

Note that in the same basis we can write a simple root α_i as

$$\begin{aligned} \alpha_i &= (\langle \alpha_i, H_{\alpha_1} \rangle, \langle \alpha_i, H_{\alpha_2} \rangle, \ldots, \langle \alpha_i, H_{\alpha_r} \rangle) \\ &= (\mathcal{C}_{i1}, \mathcal{C}_{i2}, \ldots, \mathcal{C}_{ir}) \,, \end{aligned} \tag{4.3}$$

where $\mathcal{C}_{ij}$ are the entries of the Cartan matrix.

• Remark: recall the definition of root:

$$\mathrm{ad}_H v = \langle \alpha, H \rangle v \tag{4.4}$$

for all $H \in \mathfrak{h}$, one can see that the weight system of the adjoint representation ad of any semisimple Lie algebra is identical to its root system R.

• Now let us define the "higher" ordering $\succ$ between two weights μ_1 and μ_2: $\mu_1 \succ \mu_2$ if and only if

$$\mu_1 - \mu_2 = \sum_{i=1}^{r} a_i \alpha_i \,, \tag{4.5}$$

where $a_i \geq 0$, and they are not identically zero. Similarly we can define the notion "lower".

• The *highest weight* of a representation (π, V) is thus the highest weight μ of its weight system. Similarly one can introduce the notion of the *lowest weight*.

Now let us discuss some properties of the weight system:

• If (π, V) is a finite-dimensional representation of $\mathfrak{g}$, every weight λ of π is *integral*, which means that $\langle \lambda, H_{\alpha_i} \rangle \in \mathbb{Z}$ for any H_{α_i}.

Proof: for each root, consider the subalgebra $\mathfrak{s}_\alpha = \langle X_\alpha, Y_\alpha, H_\alpha \rangle \cong \mathfrak{sl}(2, \mathbb{C})$ of $\mathfrak{g}$. If v is a weight vector of the weight λ, then

$$\pi(H_\alpha) v = \langle \lambda, H_\alpha \rangle v \,. \tag{4.6}$$

Using the properties of $\mathfrak{sl}(2, \mathbb{C})$ representations, applied to the restriction of π in this case, we see that $\langle \lambda, H_\alpha \rangle \in \mathbb{Z}$.

• A weight λ is called *dominant integral* if $\langle \lambda, H_{\alpha_i} \rangle \geq 0$ for all H_{α_i}. Now there are the following theorems (see p. 243 of B. Hall):

(1) Every irreducible, finite-dimensional representation of $\mathfrak{g}$ has a highest weight μ, and μ is dominant integral;

(2) Two irreducible, finite-dimensional representations of $\mathfrak{g}$ with the same highest weight are isomorphic;

(3) If μ is a dominant integral element, there exists an irreducible, finite-dimensional representation of $\mathfrak{g}$ with highest weight μ.

Now in order to write down all the weights λ in the weight system of a representation π with highest weight μ, the idea is to multiply a weight vector with the (simple) roots:

• let $\alpha = (a_1, a_2, \ldots, a_r)$ be a root ($a_i := \langle \alpha, H_{\alpha_i} \rangle$), and $Z_\alpha \in \mathfrak{g}_\alpha$ be a corresponding root vector. π is a representation of $\mathfrak{g}$ and $\mu = (m_1, m_2, \ldots, m_r)$ is a weight of π, and let $v \neq 0$ be a corresponding weight vector. Now we have

$$\pi(H_{\alpha_i}) \pi(Z_\alpha) v = (m_i + a_i) \pi(Z_\alpha) v \,, \tag{4.7}$$

which means that either $\pi(Z_\alpha) v = 0$ or $\pi(Z_\alpha) v$ is a new weight vector with weight

$$\mu + \alpha = (m_1 + a_1, m_2 + a_2, \ldots, m_i + a_i) \,. \tag{4.8}$$

Proof: from the definition of root, we have

$$[H_{\alpha_i}, Z_\alpha] = a_i Z_\alpha \,, \tag{4.9}$$

hence

$$\begin{aligned} \pi(H_{\alpha_i})\pi(Z_\alpha)v &= (\pi(Z_\alpha)\pi(H_{\alpha_i}) + a_i\pi(Z_\alpha))v \\ &= \pi(Z_\alpha)(m_i v) + a_i\pi(Z_\alpha)v \\ &= (m_i + a_i)\pi(Z_\alpha)v \,. \end{aligned} \tag{4.10}$$

Using this proposition, we can prove the following basic statement:

• Theorem: for every irreducible representation (π, V) of $\mathfrak{g}$, V is the direct sum of its weight spaces.

Proof: let W be the sum of weight spaces of π, which is a subspace of V.That is, W is also the space of all vectors $w \in V$ that can be written as a linear combination of simultaneous eigenvectors for $\pi(H_{\alpha_i})$ $(i = 1, \ldots, r)$. Obviously W is invariant under the action of H_{α_i}.

If Z_α is a root vector corresponding to a root α, $\pi(Z_\alpha)$ maps the weight space of μ into the weight space of $\mu+\alpha$. This means that W is also invariant under all root vectors Z_α. Hence W is invariant under all elements of $\mathfrak{g}$. By irreducibility of π, we conclude that $V = W$.

• (B.Hall p. 269, Lemma 10.3) Suppose λ is a weight of (π, V) and $\langle\lambda, \alpha\rangle > 0$ for some root α, we define

$$j = \langle\lambda, H_\alpha\rangle = \frac{2\langle\lambda, \alpha\rangle}{\langle\alpha, \alpha\rangle} \,, \tag{4.11}$$

then $\lambda - k\alpha$, $(k = 1, \ldots, j)$ are all weights of (π, V).

• Now we describe the procedure of generating all weights of a representation π. Starting from the highest weight

$$\mu = (\mu_1, \mu_2, \ldots, \mu_r) \,, \tag{4.12}$$

for any $\mu_i > 0$, we add $\mu - k\alpha_i$, $(k = 1, \ldots, \mu_i)$ into the weight system of π (α_i are the simple roots). The procedure is also applied to each new weight λ, until no new weight can be added.

• For the dual representation, we have the following proposition: if (π, V) is a representation of $\mathfrak{g}$, and $\pi^*(X) = -\pi(X)^T$ is its dual representation, then μ is a weight of π if and only if $-\mu$ is a weight of π^*.

Proof: in the definition of weight μ:

$$\pi(H)v = \langle\mu, H\rangle v \,, \tag{4.13}$$

we replace $\pi(H)$ with $-\pi^*(H)^T$. Since $\pi(H)$ can be simultaneously diagonalized, we have $\pi^*(H)^T = \pi^*(H)$. Now we have

$$\pi^*(H)v = -\langle \mu, H\rangle v\,, \tag{4.14}$$

which means that $-\mu$ is a weight of π^*.

• Examples:

(1) Trivial representation $\mathbf{1}$ for $\mathfrak{g}$: $\mu = (0, 0, \ldots, 0)$. Only one weight and $\dim(\mathbf{1}) = 1$.

(2) $\mathfrak{g} = A_1 = \mathfrak{su}(2)$, the simple root is $\alpha = 2$. For a representation π_m with highest weight m, the weight system is $\{m, m-2, \ldots, -(m-2), -m\}$. Each weight has multiplicity 1. Hence the dimension of π_m is $(m+1)$, consistent with what we learned before.

(3) $\mathfrak{g} = A_2 = \mathfrak{su}(3)$, the simple roots are $\alpha_1 = (2, -1)$, $\alpha_2 = (-1, 2)$.

(a) Fundamental representation $\mathbf{3}$, with highest weight $\mu = (\langle \mu, H_1\rangle, \langle \mu, H_2\rangle) = (1, 0)$.

$$(1,0) \xrightarrow{-\alpha_1} (-1,1) \xrightarrow{-\alpha_2} (0,-1) \tag{4.15}$$

This kind of diagram is called the *weight diagram*.

(b) Anti-fundamental representation $\overline{\mathbf{3}}$, with highest weight $\mu = (0, 1)$.

$$(0,1) \xrightarrow{-\alpha_2} (1,-1) \xrightarrow{-\alpha_1} (-1,0) \tag{4.16}$$

Note that the lowest weight of a representation is the opposite of the highest weight of its conjugate (dual) representation.

(c) Adjoint representation $\mathbf{8}$, with highest weight (1,1)

$$\begin{tikzcd}
 & (1,1) \arrow[dl] \arrow[dr] & \\
(-1,2) \arrow[dr] & & (2,-1) \arrow[dl] \\
 & (0,0) \arrow[dl] \arrow[dr] & \\
(-2,1) \arrow[dr] & & (1,-2) \arrow[dl] \\
 & (-1,-1) &
\end{tikzcd} \tag{4.17}$$

Note that the weight system of $\mathbf{8}$ is identical to the root system of A_2.

The conjugate representation of the adjoint representation is itself, and it is a real representation.

Also note that the weight $(0,0)$ has multiplicity two, the weight space of $(0,0)$ is spanned by $\pi(Z_{-\alpha_1})\pi(Z_{-\alpha_2})v_{(1,1)}$ and $\pi(Z_{-\alpha_2})\pi(Z_{-\alpha_1})v_{(1,1)}$, where $v_{1,1}$ is a weight vector of $(1,1)$. This can also be derived from the fact that the weight space of $(0,0)$ is equal to the Cartan subalgebra of A_2.

(d) Rank-2 symmetric representation **6**, with highest weight (2,0)

$$\begin{tikzcd}
 & (2,0) \arrow[dr] & \\
 & & (0,1) \arrow[dl] \arrow[dr] & \\
 & (1,-1) \arrow[dr] & & (-2,2) \arrow[dl] \\
 & & (-1,0) \arrow[dl] & \\
 & (0,-2) & &
\end{tikzcd} \tag{4.18}$$

(4) $\mathfrak{g} = A_{n-1} = \mathfrak{su}(n)$. The fundamental representation $\mathbf{n}$ has highest weight $(1,0,\ldots,0)$, while the anti-fundamental representation $\overline{\mathbf{n}}$ has highest weight $(0,\ldots,0,1)$. The adjoint representation has highest weight $(1,0,\ldots,0,1)$.

More generally, an irreducible representation with highest weight $(\mu_1,\mu_2,\ldots,\mu_{n-1})$ can be equivalently denoted by a *Young diagram* $[l_1;l_2;\ldots;l_{n-1}]$, where the i-th row has l_i boxes.

$$l_i = \mu_i + \mu_{i+1} + \cdots + \mu_{n-1}\,. \tag{4.19}$$

For example, the fundamental representation with highest weight $(1,0,\ldots,0)$ is

$$\begin{array}{|c|}\hline \\ \hline \end{array} \tag{4.20}$$

the adjoint representation of $\mathfrak{su}(3)$, with highest weight $(1,1)$ is

$$\begin{array}{|c|c|}\hline & \\ \hline & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \tag{4.21}$$

For a representation π with highest weight $(\mu_1,\mu_2,\ldots,\mu_{n-1})$, its dual representation π^* has the highest weight $(\mu_{n-1},\ldots,\mu_2,\mu_1)$. This is because the

lowest weight of π is $(-\mu_{n-1}, \ldots, -\mu_2, -\mu_1)$, which implies that $(\mu_{n-1}, \ldots, \mu_2, \mu_1)$ is the highest weight of π^*.

(5) $\mathfrak{g} = B_n = \mathfrak{so}(2n+1)$. The vector representation has highest weight $(1, 0, \ldots, 0)$, whose dimension is $(2n+1)$. The weight diagram is

$$\begin{gathered}(1,0,0,\ldots,0) \to (-1,1,0,\ldots,0) \to (0,-1,1,\ldots,0) \to \cdots \to \\ (0,\ldots,-1,1,0) \to (0,\ldots,0,-1,2) \to (0,\ldots,0) \to \\ (0,\ldots,0,1,-2) \to (0,\ldots,1,-1,0) \to \cdots \to (-1,0,\ldots,0)\end{gathered} \tag{4.22}$$

Another interesting representation of B_n is the spinor representation, with highest weight $(0, 0, \ldots, 1)$ and dimension 2^n.

The adjoint representation of $B_n (n > 2)$ has highest weight $(0, 1, \ldots, 0)$ and dimension $n(2n+1)$. For B_2, the adjoint representation has highest weight $(0, 2)$.

All representations of B_n are self-conjugate.

(6) $\mathfrak{g} = C_n = \mathfrak{sp}(n)$. The vector representation has highest weight $(1, 0, \ldots, 0)$, whose dimension is $2n$. The weight diagram is

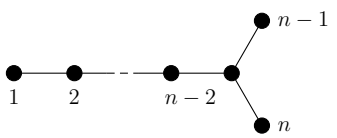
$$\begin{gathered}(1,0,0,\ldots,0) \to (-1,1,0,\ldots,0) \to (0,-1,1,\ldots,0) \to \cdots \to \\ (0,\ldots,0,-1,1) \to (0,\ldots,0,1,-1) \to (0,\ldots,1,-1,0) \to (-1,0,\ldots,0)\end{gathered} \tag{4.23}$$

The adjoint representation of C_n has highest weight $(2, 0, \ldots, 0)$ and dimension $n(2n+1)$.

All representations of C_n are self-conjugate.

(7) $\mathfrak{g} = D_n = \mathfrak{so}(2n)$.

The simple roots of D_n are labeled as



(4.24)

The vector representation has highest weight $(1, 0, \ldots, 0)$, whose dimension is $2n$. The weight diagram is

$$\begin{gathered}(1,0,0,\ldots,0) \to (-1,1,0,\ldots,0) \to (0,-1,1,\ldots,0) \to \cdots \to (0,0,\ldots,1,-1,-1) \\ \to (0,\ldots,1,-1,0,0) \to (1,-1,\ldots,0) \to (-1,0,\ldots,0)\end{gathered} \tag{4.25}$$

There are two inequivalent spinor representations with highest weight $(0, 0, \ldots, 1, 0)$ and $(0, 0, \ldots, 0, 1)$ and dimension 2^{n-1}, which are conjugate to

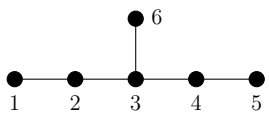
each other. They correspond to the chiral and anti-chiral spinor representations of the Lorentz group in physics.

The adjoint representation has highest weight $(0, 1, \ldots, 0)$ and dimension $n(2n-1)$.

In general, the dual representation of the one with highest weight $(\mu_1, \mu_2, \ldots, \mu_{n-2}, \mu_{n-1}, \mu_n)$ is the one with $(\mu_1, \mu_2, \ldots, \mu_{n-2}, \mu_n, \mu_{n-1})$.

(8) $\mathfrak{g} = E_6$.

The simple roots are labeled as



(4.26)

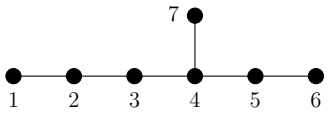
The fundamental representation **27** has highest weight $(1, 0, 0, 0, 0, 0)$. Its conjugate representation $\overline{\mathbf{27}}$ has highest weight $(0, 0, 0, 0, 1, 0)$.

The adjoint representation **78** has highest weight $(0, 0, 0, 0, 0, 1)$.

In general, the dual representation of the one with highest weight $(\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6)$ is the one with $(\mu_5, \mu_4, \mu_3, \mu_2, \mu_1, \mu_6)$.

(9) $\mathfrak{g} = E_7$.

The simple roots are labeled as



(4.27)

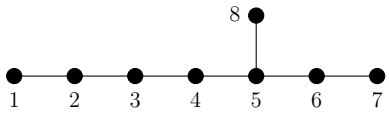
The fundamental representation **56** has highest weight $(1, 0, 0, 0, 0, 0, 0)$.

The adjoint representation **133** has highest weight $(0, 0, 0, 0, 0, 1, 0)$.

All representations of E_7 are self-conjugate.

(10) $\mathfrak{g} = E_8$.

The simple roots are labeled as

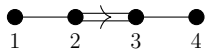


(4.28)

The adjoint representation **248** has highest weight $(1, 0, 0, 0, 0, 0, 0, 0)$, which is also the smallest non-trivial representation of E_8. All representations of E_8 are real.

(11) $\mathfrak{g} = F_4$.

The simple roots are labeled as



(4.29)

The fundamental representation **26** has highest weight $(0,0,0,1)$.

The adjoint representation **52** has highest weight $(1,0,0,0)$.

All representations of F_4 are real.

(12) $\mathfrak{g} = G_2$.

The fundamental representation **7** has highest weight $(0,1)$, the weight diagram is

$$(0,1) \to (1,-1) \to (-1,2) \to (0,0) \to (1,-2) \to (-1,1) \to (0,-1) \quad (4.30)$$

The adjoint representation **14** has highest weight $(1,0)$, the (half of) weight diagram is

$$\begin{tikzcd}
 & (1,0) \arrow[dr] & \\
 & & (-1,3) \arrow[dl] \\
 & (0,1) \arrow[dl] & \\
(1,-1) \arrow[dl] \arrow[dr] & & \\
(2,-3) \arrow[dr] & & (-1,2) \arrow[dl] \\
 & (0,0) &
\end{tikzcd} \quad (4.31)$$

All representations of G_2 are real.

(13) $\mathfrak{g} = A_1 \oplus A_1$

The Cartan matrix is

$$\mathcal{C}_{ij} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} , \quad (4.32)$$

hence the simple roots are $\alpha_1 = (2,0)$, $\alpha_2 = (0,2)$.

The irreducible representations have the highest weight $\mu = (\mu_1, \mu_2)$. In fact, the representation $\pi_\mu = (\pi_{\mu_1}, \pi_{\mu_2}) \equiv (\mu_{\mathbf{1}} + \mathbf{1}, \mu_{\mathbf{2}} + \mathbf{1})$, which is the tensor product representation of the A_1 representations π_{μ_1} and π_{μ_2}. For example, the representation with the highest weight $\mu = (1,1)$ has the weight diagram

$$
\begin{tikzcd}
 & (1,1) \arrow[dl] \arrow[dr] & \\
(1,-1) \arrow[dr] & & (-1,1) \arrow[dl] \\
 & (-1,-1) &
\end{tikzcd}
\tag{4.33}
$$

This is the bifundamental representation $(\mathbf{2},\mathbf{2})$ of $\mathfrak{su}(2)\oplus\mathfrak{su}(2)$.

• Note that the adjoint representation of $\mathfrak{su}(2)\oplus\mathfrak{su}(2)$ is $(\mathbf{3},\mathbf{1})\oplus(\mathbf{1},\mathbf{3})$, which is not an irreducible representation.

A brief summary:

(1) Each irrep. π_μ of $\mathfrak{g}$ is uniquely characterized by the highest weight $\mu=(\mu_1,\ldots,\mu_r)$, $(\mu_i\geq 0)$

(2) From the highest weight μ, subtract roots (rows of Cartan matrix) from it recursively to generate the weight system of π_μ.

(3) For a rep. π_μ with weights $(\lambda_1,\lambda_2,\ldots,\lambda_n)$, its dual rep. π_μ^* has weights $(-\lambda_1,-\lambda_2,\ldots,-\lambda_n)$.

(4) The trivial rep. always correspond to $\mu=(0,\ldots,0)$.

(5) The adjoint rep. is always self-dual (self-conjugate), and its highest weight can be read off using the extended Dynkin diagram $\hat{\mathfrak{g}}$.

4.2 Weyl character formula and dimensions

We present some simple ways to compute the dimension of a representation (π, V) of $\mathfrak{g}$, without writing out the full weight diagram. We first define the *character* of π to be the function $\chi_\pi : \mathfrak{g} \to \mathbb{C}$:

$$\chi_\pi(X) = tr(e^{\pi(X)}) \,. \tag{4.34}$$

• Suppose that V can be decomposed as a direct sum of weight spaces V_λ with multiplicity $\text{mult}(\lambda)$, then for $H \in \mathfrak{h}$ we have

$$\chi_\pi(H) = \sum_\lambda \text{mult}(\lambda) e^{\langle \lambda, H \rangle} \,. \tag{4.35}$$

• Moreover, the dimension of V (thus the dimension of the representation) is equal to the value of χ_π at the origin:

$$\dim(\pi) := \dim(V) = \chi_\pi(0) \,. \tag{4.36}$$

For example, let π to be the irrep $\mathbf{m+1}$ of $\mathfrak{sl}(2, \mathbb{C})$, and let

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \,, \tag{4.37}$$

then

$$\begin{aligned} \chi_\pi(aH) &= \chi_\pi(\begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}) \\ &= e^{ma} + e^{(m-2)a} + \cdots + e^{-ma} \\ &= \frac{\sinh((m+1)a)}{\sinh(a)} \,. \end{aligned} \tag{4.38}$$

Take the limit of $a \to 0$, we recover the dimension of $\mathbf{m+1}$. From the sum of exponentials, we also confirm that the multiplicity of each weight equals to 1.

• Now we present the *Weyl Character Formula* for a general $\mathfrak{g}$ and (π, V) with highest weight μ:

$$\chi_\pi(H) = \frac{\sum_{w \in W} \epsilon(w) e^{\langle w \cdot (\mu + \delta), H \rangle}}{\sum_{w \in W} \epsilon(w) e^{\langle w \cdot \delta, H \rangle}} \tag{4.39}$$

The sum is over all the elements w of the Weyl group W. $\epsilon(w)$ is the determinant of the Weyl group action, or equivalently, defined to be

$$\epsilon(w) = (-1)^{s(w)} \,, \tag{4.40}$$

where $s(w)$ is minimal number of s_{α_i} elements (3.29) in the multiplicative decomposition of w.

δ is half of the sum of positive roots of $\mathfrak{g}$:

$$\delta := \frac{1}{2} \sum_{\alpha_+ \in R_+} \alpha_+ \,. \tag{4.41}$$

Now to compute the dimension of (π, V), we set $H = t\delta$ and simplify

$$\chi_\pi(t\delta) = \prod_{\alpha \in R^+} \frac{\sinh\langle \alpha, (\mu+\delta)t/2\rangle}{\sinh\langle \alpha, \delta t/2\rangle} \tag{4.42}$$

We get

$$\dim(\pi) = \lim_{t\to 0} \chi_\pi(t\delta) = \prod_{\alpha \in R^+} \frac{\langle \mu+\delta, \alpha\rangle}{\langle \delta, \alpha\rangle} \,. \tag{4.43}$$

• Example: (1) $\mathfrak{su}(3)$, the representation with highest weight $\mu = (m_1, m_2)$.

The simple roots are $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, and the other positive root is $\alpha_3 = \alpha_1 + \alpha_2 = e_1 - e_3$ (e_is are the orthonormal basis of $\mathbb{R}^3$). Hence

$$\delta = \frac{1}{2}(\alpha_1 + \alpha_2 + \alpha_3) = \alpha_1 + \alpha_2 \,. \tag{4.44}$$

We can compute

$$\begin{aligned} &\langle \alpha_1, \mu+\delta\rangle = m_1 + 1 \ , \ \langle \alpha_2, \mu+\delta\rangle = m_2 + 1 \ , \ \langle \alpha_3, \mu+\delta\rangle = m_1 + m_2 + 2 \\ &\langle \delta, \alpha_1\rangle = 1 \ , \ \langle \delta, \alpha_2\rangle = 1 \ , \ \langle \delta, \alpha_3\rangle = 2 \,. \end{aligned} \tag{4.45}$$

Hence

$$\dim(\pi_\mu) = \frac{1}{2}(m_1+1)(m_2+1)(m_1+m_2+2) \,. \tag{4.46}$$

(2) $\mathfrak{su}(4)$, the simple roots are

$$\alpha_1 = e_1 - e_2 \ , \ \alpha_2 = e_2 - e_3 \ , \ \alpha_3 = e_3 - e_4 \,, \tag{4.47}$$

and the other positive roots are

$$\alpha_1 + \alpha_2 \ , \ \alpha_2 + \alpha_3 \ , \ \alpha_1 + \alpha_2 + \alpha_3 \,. \tag{4.48}$$

$$\delta = \frac{3}{2}\alpha_1 + 2\alpha_2 + \frac{3}{2}\alpha_3 \tag{4.49}$$

For the representation μ with highest weight (m_1, m_2, m_3), we compute

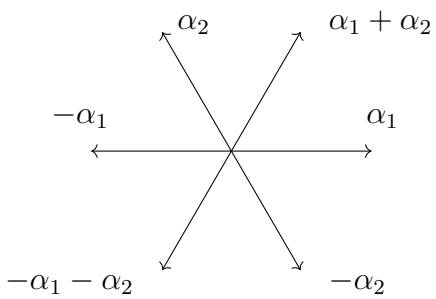
$$\begin{aligned}
&\langle \alpha_1, \mu+\delta\rangle = m_1+1\ ,\ \langle \alpha_2, \mu+\delta\rangle = m_2+1\ ,\ \langle \alpha_3, \mu+\delta\rangle = m_3+1\\
&\langle \alpha_1+\alpha_2, \mu+\delta\rangle = m_1+m_2+2\ ,\ \langle \alpha_2+\alpha_3, \mu+\delta\rangle = m_2+m_3+2\\
&\langle \alpha_1+\alpha_2+\alpha_3, \mu+\delta\rangle = m_1+m_2+m_3+3\\
&\langle \delta, \alpha_1\rangle = 1\ ,\ \langle \delta, \alpha_2\rangle = 1\ ,\ \langle \delta, \alpha_3\rangle = 1\ ,\ \langle \delta, \alpha_1+\alpha_2\rangle = 2\\
&\langle \delta, \alpha_2+\alpha_3\rangle = 2\ ,\ \langle \delta, \alpha_1+\alpha_2+\alpha_3\rangle = 3\,.
\end{aligned} \tag{4.50}$$

The dimension formula is

$$\dim(\pi_\mu) = \frac{1}{12}(m_1+1)(m_2+1)(m_3+1)(m_1+m_2+2)(m_2+m_3+2)(m_1+m_2+m_3+3)\,. \tag{4.51}$$

• Let us show an example of using the Weyl Character formula to compute the multiplicity of a weight λ in a highest weight representation.

Take $\mathfrak{g} = \mathfrak{su}(3)$, and the two simple roots are $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$. The root system of $\mathfrak{g}$ is



(4.52)

The half of the sum of positive roots is

$$\delta = \alpha_1 + \alpha_2 = e_1 - e_3\,. \tag{4.53}$$

We take a generic element of the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{su}(3)$ as

$$H = aH_{\alpha_1} + bH_{\alpha_2}\,. \tag{4.54}$$

Now for the action of Weyl group, note that the Weyl group of $\mathfrak{su}(3)$ is the permutation group S_3 which permutes 3 elements: $\{e_1, e_2, e_3\}$.

The list of Weyl group elements w and ϵ_w are:

w	$\epsilon(w)$
1	1
$e_1 \leftrightarrow e_2$	-1
$e_2 \leftrightarrow e_3$	-1
$e_1 \leftrightarrow e_3$	-1
$e_1e_2e_3 \leftrightarrow e_2e_3e_1$	1
$e_1e_2e_3 \leftrightarrow e_3e_1e_2$	1

(4.55)

Now let us compute each term in

$$\chi_\pi(H) = \frac{\sum_{w\in W} \epsilon(w) e^{\langle w\cdot(\mu+\delta),H\rangle}}{\sum_{w\in W} \epsilon(w) e^{\langle w\cdot\delta,H\rangle}}\,, \tag{4.56}$$

for a highest weight $\mu = (m_1, m_2)$ ($m_i = \langle \mu, H_{\alpha_i}\rangle$).

As a result, we have the following character formula for a generic $\mathfrak{su}(3)$ highest weight rep.:

$$\begin{aligned}\chi_{(m_1,m_2)}(aH_{\alpha_1} + bH_{\alpha_2}) = [(&e^{(m_1+1)a+(m_2+1)b} + e^{(m_1+1)b-(m_1+m_2+2)a} + e^{(m_2+1)a-(m_1+m_2+2)b} \\ &- e^{(m_1+m_2+2)b-(m_1+1)a} - e^{(m_1+m_2+2)a-(m_2+1)b} - e^{-(m_2+1)a-(m_1+1)b}] \\ &/(e^{a+b} + e^{b-2a} + e^{a-2b} - e^{2a-b} - e^{2b-a} - e^{-a-b})\end{aligned} \tag{4.57}$$

Let us check a few examples:

(1) $(m_1, m_2) = (0, 0)$: $\chi_{(0,0)}(aH_{\alpha_1} + bH_{\alpha_2}) = 1$

(2) $(m_1, m_2) = (1, 0)$: $\chi_{(1,0)}(aH_{\alpha_1} + bH_{\alpha_2}) = e^a + e^{b-a} + e^{-b}$, corresponding to the weights $(1, 0)$, $(-1, 1)$ and $(0, -1)$

(3) $(m_1, m_2) = (1, 1)$:

$$\chi_{(1,1)}(aH_{\alpha_1} + bH_{\alpha_2}) = e^{a+b} + e^{-a+2b} + e^{-b+2a} + 2 + e^{a-2b} + e^{b-2a} + e^{-a-b}\,. \tag{4.58}$$

One can see that the weight $(0, 0)$ indeed has multiplicity two.

4.3 Casimir operators

In this section, we introduce the concept of Casimir operators of a semisimple Lie algebra $\mathfrak{g}$. They are invariant operators of $\mathfrak{g}$, whose eigenvalues characterize a representation π.

We first introduce the notion of the universal enveloping algebra of $\mathfrak{g}$, which contains the Casimir operators.

• We first define the *tensor algebra* of $\mathfrak{g}$ to be

$$\mathcal{T}(\mathfrak{g}) = \bigoplus_{k=0}^{\infty} \mathfrak{g}^{\otimes k} \tag{4.59}$$

which is equipped with the associative multiplication

$$(v_1 \otimes v_2 \otimes \cdots \otimes v_k) \cdot (w_1 \otimes w_2 \otimes \ldots w_l) = v_1 \otimes \cdots \otimes v_k \otimes w_1 \otimes \cdots \otimes w_l \,. \tag{4.60}$$

Now we consider an associative subalgebra (two-sided ideal) $\mathcal{J} \subset \mathcal{T}(\mathfrak{g})$, generated by elements of the form $x \otimes y - y \otimes x - [x, y]$:

$$\mathcal{J} = \mathrm{span}\{A \otimes (x \otimes y - y \otimes x - [x, y]) \otimes B | x, y \in \mathfrak{g},\ A, B \in \mathcal{T}(\mathfrak{g})\} \,. \tag{4.61}$$

The quotient algebra

$$U(\mathfrak{g}) = \mathcal{T}(\mathfrak{g})/\mathcal{J} \,, \tag{4.62}$$

which can also be written as

$$U(\mathfrak{g}) = \mathcal{T}(\mathfrak{g})/(x \otimes y - y \otimes x - [x, y]) \,, \tag{4.63}$$

is called the *universal enveloping algebra* of $\mathfrak{g}$.

Now the Casimir operators are the generators of the center $Z(U(\mathfrak{g}))$, which commute with any element in $U(\mathfrak{g})$.

• For all semisimple Lie algebra $\mathfrak{g}$, there exists a *quadratic Casimir operator* defined in the following way:

$$C_2 = -\sum_j X_j^2 \,, \tag{4.64}$$

where X_j is the orthonormal basis of $\mathfrak{r} \subset \mathfrak{g}$ (real part of the complex Lie algebra).

• We prove that C_2 is indeed a center element that commutes with any $X_j \in \mathfrak{g}$:

$$\begin{aligned} -[X_j, C_2] &= \sum_k [X_j, X_k^2] \\ &= \sum_k ([X_j, X_k]X_k + X_k[X_j, X_k]) \\ &= \sum_{k,l} c_{jkl} X_l X_k + \sum_{k,l} c_{jkl} X_k X_l \\ &= \sum_{k,l} (c_{jkl} + c_{jlk}) X_l X_k \,. \end{aligned} \tag{4.65}$$

Now note that in the basis X_j, the operator ad_{X_j} is represented by

$$(\mathrm{ad}_{X_j})_{kl} = c_{jlk} \,. \tag{4.66}$$

From the definition of inner product

$$\langle \mathrm{ad}_X(Y), Z \rangle = -\langle Y, \mathrm{ad}_X(Z) \rangle = \langle Y, \mathrm{ad}_{X^*}(Z) \rangle \,, \tag{4.67}$$

the matrix operator ad_{X_j} should be anti-symmetric, which means that

$$c_{jlk} = -c_{jkl} \,. \tag{4.68}$$

Hence we have proved that $[X_j, C_2] = 0$.

• Now let's act C_2 on a highest weight representation (π, V) of $\mathfrak{g}$ with highest weight μ. We have

$$\pi(C_2) = -\sum_j \pi(X_j)^2 = c_\mu I \,. \tag{4.69}$$

c_μ is a constant number. Since C_2 is in the center of $U(\mathfrak{g})$, $\pi(C_2)$ commutes with each $\pi(X)$, $X \in \mathfrak{g}$. From Schur's lemma, $\pi(C_2)$ is a multiple of the identity operator.

The number c_μ can be computed as

$$c_\mu = \langle \mu, \mu + 2\delta \rangle \,, \tag{4.70}$$

where δ is half of the sum of positive roots of $\mathfrak{g}$:

$$\delta := \frac{1}{2} \sum_{\alpha_+ \in R_+} \alpha_+ \,. \tag{4.71}$$

• Proof: let us take an orthonormal basis of $\mathfrak{r} \subset \mathfrak{g}$ as follows: (1) take an orthonormal basis H_i for the (real part of) Cartan subalgebra $\mathfrak{t} \subset \mathfrak{h} \subset \mathfrak{g}$; (2)

for each $\alpha \in R_+$, choose a unit vector X_α in $\mathfrak{g}_\alpha$, so that X_α^* is a unit vector in $\mathfrak{g}_{-\alpha}$, and we define

$$Y_\alpha := \frac{1}{\sqrt{2}i}(X_\alpha + X_\alpha^*)\ ,\ Z_\alpha := \frac{1}{\sqrt{2}}(X_\alpha - X_\alpha^*)\,. \tag{4.72}$$

They satisfy $Y_\alpha^* = -Y_\alpha$, $Z_\alpha^* = -Z_\alpha$, and they are unit vectors orthogonal to each other.

We use H_i $(i = 1, \ldots, r)$, Y_α, Z_α, $\alpha \in R_+$ as an orthonormal basis of $\mathfrak{r}$.

We can compute

$$\begin{aligned} -Y_\alpha^2 - Z_\alpha^2 &= X_\alpha X_\alpha^* + X_\alpha^* X_\alpha \\ &= 2X_\alpha^* X_\alpha + [X_\alpha, X_\alpha^*]\,. \end{aligned} \tag{4.73}$$

Hence

$$C_2 = \sum_{\alpha \in R_+} (2X_\alpha^* X_\alpha + [X_\alpha, X_\alpha^*]) - \sum_{j=1}^r H_j^2\,. \tag{4.74}$$

Suppose that v is a highest weight vector with weight μ, we compute

$$\pi(C_2)v = -\sum_{j=1}^r \pi(H_j)^2 v + 2\sum_{\alpha \in R^+} \pi(X_\alpha^*)\pi(X_\alpha)v + \sum_{\alpha \in R^+} \pi([X_\alpha, X_\alpha^*])v\,. \tag{4.75}$$

The first term is

$$\begin{aligned} -\sum_{j=1}^r \pi(H_j)^2 v &= -\sum_{j=1}^r \langle \mu, H_j\rangle^2 v \\ &= \sum_{j=1}^r |\langle \mu, H_j\rangle|^2 v \\ &= \langle \mu, \mu\rangle v\,. \end{aligned} \tag{4.76}$$

The first line to the second line is because $\langle \mu, H_j\rangle$ is a pure imaginary number. The reason is that since ad_H is a skew self-adjoint operator, from the definition of inner product, for an orthonormal basis H_i, actually all $\langle \alpha, H_i\rangle$ are imaginary numbers. Since $\langle \alpha, \mu\rangle$ are always real for any weight μ, $\langle \mu, H_j\rangle$ also has to be an imaginary number.

The second term vanishes since $\pi(X_\alpha)v = 0$ for the highest weight vector v.

For the third term, note that we have

$$\begin{aligned} \langle [X_\alpha, X_\alpha^*], H_\alpha\rangle &= \langle \mathrm{ad}_{X_\alpha}(X_\alpha^*), H_\alpha\rangle \\ &= \langle X_\alpha^*, \mathrm{ad}_{X_\alpha^*}(H_\alpha)\rangle \\ &= -\langle X_\alpha^*, [H_\alpha, X_\alpha^*]\rangle \\ &= -\langle X_\alpha^*, \langle -\alpha, H_\alpha\rangle X_\alpha^*\rangle \\ &= \langle \alpha, H_\alpha\rangle\,. \end{aligned} \tag{4.77}$$

Which implies

$$[X_\alpha, X_\alpha^*] = \alpha\,. \tag{4.78}$$

Thus

$$\sum_{\alpha\in R_+} \pi([X_\alpha, X_\alpha^*])v = \sum_{\alpha\in R_+} \langle\mu,\alpha\rangle v \tag{4.79}$$

and we get the final result

$$c_\mu = \langle\mu, \mu + 2\delta\rangle\,. \tag{4.80}$$

- Example: $\mathfrak{so}(3)$, with generators

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}\ ,\ L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}\ ,\ L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\,. \tag{4.81}$$

Since the inner product of $\mathfrak{so}(n)$ is defined as

$$\langle X, Y\rangle = \frac{1}{2} tr(X^\dagger Y)\,, \tag{4.82}$$

L_i is an orthonormal basis.

The quadratic Casimir operator is

$$C_2 = -(L_1^2 + L_2^2 + L_3^2) \tag{4.83}$$

For the representation $\mathbf{2l+1}$, labeled by the highest weight μ, where

$$\langle\mu, H_\alpha\rangle = 2l\ ,\ \mu = l\alpha\,. \tag{4.84}$$

The eigenvalue of $\pi_{\mathbf{2l+1}}(C_2)$ is

$$c_\mu = \langle\mu, \mu + 2\delta\rangle = l(l+1)\langle\alpha,\alpha\rangle\,. \tag{4.85}$$

Now for $\langle\alpha,\alpha\rangle$, note that from the definition of root α:

$$[H_\alpha, X_\alpha] = \langle\alpha, H_\alpha\rangle X_\alpha = 2X_\alpha\,, \tag{4.86}$$

we can choose $X_\alpha = L_1 + iL_2$, $H_\alpha = 2iL_3$, $\alpha = iL_3$, and thus $\langle\alpha,\alpha\rangle = 1$ is fixed.

Hence

$$c_\mu = l(l+1)\,, \tag{4.87}$$

which matches the eigenvalue of operator L^2 for a quantum mechanics system with angular momentum l.

As a consistency check, take $l = 1$, which corresponds to the 3d standard representation of $\mathfrak{so}(3)$ with $\pi(L_i) = L_i$,

$$\pi_{\mathbf{3}}(C_2) = C_2 = 2I\,. \tag{4.88}$$

• In fact, for a general semisimple Lie algebra $\mathfrak{g}$, the are other Casimir operators than C_2, which are invariant tensors of $\mathfrak{g}$ as well. For a Casimir operator $C_n \in \mathfrak{g}^{\otimes n}$ with degree n, it also satisfies $[X_j, C_n] = 0$ for all $X_j \in \mathfrak{g}$. We list the degrees of Casimir operator for a simple Lie algebra in the following table:

$\mathfrak{g}$	Degree of Casimirs
A_n	$2, 3, \ldots, n+1$
B_n	$2, 4, \ldots, 2n$
C_n	$2, 4, \ldots, 2n$
D_n	$2, 4, \ldots, 2n-2, n$
E_6	$2, 5, 6, 8, 9, 12$
E_7	$2, 6, 8, 10, 12, 14, 18$
E_8	$2, 8, 12, 14, 18, 20, 24, 30$
F_4	$2, 6, 8, 12$
G_2	$2, 6$

(4.89)

4.4 Tensor product decomposition

4.4.1 General procedure

We discuss the tensor product decomposition of two irreducible representations (π_1, V_1) and (π_2, V_2) of $\mathfrak{g}$ into the direct sum of irreps:

$$\pi_1 \otimes \pi_2 = \bigoplus \pi_i^{\oplus m_i} \,. \tag{4.90}$$

Here m_i is the multiplicity of the irrep π_i in the decomposition.

Denote the weight system of π_1 and π_2 by W_1 and W_2. Since V_1 and V_2 can be decomposed into the direct sum of the weight spaces V_i^λ respectively:

$$V_1 = \bigoplus_{\lambda \in W_1} V_1^\lambda \,, \; V_2 = \bigoplus_{\mu \in W_2} V_2^\mu \,, \tag{4.91}$$

the vector space of the product rep. $\pi_1 \otimes \pi_2$ is

$$V_{\pi_1 \otimes \pi_2} = V_1 \otimes V_2 = \bigoplus_{\lambda \in W_1, \mu \in W_2} V_1^\lambda \otimes V_2^\mu \,. \tag{4.92}$$

Now we take any weight vector $v_1 \in V_1^\lambda$, $v_2 \in V_2^\mu$.

From the definition of the tensor product representation, we have for any $H \in \mathfrak{h}$

$$\begin{aligned} (\pi_1 \otimes \pi_2)(H)(v_1 \otimes v_2) &= (\pi_1(H)v_1) \otimes v_2 + v_1 \otimes (\pi_2(H)v_2) \\ &= \langle \lambda + \mu, H \rangle v_1 \otimes v_2 \,. \end{aligned} \tag{4.93}$$

Hence for any weight $\lambda \in W_1$, $\mu \in W_2$, $\lambda + \mu$ is a weight of $\pi_1 \otimes \pi_2$.

- Now we have the following procedure to calculate (4.90):

1. Write down the weight systems W_1 and W_2 of π_1 and π_2.

2. Write down all the possible sum of weights $\lambda + \mu$, where $\lambda \in W_1$, $\mu \in W_2$. Keep in mind that the weights can have multiplicities. The weight $\lambda + \mu$ in $\pi_1 \otimes \pi_2$ has multiplicity $\mathrm{mult}(\lambda) \cdot \mathrm{mult}(\mu)$. More precisely, since there can be multiple ways to get a weight Λ of $\pi_1 \otimes \pi_2$. The multiplicity of Λ is
$$\mathrm{mult}(\Lambda) = \sum_{\lambda \in W_1, \mu \in W_2, \lambda + \mu = \Lambda} \mathrm{mult}(\lambda) \cdot \mathrm{mult}(\mu) \,. \tag{4.94}$$

3. Try to reorganize these weights into irreps of $\mathfrak{g}$.

• Examples: (1) $\mathfrak{su}(2)$, $\pi_1 = \mathbf{m}$, $\pi_2 = \mathbf{n}$.
The weights of π_1 are

$$m-1, m-3, \ldots, -(m-1)\,. \tag{4.95}$$

The weights of π_2 are

$$n-1, n-3, \ldots, -(n-1)\,. \tag{4.96}$$

The weights of $\pi_1 \otimes \pi_2$ are the sums of the weights above, with the highest weight $m+n-2$ and the lowest weight $-(m+n-2)$. All the other weights have a non-trivial multiplicity. After reorganizing them into irreps of $\mathfrak{su}(2)$, we get the following decomposition

$$\mathbf{m} \otimes \mathbf{n} = (\mathbf{m}+\mathbf{n}-\mathbf{1}) \oplus (\mathbf{m}+\mathbf{n}-\mathbf{3}) \oplus \cdots \oplus (\mathbf{m}-\mathbf{n}+\mathbf{1})\,. \tag{4.97}$$

(2) $\mathfrak{su}(3)$, $\pi_1 = \mathbf{3}$, $\pi_2 = \bar{\mathbf{3}}$.
The weights of π_1 are

$$(1,0)\ ,\ (-1,1)\ ,\ (0,-1)\,. \tag{4.98}$$

The weights of π_2 are

$$(0,1)\ ,\ (1,-1)\ ,\ (-1,0)\,. \tag{4.99}$$

The weights of $\pi_1 \otimes \pi_2$ are

$$(1,1)\ ,\ (-1,2)\ ,\ 3\cdot(0,0)\ ,\ (2,-1)\ ,\ (1,-2)\ ,\ (-2,1)\ ,\ (-1,-1)\,. \tag{4.100}$$

Now apart from one weight $(0,0)$, the other weights exactly form the adjoint representation $\mathbf{8}$. Hence we have

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}\,. \tag{4.101}$$

(3) $\mathfrak{su}(3)$, $\pi_1 = \mathbf{3}$, $\pi_2 = \mathbf{3}$.
The weights of π_1 and π_2 are

$$(1,0)\ ,\ (-1,1)\ ,\ (0,-1)\,. \tag{4.102}$$

The weights of $\pi_1 \otimes \pi_2$ are

$$(2,0)\ ,\ 2\cdot(0,1)\ ,\ (-2,2)\ ,\ 2\cdot(1,-1)\ ,\ 2\cdot(-1,0)\ ,\ (0,-2)\,. \tag{4.103}$$

They can be organized into a $\mathbf{6}$ representation with highest weight $(2,0)$ and a $\bar{\mathbf{3}}$ representation with highest weight $(0,1)$. Hence

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}\,. \tag{4.104}$$

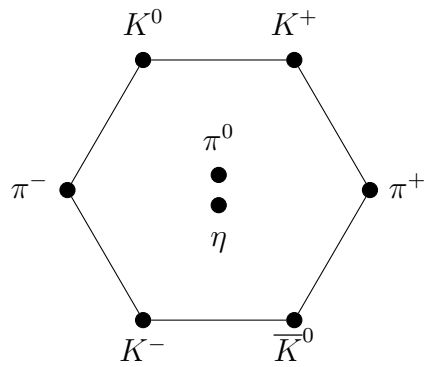
• Now we discuss some applications in particle physics. Consider the $SU(3)$ flavor symmetry in the massless limit of QCD, which rotates the three lightest quarks u, d, s. If the representation $\mathbf{3}$ is considered as a quark, where $SU(3)$ acts on the 3d vector space (u, d, s), and $\bar{\mathbf{3}}$ is considered as the anti-quark which acts on $(\bar{u}, \bar{d}, \bar{s})$. Then we have the following important $SU(3)$ representations from the tensor product decomposition:

(1) A meson consists of a quark and an anti-quark, from

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}\,, \tag{4.105}$$

the representation $\mathbf{1}$ corresponds to the "meson singlet" while $\mathbf{8}$ corresponds to the "meson octet".

• Meson octet:



(4.106)

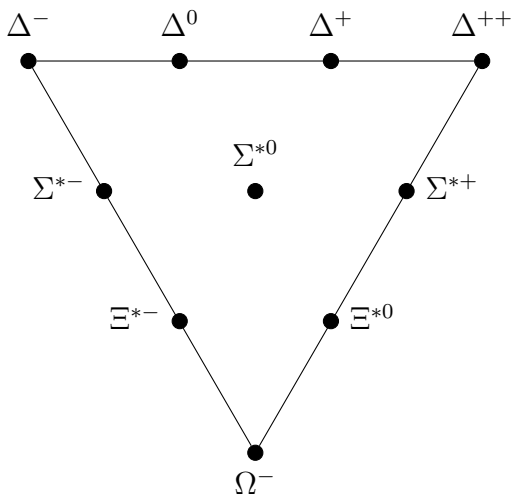
• Meson singlet η'

(2) A baryon consists of three quarks, from

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}\,, \tag{4.107}$$

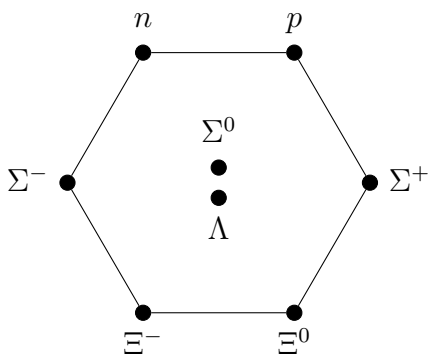
the representation $\mathbf{10}$ corresponds to the "baryon decuplet", and the two $\mathbf{8}$ corresponds to the "baryon octet" including the proton and neutron.

- Baryon decuplet:



(4.108)

- Baryon octet:



(4.109)

4.4.2 Tensor representations of $SU(N)$ and Young diagram

Now discuss the irreducible tensor representations of $SU(N)$ group (A_{n-1} Lie algebra) from the tensor product decomposition of fundamental and anti-fundamental reps.

A representation (Π, V) of $SU(N)$ acts on the N-dimensional vector space V as

$$v' = \Pi(g)v\,, \tag{4.110}$$

for $g \in SU(N)$, $v, v' \in V$.

For the fundamental rep. $\mathbf{N}$, we write out the components

$$v'_i = (\Pi(g))_{ij} v_j\,. \tag{4.111}$$

For anti-fundamental rep. $\overline{\mathbf{N}}$, it is

$$(v')^*_i = (\Pi^*(g))_{ij} v^*_j = (\overline{\Pi(g)})_{ij} v^*_j\,. \tag{4.112}$$

Now we rewrite them using upper and lower indices:

$$v_i \to v^i\ ,\ v^*_i \to v_i\ ,\ (\Pi(g))_{ij} \to U^i{}_j\ ,\ (\Pi^*(g))_{ij} = (\overline{\Pi(g)})_{ij} \to U_i{}^j\,. \tag{4.113}$$

U is just a unitary $N \times N$ matrix in this notation.

Using this notation, v^i is a contravariant tensor (with rank $(1,0)$) while v_i is a covariant tensor (with rank $(0,1)$):

$$v'^i = U^i{}_j v^j\ ,\ v'_i = U_i{}^j v_j\,. \tag{4.114}$$

From $UU^\dagger = I$, we have

$$U^i{}_j U_k{}^j = \delta^i_k\,. \tag{4.115}$$

We can also define a mixed tensor of rank (n, m), which transforms as

$$v'^{i_1 i_2 \ldots i_n}_{j_1 j_2 \ldots j_m} = (U^{i_1}{}_{k_1} \ldots U^{i_n}{}_{k_n})(U_{j_1}{}^{l_1} \ldots U_{j_m}{}^{l_m}) v^{k_1 \ldots k_n}_{l_1 \ldots l_m}\,. \tag{4.116}$$

In fact, this tensor representation is exactly the tensor product representation $\mathbf{N}^{\otimes n} \otimes \overline{\mathbf{N}}^{\otimes m}$ of $SU(N)$, which is generally reducible.

Take the example of the rank $(2,0)$ tensor v^{ij}, which transforms as

$$v'^{ij} = U^i{}_k U^j{}_l v^{kl} \tag{4.117}$$

Note that this representation is reducible into the symmetric and antisymmetric part

$$S^{ij} = \frac{1}{2}(v^{ij} + v^{ji})\ ,\ A^{ij} = \frac{1}{2}(v^{ij} - v^{ji})\,, \tag{4.118}$$

which are by themselves irreducible representations of $SU(N)$:

$$\begin{aligned} S'^{ij} &= \frac{1}{2}(U^i{}_k U^j{}_l v^{kl} + U^j{}_k U^i{}_l v^{kl}) \\ &= U^i{}_k U^j{}_l \cdot \frac{1}{2}(v^{kl} + v^{lk}) \\ &= U^i{}_k U^j{}_l S^{kl} \end{aligned} \tag{4.119}$$

$$\begin{aligned} A'^{ij} &= \frac{1}{2}(U^i{}_k U^j{}_l v^{kl} - U^j{}_k U^i{}_l v^{kl}) \\ &= U^i{}_k U^j{}_l \cdot \frac{1}{2}(v^{kl} - v^{lk}) \\ &= U^i{}_k U^j{}_l A^{kl} \,. \end{aligned} \tag{4.120}$$

Since they form invariant subspaces of the rank $(2,0)$ tensor rep. $\mathbf{N} \otimes \mathbf{N}$, we have the tensor product decomposition

$$\mathbf{N} \otimes \mathbf{N} = \frac{\mathbf{N}(\mathbf{N}-\mathbf{1})}{\mathbf{2}} \oplus \frac{\mathbf{N}(\mathbf{N}+\mathbf{1})}{\mathbf{2}} \,. \tag{4.121}$$

For example, for $SU(2)$ this means

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3} \,. \tag{4.122}$$

For $SU(3)$ it means

$$\mathbf{3} \otimes \mathbf{3} = \overline{\mathbf{3}} \oplus \mathbf{6} \,. \tag{4.123}$$

Note that a rank-2 antisymmetric tensor rep. A^{ij} is equivalent to the anti-fundamental rep. v_i in the $SU(3)$ case.

• One can also generalize this and define the rank-n symmetric tensor representation of $SU(N)$:

$$v^{(i_1 i_2 \ldots i_n)} = \frac{1}{n!} \sum_{\sigma \in S_n} (v^{i_{\sigma_1} i_{\sigma_2} \ldots i_{\sigma_n}}) \,. \tag{4.124}$$

Here σ is an element of the permutation group S_n, which describes a permutation of n elements $\{1, \ldots, n\} \to \{\sigma_1, \ldots, \sigma_n\}$.

There's also the rank-n antisymmetric tensor representation of $SU(N)$.

$$v^{[i_1 i_2 \ldots i_n]} = \frac{1}{n!} \sum_{\sigma \in S_n} (\epsilon(\sigma) v^{i_{\sigma_1} i_{\sigma_2} \ldots i_{\sigma_n}}) \,. \tag{4.125}$$

$\epsilon(\sigma)$ is equal to 1 for an even permutation σ, and (-1) for an odd permutation σ. The dimension of rank-n antisymmetric rep. of $SU(N)$ is equal to

$$C_N^n = \frac{N!}{n!(N-n)!} \,. \tag{4.126}$$

In general, one can construct new tensor rep. from old ones using the *invariant tensors* of $SU(N)$. They are defined to be invariant under the $SU(N)$ transformations, which include:

(1) δ^i_j, which is invariant under

$$\delta^i_j = U^i{}_k U_j{}^l \delta^k_l \,. \tag{4.127}$$

(2) The Levi-Civita symbol $\epsilon_{i_1 i_2 \ldots i_N}$ and $\epsilon^{i_1 i_2 \ldots i_N}$. They are defined as

$$\epsilon_{i_1 i_2 \ldots i_N} = \epsilon^{i_1 i_2 \ldots i_N} = \begin{cases} 1 & \text{if } (i_1 i_2 \ldots i_N) \text{ is an even permutation of } (12\ldots N) \\ -1 & \text{if } (i_1 i_2 \ldots i_N) \text{ is an odd permutation of } (12\ldots N) \\ 0 & \text{if otherwise} \end{cases} \tag{4.128}$$

Under the $SU(N)$ transformation, $\epsilon_{i_1 i_2 \ldots i_N}$ is transformed as

$$\epsilon'_{i_1 i_2 \ldots i_N} = U_{i_1}{}^{j_1} U_{i_2}{}^{j_2} \ldots U_{i_N}{}^{j_N} \epsilon_{j_1 j_2 \ldots j_N} \,. \tag{4.129}$$

Since the r.h.s. is equal to

$$\det(U) \epsilon_{i_1 i_2 \ldots i_N} = \epsilon_{i_1 i_2 \ldots i_N} \,, \tag{4.130}$$

we see that $\epsilon_{i_1 i_2 \ldots i_N}$ is indeed an invariant tensor of $SU(N)$.

Using the Levi-Civita tensor, we can transform the rank-$(N-1)$ anti-symmetric tensor rep. into the anti-fundamental rep.:

$$\epsilon_{i_1 i_2 \ldots i_N} v^{[i_2 \ldots i_N]} = v_{i_1} \,. \tag{4.131}$$

For example, the rank-2 anti-symmetric rep. of $SU(3)$ is equivalent to the anti-fundamental representation $\bar{\mathbf{3}}$.

• We also describe the tensor product decomposition $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$.

The fundamental rep. $\mathbf{3}$ corresponds to a contravariant tensor u^i and the anti-fundamental rep. $\bar{\mathbf{3}}$ corresponds to a covariant tensor v_j. We then have the following decomposition:

$$\underbrace{u^i v_j}_{\mathbf{3}\otimes\bar{\mathbf{3}}} = \underbrace{(u^i v_j - \frac{1}{3}\delta^i_j u^k v_k)}_{\mathbf{8}} + \underbrace{\frac{1}{3}\delta^i_j u^k v_k}_{\mathbf{1}} \,. \tag{4.132}$$

As one can see, the part $\frac{1}{3}\delta^i_j u^k v_k$ is invariant under the $SU(3)$ transformation. So does the other part $(u^i v_j - \frac{1}{3}\delta^i_j u^k v_k)$ which describes a traceless tensor T^i_j with $tr(T^i_j) = 0$. Hence T^i_j is an element of the Lie algebra $\mathfrak{su}(3)$, which is in the adjoint rep. $\mathbf{8}$ of the $SU(3)$ group.

Let us apply to the physics of meson octet! Denote the 3d vector of $\mathbf{3}$ as (u, d, s), where u, d, s correspond to the weight spaces of $(1, 0)$, $(-1, 1)$

and $(0,-1)$. Similarly, the 3d vector of $\bar{\mathbf{3}}$ is denoted as $(\bar{u},\bar{d},\bar{s})$, where $\bar{u}$, $\bar{d}$, $\bar{s}$ correspond to the weight spaces of $(-1,0)$, $(1,-1)$ and $(0,1)$. After the tensor product, the weights of $\mathbf{8}$ have the following correspondence with the mesons (tensor products of 1d vector spaces)

$$\begin{aligned} u\bar{s}(K^+): &\quad (1,1) \\ u\bar{d}(\pi^+): &\quad (2,-1) \\ d\bar{u}(\pi^-): &\quad (-2,1) \\ s\bar{u}(K^-): &\quad (-1,-1) \\ s\bar{d}(\overline{K}^0): &\quad (1,-2) \\ d\bar{s}(K^0): &\quad (-1,2)\,. \end{aligned} \tag{4.133}$$

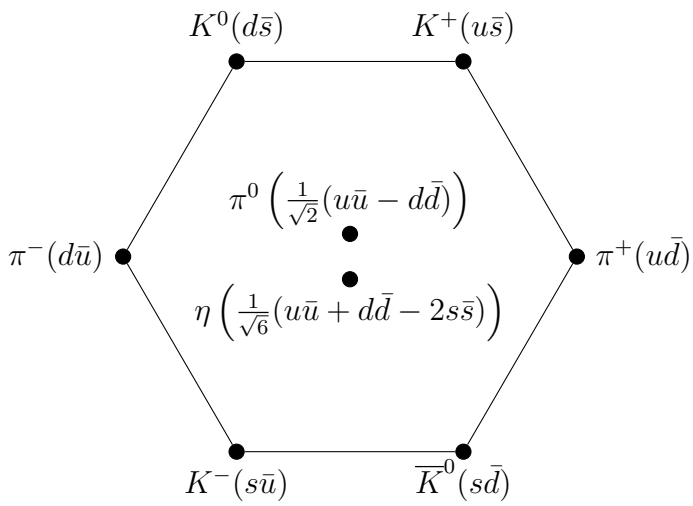
For the weights $(0,0)$, note that the traceless part is spanned by $\frac{1}{3}(2u\bar{u} - d\bar{d} - s\bar{s})$ and $\frac{1}{3}(2d\bar{d} - u\bar{u} - s\bar{s})$. We choose the following (normalized) linear combinations

$$\begin{aligned} \pi_0 &= \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}) \\ \eta &= \frac{1}{\sqrt{6}}(u\bar{u} + d\bar{d} - 2s\bar{s})\,. \end{aligned} \tag{4.134}$$

Finally, the meson singlet is

$$\eta' = \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s})\,. \tag{4.135}$$

We label it in the Meson octet diagram



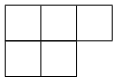
(4.136)

This story can be generalized to $SU(N)$ group as well:

$$\mathbf{N} \otimes \overline{\mathbf{N}} = adj. \oplus \mathbf{1}\,. \tag{4.137}$$

$$\underbrace{u^i v_j}_{\mathbf{N}\otimes\overline{\mathbf{N}}} = \underbrace{(u^i v_j - \frac{1}{N}\delta^i_j u^k v_k)}_{\mathbf{N^2-1}} + \underbrace{\frac{1}{N}\delta^i_j u^k v_k}_{\mathbf{1}} \,. \tag{4.138}$$

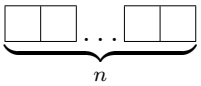
Now we introduce the *Young diagram* corresponding to an arbitrary irreducible tensor representation of $SU(N)$ (A_{N-1}). For a Young diagram such as



, (4.139)

each box denotes a contravariant (upper) tensor index, each column means the antisymmetrization of the indices, and each row means the symmetrization of the indices.

Hence



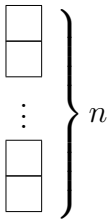
(4.140)

corresponds to a rank-n symmetric tensor representation of $SU(N)$:

$$v^{(i_1 i_2 \ldots i_n)} = \frac{1}{n!}\sum_{\sigma\in S_n}(v^{i_{\sigma 1} i_{\sigma 2}\ldots i_{\sigma n}})\,. \tag{4.141}$$

Here σ is an element of the permutation group S_n, which describes a permutation of n elements $\{1,\ldots,n\}\to\{\sigma_1,\ldots,\sigma_n\}$.

On the other hand,



(4.142)

corresponds to a rank-n antisymmetric tensor representation of $SU(N)$.

$$v^{[i_1 i_2 \ldots i_n]} = \frac{1}{n!}\sum_{\sigma\in S_n}(\epsilon(\sigma) v^{i_{\sigma 1} i_{\sigma 2}\ldots i_{\sigma n}})\,. \tag{4.143}$$

$\epsilon(\sigma)$ is equal to 1 for an even permutation σ, and (-1) for an odd permutation σ.

More generally, for an irreducible representation of $\mathfrak{su}(n)$ with highest weight $(\mu_1,\mu_2,\ldots,\mu_{n-1})$, it can be equivalently denoted by the Young diagram $[l_1;l_2;\ldots;l_{n-1}]$, where the i-th row has l_i boxes.

$$l_i = \mu_i + \mu_{i+1} + \cdots + \mu_{n-1}\,. \tag{4.144}$$

For example, the fundamental representation with highest weight $(1, 0, \ldots, 0)$ is

$$\begin{array}{|c|}\hline \\ \hline \end{array} \tag{4.145}$$

the adjoint representation of $\mathfrak{su}(3)$, with highest weight $(1, 1)$ is

$$\begin{array}{|c|c|}\hline & \\ \hline & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} \tag{4.146}$$

In general, a rank-n symmetric representation of $SU(N)$ has Young diagram

$$\underbrace{\begin{array}{|c|c|}\hline & \\ \hline \end{array}\ldots\begin{array}{|c|c|}\hline & \\ \hline \end{array}}_{n} \tag{4.147}$$

which corresponds to the highest weight $(n, 0, \ldots, 0)$.

A rank-n anti-symmetric representation of $SU(N)$ has Young diagram

$$\left.\begin{array}{|c|}\hline \\ \hline \\ \hline \end{array}\right\} n \tag{4.148}$$

$$\vdots$$

which corresponds to the highest weight $(0, \ldots, 0, 1, 0 \ldots, 0)$, where the 1 is at the n-th entry.

A good way to keep track the dimension of representation is to use the Young tableaux of a Young diagram. For a Young diagram of a representation of $SU(N)$, one fills integers $1 \sim N$ into the boxes. The rule is

1. For each row, the integers are non-decreasing from left to the right.

2. For each column, the integers are increasing from top to the bottom.

The number of Young tableaux associated to a Young diagram is then equal to the dimension of the irrep.

• For example: (1) rank-3 symmetric rep. $\begin{array}{|c|c|c|}\hline & & \\ \hline \end{array}$ of $\mathfrak{su}(3)$:

$$\begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 1 & 1 & 2 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 1 & 1 & 3 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 1 & 2 & 2 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 1 & 3 & 3 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 2 & 2 & 2 \\ \hline \end{array}$$
$$\begin{array}{|c|c|c|}\hline 2 & 2 & 3 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 2 & 3 & 3 \\ \hline \end{array}\ \begin{array}{|c|c|c|}\hline 3 & 3 & 3 \\ \hline \end{array} \tag{4.149}$$

The rank-3 symmetric rep. of $\mathfrak{su}(3)$ is indeed 10-dimensional.

Each Young tableaux $\boxed{i}\boxed{j}\boxed{k}$ corresponds to a symmetric tensor element $v^{(ijk)}$, for example

$$v^{(111)} = v^{111}\ ,\ v^{(112)} = \frac{1}{3}(v^{112}+v^{121}+v^{211})\ ,\ v^{(123)} = \frac{1}{6}(v^{123}+v^{132}+v^{213}+v^{231}+v^{312}+v^{321})\ . \tag{4.150}$$

(2) rank-2 anti-symmetric rep. of $\mathfrak{su}(4)$

$$\begin{array}{|c|}\hline 1\\\hline 2\\\hline\end{array}\ \begin{array}{|c|}\hline 1\\\hline 3\\\hline\end{array}\ \begin{array}{|c|}\hline 1\\\hline 4\\\hline\end{array}\ \begin{array}{|c|}\hline 2\\\hline 3\\\hline\end{array}\ \begin{array}{|c|}\hline 2\\\hline 4\\\hline\end{array}\ \begin{array}{|c|}\hline 3\\\hline 4\\\hline\end{array}\ . \tag{4.151}$$

Each Young tableaux $\begin{array}{|c|}\hline i\\\hline j\\\hline\end{array}$ corresponds to an antisymmetric tensor element $v^{[ij]}$.

(3) adjoint rep. of $\mathfrak{su}(3)$:

$$\begin{array}{|c|c|}\hline 1 & 1\\\hline 2 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 1 & 2\\\hline 2 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 1 & 3\\\hline 2 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 1 & 1\\\hline 3 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 1 & 2\\\hline 3 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 1 & 3\\\hline 3 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 2 & 2\\\hline 3 & \\\hline\end{array}\ \begin{array}{|c|c|}\hline 2 & 3\\\hline 3 & \\\hline\end{array} \tag{4.152}$$

The adjoint rep. of $\mathfrak{su}(3)$ is indeed 8-dimensional.

For the mixed tensor elements of an $\mathbf{8}$, one can write down two different mixed tensors:

$$\mathbf{8}_1 : \begin{array}{|c|c|}\hline i & j\\\hline k & \\\hline\end{array} : \tfrac{1}{4}(v^{ijk} + v^{jik} - v^{kji} - v^{kij}) \tag{4.153}$$

$$\mathbf{8}_2 : \begin{array}{|c|c|}\hline i & k\\\hline j & \\\hline\end{array} : \tfrac{1}{4}(v^{ijk} + v^{kji} - v^{jik} - v^{jki}) \tag{4.154}$$

In fact, these are precisely the ones appearing in the tensor product decomposition $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8}_1 \oplus \mathbf{8}_2 \oplus \mathbf{10}$.

4.4.3 CG coefficients

Suppose that we have the tensor product decomposition

$$\pi_1 \otimes \pi_2 = \bigoplus_{\mu} \pi_\mu^{\oplus m_\mu} \tag{4.155}$$

μ is the highest weight of an irrep π_μ, and m_μ is the multiplicity of π_μ in the decomposition.

The underlying vector space has a similar form

$$V_1 \otimes V_2 = \bigoplus_{\mu,\beta} V_{\mu,\beta}\,, \tag{4.156}$$

where $\beta = 1, \ldots, m_\mu$ is a label to distinguish different copies of π_μ.

Now take a orthonormal basis of V_1, V_2 to be $\left| \begin{matrix} \mu_1 \\ \lambda_1\kappa_1 \end{matrix} \right\rangle$, $\left| \begin{matrix} \mu_2 \\ \lambda_2\kappa_2 \end{matrix} \right\rangle$. μ_i is the highest weight of the representation π_i, λ_i is a weight in the weight system of π_i, and κ_i is a label to distinguish different weight vectors with the same weight λ_i.

The orthonormal basis of $V_{\mu,\beta}$ is denoted as $\left| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle$, where λ and κ are defined in a similar way as above.

Beside this basis, we also have the trivial decoupled basis of $V_1 \otimes V_2$, in form of

$$\left| \begin{matrix} \mu_1 \\ \lambda_1\kappa_1 \end{matrix} \right\rangle \otimes \left| \begin{matrix} \mu_2 \\ \lambda_2\kappa_2 \end{matrix} \right\rangle = \left| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle. \tag{4.157}$$

The two sets of basis satisfy non-trivial relations:

$$\left| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle = \sum_{\beta\mu\lambda\kappa} \left\langle \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \middle| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle \left| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle \tag{4.158}$$

$$\left| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle = \sum_{\lambda_1\lambda_2\kappa_1\kappa_2} \left\langle \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \middle| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle \left| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle \tag{4.159}$$

The coefficients $\left\langle \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \middle| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle$, $\left\langle \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \middle| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle$ are called *Clebsch-Gordan coefficients.*

They obviously satisfy the orthonormal conditions

$$\sum_{\beta\mu\lambda\kappa} \left\langle \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1'\kappa_1' & \lambda_2'\kappa_2' \end{matrix} \middle| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle \left\langle \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \middle| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle = \delta_{\lambda_1'\lambda_1}\delta_{\lambda_2'\lambda_2}\delta_{\kappa_1'\kappa_1}\delta_{\kappa_2'\kappa_2}\,, \tag{4.160}$$

$$\sum_{\lambda_1\kappa_1\lambda_2\kappa_2} \left\langle \beta' \begin{matrix} \mu' \\ \lambda'\kappa' \end{matrix} \middle| \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \right\rangle \left\langle \begin{matrix} \mu_1 & \mu_2 \\ \lambda_1\kappa_1 & \lambda_2\kappa_2 \end{matrix} \middle| \beta \begin{matrix} \mu \\ \lambda\kappa \end{matrix} \right\rangle = \delta_{\beta'\beta}\delta_{\mu'\mu}\delta_{\lambda'\lambda}\delta_{\kappa'\kappa} \,. \tag{4.161}$$

And there's the selection rule: the CG coefficients are non-zero only if $\lambda = \lambda_1 + \lambda_2$.

As an example, we consider the quantum mechanics problem of decomposing the tensor product of two spin $s = \frac{1}{2}$ states into the singlet and triplet. In the language of Lie algebra, this is the tensor product decomposition of representations of $\mathfrak{su}(2)$:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3} \,. \tag{4.162}$$

We use the following 2-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$:

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \ , \ X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \ , \ Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \,. \tag{4.163}$$

The 2-dimensional vector space it acts on is denoted as $(|\uparrow\rangle, |\downarrow\rangle)^T$. Hence

$$\begin{aligned} &H|\uparrow\rangle = |\uparrow\rangle \ , \ H|\downarrow\rangle = -|\downarrow\rangle \\ &X|\uparrow\rangle = 0 \ , \ X|\downarrow\rangle = |\uparrow\rangle \\ &Y|\uparrow\rangle = |\downarrow\rangle \ , \ Y|\downarrow\rangle = 0 \,. \end{aligned} \tag{4.164}$$

Let us consider two spin $s = \frac{1}{2}$ systems, where π_1 and π_2 acts on the 2d vector space in the same way as (4.163).

In the tensor product representation $\pi_1 \otimes \pi_2$, the 4-dimensional vector space is $(|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle)^T$. We can compute the action of $\mathfrak{sl}(2, \mathbb{C})$ generators as:

$$\begin{aligned} (\pi_1 \otimes \pi_2)(H) &= \pi_1(H) \otimes I_2 + I_1 \otimes \pi_2(H) \\ &= \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} \,, \end{aligned} \tag{4.165}$$

$$\begin{aligned} (\pi_1 \otimes \pi_2)(X) &= \pi_1(X) \otimes I_2 + I_1 \otimes \pi_2(X) \\ &= \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \,, \end{aligned} \tag{4.166}$$

$$
\begin{aligned}
(\pi_1 \otimes \pi_2)(Y) &= \pi_1(Y) \otimes I_2 + I_1 \otimes \pi_2(Y) \\
&= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} .
\end{aligned} \tag{4.167}
$$

Now we decompose the 4d vector space into the 3d irrep. (known as the triplet state) and the 1d irrep. (the singlet state). For the triplet, its vector space $V_{\mathbf{3}}$ is spanned by

$$
|\uparrow\uparrow\rangle \ , \ \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \ , \ |\downarrow\downarrow\rangle \, . \tag{4.168}
$$

Note that we have made them orthonormal. Using this basis, the action of $\mathfrak{sl}(2,\mathbb{C})$ generators is

$$
\pi_{\mathbf{3}}(H) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} , \pi_{\mathbf{3}}(X) \ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} , \pi_{\mathbf{3}}(Y) = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix} . \tag{4.169}
$$

In particular, $|\uparrow\uparrow\rangle$, $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$, $|\downarrow\downarrow\rangle$ are eigenstates of $\pi_{\mathbf{3}}(H)$, with eigenvalues 2, 0 and -2 respectively. They are exactly equal to the weights of the representation $\mathbf{3}$.

Finally, the vector space of the singlet is spanned by $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$. Since the representation is trivial, we have

$$
\pi_{\mathbf{1}}(H) = \pi_{\mathbf{1}}(X) = \pi_{\mathbf{1}}(Y) = 0 \, , \tag{4.170}
$$

and the state has weight 0 in the representation $\mathbf{1}$.

Now let us rewrite this basis in the notations of (4.158,4.159) ($\beta, \kappa, \kappa_1, \kappa_2$ are omitted):

$$
|\uparrow\rangle = \left| \begin{matrix} 1 \\ 1 \end{matrix} \right\rangle \ , \ |\downarrow\rangle = \left| \begin{matrix} 1 \\ -1 \end{matrix} \right\rangle \tag{4.171}
$$

$$
|\uparrow\uparrow\rangle = \left| \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} \right\rangle , \ |\uparrow\downarrow\rangle = \left| \begin{matrix} 1 & 1 \\ 1 & -1 \end{matrix} \right\rangle , \ |\downarrow\uparrow\rangle = \left| \begin{matrix} 1 & 1 \\ -1 & 1 \end{matrix} \right\rangle , \ |\downarrow\downarrow\rangle = \left| \begin{matrix} 1 & 1 \\ -1 & -1 \end{matrix} \right\rangle . \tag{4.172}
$$

$$
\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = \left| \begin{matrix} 0 \\ 0 \end{matrix} \right\rangle . \tag{4.173}
$$

$$|\uparrow\uparrow\rangle = \left| \begin{matrix} 2 \\ 2 \end{matrix} \right\rangle \ , \ \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) = \left| \begin{matrix} 2 \\ 0 \end{matrix} \right\rangle \ , \ |\downarrow\downarrow\rangle = \left| \begin{matrix} 2 \\ -2 \end{matrix} \right\rangle . \tag{4.174}$$

Hence we have the following CG coefficients:

	$\left\vert \begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} \right\rangle$	$\left\vert \begin{matrix} 1 & 1 \\ 1 & -1 \end{matrix} \right\rangle$	$\left\vert \begin{matrix} 1 & 1 \\ -1 & 1 \end{matrix} \right\rangle$	$\left\vert \begin{matrix} 1 & 1 \\ -1 & -1 \end{matrix} \right\rangle$
$\left\langle \begin{matrix} 0 \\ 0 \end{matrix} \right\vert$	0	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	0
$\left\langle \begin{matrix} 2 \\ 2 \end{matrix} \right\vert$	1	0	0	0
$\left\langle \begin{matrix} 2 \\ 0 \end{matrix} \right\vert$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0
$\left\langle \begin{matrix} 2 \\ -2 \end{matrix} \right\vert$	0	0	0	1

(4.175)

Now we describe the general derivations of CG coefficients, in the example of $\mathfrak{so}(3) \equiv \mathfrak{su}(2)$ ($\mathfrak{sl}(2,\mathbb{C})$). We use the rescaled generators

$$J_0 = \frac{H}{2} \ , \ J_+ = \frac{X}{\sqrt{2}} \ , \ J_- = -\frac{Y}{\sqrt{2}} \, , \tag{4.176}$$

so that they satisfy the commutation relations

$$[J_+, J_-] = -J_0 \ , \ [J_0, J_+] = J_+ \ , \ [J_0, J_-] = J_- \, . \tag{4.177}$$

Note that we have

$$J_0^\dagger = J_0 \ , \ J_\pm^\dagger = -J_\mp \, . \tag{4.178}$$

The quadratic Casimir operator is

$$C_2 = J_0^2 - J_+ J_- - J_- J_+ \, . \tag{4.179}$$

We use the standard notation of angular momentum j for an irrep. of $\mathfrak{so}(3)$, which is related to the highest weight μ as $j = \frac{1}{2}\mu$. j can be either integer or half-integer. The orthonormal weight vectors are denoted as $|jm\rangle$, ($m = -j, -(j-1), \ldots, (j-1), j$). The Lie algebra generators act on $|jm\rangle$ as

$$J_0|jm\rangle = m|jm\rangle \ , \ J_\pm|jm\rangle = C_\pm(j,m)|j(m \pm 1)\rangle \, . \tag{4.180}$$

$C_\pm(j,m)$ is a constant depending on j, m. Of course for the highest weight state and lowest weight state, we have

$$J_+|jj\rangle = 0 \ , \ J_-|j \ -j\rangle = 0 \, . \tag{4.181}$$

Because $[J_+, J_-] = -J_0$, we have

$$\langle jm|J_+J_-|jm\rangle = \langle jm|J_-J_+|jm\rangle - \langle jm|J_0|jm\rangle \tag{4.182}$$

and

$$\langle jm|J_+|j(m-1)\rangle\langle j(m-1)J_-|jm\rangle = \langle jm|J_-|j(m+1)\rangle\langle j(m+1)J_+|jm\rangle + \langle jm|J_0|jm\rangle\,. \tag{4.183}$$

Since $J_\pm^\dagger = -J_\mp$, we have

$$|\langle j(m-1)|J_-|jm\rangle|^2 = |\langle jm|J_-|j(m+1)\rangle|^2 + m\,. \tag{4.184}$$

Consider the Casimir operator C_2, we have

$$\begin{aligned}\langle jm|C_2|jm\rangle &= \langle jm|J_0^2|jm\rangle - \langle jm|J_+J_-|jm\rangle - \langle jm|J_-J_+|jm\rangle \\ &= m^2 + |\langle j(m-1)|J_-|jm\rangle|^2 + |\langle jm|J_-|j(m+1)\rangle|^2\,.\end{aligned} \tag{4.185}$$

On the other hand, since we know that

$$\langle jm|C_2|jm\rangle = j(j+1)\,, \tag{4.186}$$

we can solve

$$|\langle j(m-1)|J_-|jm\rangle|^2 = \frac{1}{2}\left(j(j+1) - m(m-1)\right)\,. \tag{4.187}$$

Now to fix the phase factor, we choose

$$\begin{aligned}C_-(j,m) &= \langle j(m-1)|J_-|jm\rangle = \sqrt{\frac{(j+m)(j-m+1)}{2}} \\ C_+(j,m) &= \langle j(m+1)|J_+|jm\rangle = -\sqrt{\frac{(j-m)(j+m+1)}{2}}\,.\end{aligned} \tag{4.188}$$

(Note that the results on quantum mechanics book may differ by a $1/\sqrt{2}$ factor, due to this additional factor in J_+ and J_-.)

Now let us consider the tensor product decomposition

$$j_1 \otimes j_2 = (j_1 + j_2) \oplus (j_1 + j_2 - 1) \oplus \cdots \oplus |j_1 - j_2|\,. \tag{4.189}$$

The decoupled states in $V_1 \otimes V_2$ are $|j_1m_1j_2m_2\rangle = |j_1m_1\rangle|j_2m_2\rangle$, while the

coupled states are denoted as $|jm\rangle$. There are relations

$$
\begin{aligned}
|j_1m_1j_2m_2\rangle &= \sum_{j,m}\langle jm|j_1m_1j_2m_2\rangle|jm\rangle\,,\\
|jm\rangle &= \sum_{m_1,m_2}\langle j_1m_1j_2m_2|jm\rangle|j_1m_1j_2m_2\rangle\,,\\
&\sum_{m_1,m_2}\langle j'm'|j_1m_1j_2m_2\rangle\langle j_1m_1j_2m_2|jm\rangle = \delta_{jj'}\delta_{mm'}\,,\\
&\sum_{j_1'm_1'j_2'm_2'}\langle j_1'm_1'j_2'm_2'|jm\rangle\langle jm|j_1m_1j_2m_2\rangle = \delta_{j_1j_1'}\delta_{j_2j_2'}\delta_{m_1m_1'}\delta_{m_2m_2'}\,.
\end{aligned}
\tag{4.190}
$$

Now let us consider the expression

$$
\langle jm|J_-|j_1m_1j_2m_2\rangle = \langle jm|J_-|j(m+1)\rangle\langle j(m+1)|j_1m_1j_2m_2\rangle\,. \tag{4.191}
$$

On the other hand, we have

$$
\begin{aligned}
\langle jm|J_-|j_1m_1j_2m_2\rangle &= \sum_{m_1',m_2'}\langle jm|j_1m_1'j_2m_2'\rangle\langle j_1m_1'j_2m_2'|J_-|j_1m_1j_2m_2\rangle\\
&= \sum_{m_1',m_2'}\langle jm|j_1m_1'j_2m_2'\rangle[\langle j_1m_1'|J_-|j_1m_1\rangle\langle j_2m_2'|j_2m_2\rangle\\
&+ \langle j_2m_2'|J_-|j_2m_2\rangle\langle j_1m_1'|j_1m_1\rangle]\\
&= \langle jm|j_1(m_1-1)j_2m_2\rangle\langle j_1(m_1-1)|J_-|j_1m_1\rangle\\
&+ \langle jm|j_1m_1j_2(m_2-1)\rangle\langle j_2(m_2-1)|J_-|j_2m_2\rangle\,.
\end{aligned}
\tag{4.192}
$$

Comparing the r.h.s. of (4.191) and (4.192), and plug in the matrix elements for J_-, we get the equation

$$
\begin{aligned}
\sqrt{(j-m)(j+m+1)}\langle j_1m_1j_2m_2|j(m+1)\rangle &= \sqrt{(j_1+m_1)(j_1-m_1+1)}\langle j_1(m_1-1)j_2m_2|jm\rangle\\
&+ \sqrt{(j_2+m_2)(j_2-m_2+1)}\langle j_1m_1j_2(m_2-1)|jm\rangle\,.
\end{aligned}
\tag{4.193}
$$

and another equation from J_+:

$$
\begin{aligned}
\sqrt{(j+m)(j-m+1)}\langle j_1m_1j_2m_2|j(m-1)\rangle &= \sqrt{(j_1-m_1)(j_1+m_1+1)}\langle j_1(m_1+1)j_2m_2|jm\rangle\\
&+ \sqrt{(j_2-m_2)(j_2+m_2+1)}\langle j_1m_1j_2(m_2+1)|jm\rangle\,.
\end{aligned}
\tag{4.194}
$$

From this complicated set of equations, one can solve all the CG coefficients $\langle j_1m_1j_2m_2|jm\rangle$. See the expressions on Yuxin Liu's book p. 230.

4.5 Branching rules

We consider the embedding of a Lie algebra $\mathfrak{s} \subset \mathfrak{g}$, where $\mathfrak{g}$ is semi-simple. For a representation (π, V) of $\mathfrak{g}$, we want to decompose it as a direct sum of (irreducible) representations (π_i, V_i) of the subalgebra $\mathfrak{s}$. Note that $\mathfrak{s}$ can be a direct sum of simple Lie algebras and $\mathfrak{u}(1)$s.

For simplicity we are focusing on the R-type subalgebra, and we first take the example of

$$\mathfrak{s} = \mathfrak{s}' \oplus \mathfrak{u}(1)^{\oplus l}\,, \tag{4.195}$$

where $\mathfrak{s}'$ is a non-abelian subalgebra of $\mathfrak{g}$, whose simple roots are chosen from the simple roots of $\mathfrak{g}$. Namely, the Dynkin diagram of $\mathfrak{s}'$ is generated by deleting l nodes from the Dynkin diagram of $\mathfrak{g}$.

Consider a weight λ in the weight system of (π, V), with weight vector v, defined as

$$\pi(H)v = \langle \lambda, H\rangle v\ ,\ \forall H \in \mathfrak{h} \subset \mathfrak{g}\,. \tag{4.196}$$

λ is also a weight of $\bigoplus_i \pi_i$, with weight vector v, because the Cartan subalgebra of $\mathfrak{s}$ is a subalgebra of $\mathfrak{h}$.

Denote the simple roots of $\mathfrak{g}$ by $(\alpha_1, \ldots, \alpha_r)$, and the simple roots of $\mathfrak{s}'$ form a subset $(\alpha_{i_1}, \ldots, \alpha_{i_{r'}})$. For a weight λ of π, written as

$$\lambda = (\langle \lambda, H_{\alpha_1}\rangle, \ldots, \langle \lambda, H_{\alpha_r}\rangle)\,, \tag{4.197}$$

it is a weight written as

$$\lambda = (\langle \lambda, H_{\alpha_{i_1}}\rangle, \ldots, \langle \lambda, H_{\alpha_{i_{r'}}}\rangle) \tag{4.198}$$

of $\mathfrak{s}'$.

- We can compute the branching rule as follows:

1. Write down all the weights of the representation (π, V)
2. Ignore the l nodes that are deleted from the Dynkin diagram of $\mathfrak{g}$, and one can naturally see how the weights are combined into different irreps (π_i, V_i) of $\mathfrak{s}'$. For example, one can pick out some dominant integral weights λ ($\langle \lambda, H_\alpha\rangle \geq 0$ for all simple roots α of $\mathfrak{s}'$), which give rise to irreducible representations π_λ of $\mathfrak{s}'$ with highest weight λ
3. For the charges of each irrep (π_i, V_i) under $\mathfrak{u}(1)^{\oplus l}$, let us denote it by $(q_{i,1}, q_{i,2}, \ldots, q_{i,l})$. One constructs l linearly independent elements

$Q_j (j = 1, \ldots, l) \in \mathfrak{h} \subset \mathfrak{g}$ as the $\mathfrak{u}(1)$ generators. The satisfy the orthogonality conditions:

(1) $\langle Q_i, Q_j \rangle = 0$ if $i \neq j$[2].

(2) $\langle Q_i, \alpha_{j_k} \rangle = 0$ for all the simple roots α_{j_k} in $\mathfrak{s}'$.

One can easily prove that for each irrep (π_i, V_i), each charge

$$q_{i,j} = \langle Q_j, w \rangle \tag{4.199}$$

is a constant number for every weight w in (π_i, V_i) (note that here w is still an element of $\mathfrak{h} \subset \mathfrak{g}$). Since the weight system of (π_i, V_i) is generated by subtracting simple roots from the highest weight.

Let us study some examples:
(1) $\mathfrak{su}(3) \to \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.
The Dynkin diagram is embedded as

$$\bullet\!\!-\!\!\!-\!\!\times \tag{4.200}$$

For the fundamental rep. $\mathbf{3}$ of $\mathfrak{su}(3)$, its weights are

$$(1, 0)\ ,\ (-1, 1)\ ,\ (0, -1)\,. \tag{4.201}$$

Hence one can see that the first two weights form the fundamental rep. $\mathbf{2}$ of $\mathfrak{su}(2)$, while the third weight form the trivial rep. of $\mathfrak{su}(2)$. We then have

$$\mathbf{3} \to \mathbf{2} \oplus \mathbf{1}\,. \tag{4.202}$$

For the charge under $\mathfrak{u}(1)$, the generator for $\mathfrak{u}(1)$ can be chosen as

$$Q = H_{\alpha_1} + 2H_{\alpha_2}\,. \tag{4.203}$$

It is easy to check that this generator satisfies $\langle Q, \alpha_1 \rangle = 0$.

All the weights in the rep. $\mathbf{2}$ has charge 1 under Q, while the weight in the rep. $\mathbf{1}$ has charge -2 under Q. The final branching rule is then

$$\mathbf{3} \to \mathbf{2}_1 \oplus \mathbf{1}_{-2}\,. \tag{4.204}$$

More generally, for $\mathfrak{su}(n+1) \to \mathfrak{su}(n) \oplus \mathfrak{u}(1)$, we have

$$\mathbf{n+1} \to \mathbf{n}_1 \oplus \mathbf{1}_{-n}\,. \tag{4.205}$$

[2]This is usually a physical requirement.

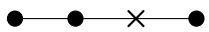
We can also apply the branching rule for more than once, for example for $\mathfrak{su}(n+1) \to \mathfrak{su}(n) \oplus \mathfrak{u}(1) \to [\mathfrak{su}(n-1) \oplus \mathfrak{u}(1)] \oplus \mathfrak{u}(1)$, we have the branching rule for $\mathbf{n+1}$:

$$\mathbf{n+1} \to \mathbf{n}_1 \oplus \mathbf{1}_{-n} \to \mathbf{n-1}_{(1,1)} \oplus \mathbf{1}_{(-(n-1),1)} \oplus \mathbf{1}_{(0,-n)} \,. \tag{4.206}$$

(2) $\mathfrak{su}(5) \to \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

This is the breaking of $\mathfrak{su}(5)$ GUT (grand unified theory) gauge algebra into the standard model gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

The Dynkin diagram of $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ is embedded as



(4.207)

For the fundamental rep. $\mathbf{5}$ of $\mathfrak{su}(5)$, its weights are

$$(1,0,0,0)\ ,\ (-1,1,0,0)\ ,\ (0,-1,1,0)\ ,\ (0,0,-1,1)\ ,\ (0,0,0,-1)\,. \tag{4.208}$$

They can be organized into the representations $(\mathbf{3},\mathbf{1})$ and $(\mathbf{1},\mathbf{2})$. For the $\mathfrak{u}(1)$ generator, it can be chosen as

$$Q = 2H_{\alpha_1} + 4H_{\alpha_2} + 6H_{\alpha_3} + 3H_{\alpha_4}\,. \tag{4.209}$$

The branching rule is then

$$\mathbf{5} \to (\mathbf{3},\mathbf{1})_2 \oplus (\mathbf{1},\mathbf{2})_{-3}\,. \tag{4.210}$$

For the rank-2 antisymmetric rep. $\begin{array}{|c|}\hline \\ \hline \\ \hline \end{array} = \mathbf{10}$ of $\mathfrak{su}(5)$, its weights are

$$\begin{aligned}&(0,1,0,0)\ ,\ (1,-1,1,0)\ ,\ (-1,0,1,0)\ ,\ (1,0,-1,1)\ ,\ (-1,1,-1,1)\\ &(1,0,0,-1)\ ,\ (0,-1,0,1)\ ,\ (-1,1,0,-1)\ ,\ (0,-1,1,-1)\ ,\ (0,0,-1,0)\,.\end{aligned} \tag{4.211}$$

They can be organized into the representations $(\bar{\mathbf{3}},\mathbf{1})$, $(\mathbf{3},\mathbf{2})$, $(\mathbf{1},\mathbf{1})$. After computing the charges of each irrep. under Q, we get the branching rule

$$\mathbf{10} \to (\bar{\mathbf{3}},\mathbf{1})_4 \oplus (\mathbf{3},\mathbf{2})_{-1} \oplus (\mathbf{1},\mathbf{1})_{-6}\,. \tag{4.212}$$

In fact, the irreps in (4.210) and (4.212) exactly form the quark and lepton contents of the particle physics standard model, known as the QUDLE. The $U(1)$ charges shown here are six times the values of hypercharge Y in physics literature. The hypercharge is defined as

$$Y = Q - T_3\,, \tag{4.213}$$

T_3 is the third component of weak isospin.

Besides the standard model fermions, there are also the following bosons in representations of $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, including the gauge bosons before electroweak symmetry breaking:

$$\begin{aligned} \text{Gluon} : (\mathbf{8}, \mathbf{1})_0 \\ W^{\pm}, W^0 : (\mathbf{1}, \mathbf{3})_0 \\ B^0 : (\mathbf{1}, \mathbf{1})_0 \end{aligned} \tag{4.214}$$

and the Higgs boson

$$H : \ (\mathbf{1}, \mathbf{2})_3 \,. \tag{4.215}$$

In the $SU(5)$ GUT, the natural way to get the standard model gauge bosons is from the $SU(5)$ gauge boson in the adjoint **24** representation, which has the branching rule

$$\mathbf{24} \to (\mathbf{8}, \mathbf{1})_0 \oplus (\mathbf{1}, \mathbf{3})_0 \oplus (\mathbf{1}, \mathbf{1})_0 \oplus (\mathbf{3}, \mathbf{2})_5 \oplus (\bar{\mathbf{3}}, \mathbf{2})_{-5} \,. \tag{4.216}$$

Thus it predicts the presence of "leptoquarks" in the representation $(\mathbf{3}, \mathbf{2})_5$ that trigger Baryon decay, and such physical process is largely constrained by the experiments.

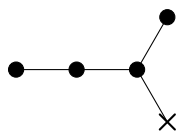
On the other hand, the way to get Higgs boson is via

$$\bar{\mathbf{5}} \to (\bar{\mathbf{3}}, \mathbf{1})_{-2} \oplus (\mathbf{1}, \mathbf{2})_3 \,. \tag{4.217}$$

However, we do not have a "Higgs boson" $(\mathbf{3}, \mathbf{1})_{-2}$ for the $SU(3)$ strong interaction in standard model . This contradiction is called the triplet-doublet splitting problem.

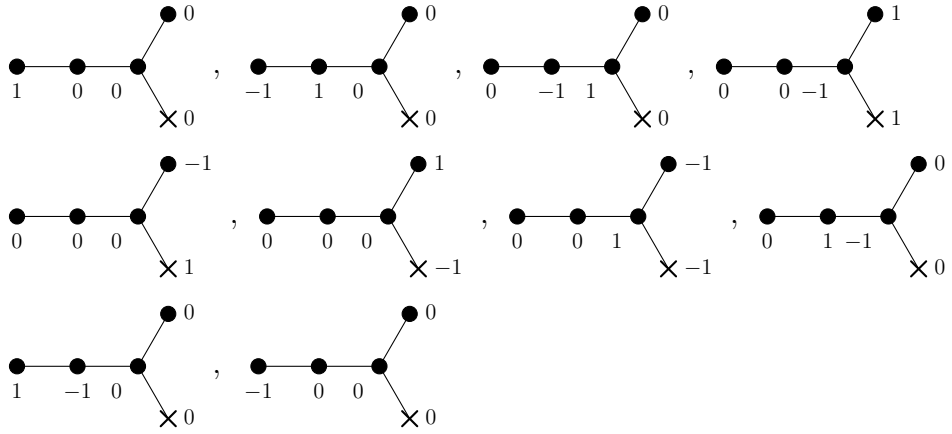
(3) $\mathfrak{so}(10) \to \mathfrak{su}(5)$.

The Dynkin diagram is embedded as



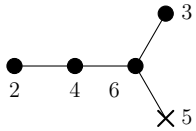
(4.218)

For the representation **10**, the weights are



(4.219)

These weights are organized into the representations $\mathbf{5} \oplus \bar{\mathbf{5}}$. The $\mathfrak{u}(1)$ charge can be taken as

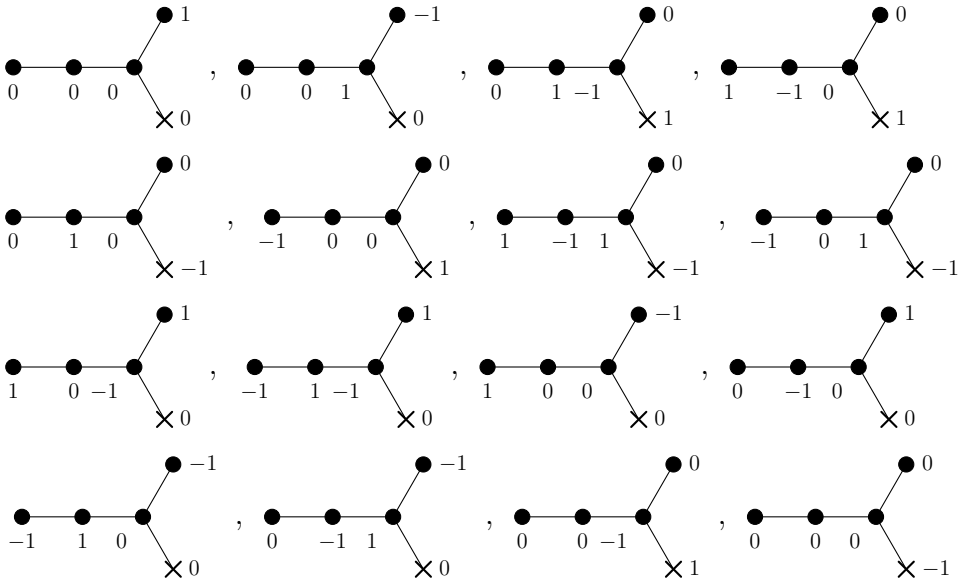


(4.220)

The branching rule is thus

$$\mathbf{10} \to \mathbf{5}_2 \oplus \bar{\mathbf{5}}_{-2}\,. \tag{4.221}$$

For the spinor representation **16** with the highest weight $(0, 0, 0, 1, 0)$, the weights are



(4.222)

The branching rule is

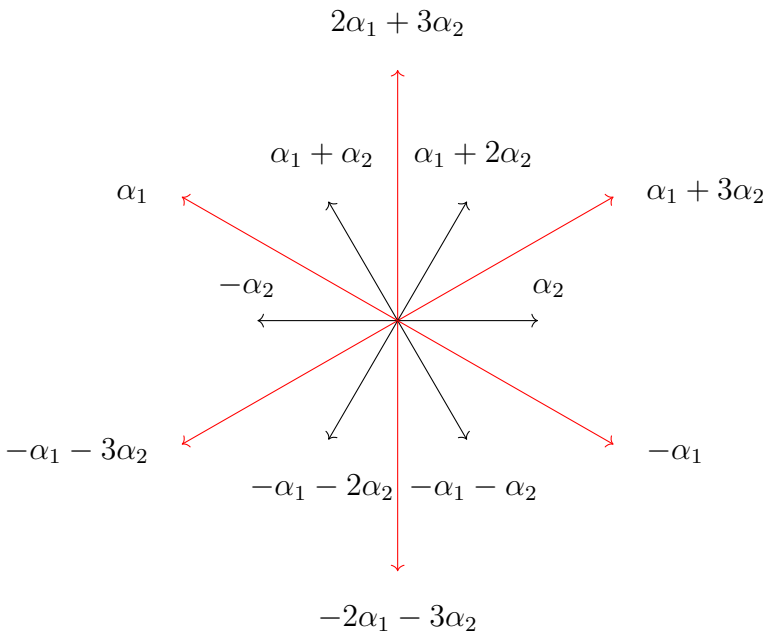
$$\mathbf{16} \to \bar{\mathbf{5}}_3 \oplus \mathbf{10}_{-1} \oplus \mathbf{1}_{-5} \tag{4.223}$$

In the $SO(10)$ GUT theory, the representation $\mathbf{16}$ contain the full QUDLE and a right-handed neutrino in the representation $\mathbf{1}_{-5}$ after breaking to $SU(5) \times U(1)$.

Now we consider the cases such that the Dynkin diagram of $\mathfrak{s}'$ cannot be embedded into the Dynkin diagram of $\mathfrak{g}$, but the root system $\mathfrak{s}'$ is still a subsystem of $\mathfrak{g}$.

(4) $\mathfrak{g}_2 \to \mathfrak{su}(3)$

We embed the roots of $A_2 = \mathfrak{su}(3)$ into the root system of G_2 as following:



(4.224)

The simple roots of G_2 are α_1 (long root) and α_2 (short root), while the simple roots of $\mathfrak{su}(3)$ can be chosen as α_1 and $\alpha_1 + 3\alpha_2$. For the coroots, we have the following relations

$$\begin{aligned} H_{\alpha_1+3\alpha_2} &= \frac{2(\alpha_1 + 3\alpha_2)}{\langle \alpha_1 + 3\alpha_2, \alpha_1 + 3\alpha_2 \rangle} \\ &= \frac{2(\alpha_1 + 3\alpha_2)}{\langle \alpha_1, \alpha_1 \rangle} \\ &= H_{\alpha_1} + H_{\alpha_2} \,. \end{aligned} \tag{4.225}$$

Hence for each weight $\lambda = (\lambda_1, \lambda_2) = (\langle \lambda, H_{\alpha_1} \rangle, \langle \lambda, H_{\alpha_2} \rangle)$ of a representation π of G_2, it corresponds to a weight of a representation of $\mathfrak{su}(3)$:

$$\begin{aligned} \lambda &= (\langle \lambda, H_{\alpha_1} \rangle, \langle \lambda, H_{\alpha_1 + 3\alpha_2} \rangle) \\ &= (\langle \lambda, H_{\alpha_1} \rangle, \langle \lambda, H_{\alpha_1} + H_{\alpha_2} \rangle) \\ &= (\lambda_1, \lambda_1 + \lambda_2) \,. \end{aligned} \tag{4.226}$$

For example, for the fundamental rep. $\mathbf{7}$ of G_2, with weights

$$(0,1)\ ,\ (1,-1)\ ,\ (-1,2)\ ,\ (0,0)\ ,\ (1,-2)\ ,\ (-1,1)\ ,\ (0,-1)\,, \tag{4.227}$$

they are mapped to the weights of $\mathfrak{su}(3)$:

$$(0,1)\ ,\ (1,0)\ ,\ (-1,1)\ ,\ (0,0)\ ,\ (1,-1)\ ,\ (-1,0)\ ,\ (0,-1)\,. \tag{4.228}$$

Recombine them into irreps. of $\mathfrak{su}(3)$, the branching rule is then

$$\mathbf{7} \to \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}\,. \tag{4.229}$$

5 Physical applications and advanced topics

5.1 Lorentz group

5.1.1 Global properties of the Lorentz group

Lorentz group is an important symmetry group of relativistic physics. Recall that the full 4d Lorentz group $O(1,3)$ is defined as the group of 4×4 real matrices M that preserve

$$M^T g M = g\,, \tag{5.1}$$

where g is the Minkowski metric with signature $(+,-,-,-)$:

$$g = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{5.2}$$

If we take a vector (t,x,y,z) and acts M on it:

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = M \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}, \tag{5.3}$$

the 4d distance is invariant

$$(t')^2 - (x')^2 - (y')^2 - (z')^2 = t^2 - x^2 - y^2 - z^2\,. \tag{5.4}$$

Now let us discuss some global properties of $O(1,3)$.

• $O(1,3)$ is non-compact.

To see this, take a particular 1-parameter subgroup of $O(1,3)$, in the form of

$$L(\beta) = \begin{pmatrix} \cosh\beta & \sinh\beta & 0 & 0 \\ \sinh\beta & \cosh\beta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\beta \in \mathbb{R}) \tag{5.5}$$

It is easy to see that the elements $L(\beta)$ are unbounded. In fact, $L(\beta)$ corresponds to a Lorentz boost in the x-direction.

Nonetheless, $O(1,3)$ has a compact $O(3)$ subgroup, consists of elements

$$\begin{pmatrix} 1 & 0 \\ 0 & M_{3\times 3} \end{pmatrix} \quad (M_{3\times 3} \in O(3))\,. \tag{5.6}$$

• $O(1,3)$ is disconnected, with four connected components.

Let us evaluate the l.h.s. of (5.1) in terms of matrix elements. The top-left element is

$$(M^T g M)_{11} = M_{11}^2 - M_{21}^2 - M_{31}^2 - M_{41}^2 \,. \tag{5.7}$$

Since

$$M_{11}^2 - M_{21}^2 - M_{31}^2 - M_{41}^2 = 1 \,, \tag{5.8}$$

we have the inequality

$$|M_{11}| \geq 1 \,. \tag{5.9}$$

From this, we can define the following four connected subsets of $O(1,3)$:
(1) $L^{\uparrow +}$: $M_{11} \geq 1$, $\det(M) = +1$;
(2) $L^{\uparrow -}$: $M_{11} \geq 1$, $\det(M) = -1$;
(3) $L^{\downarrow +}$: $M_{11} \leq -1$, $\det(M) = +1$;
(4) $L^{\downarrow -}$: $M_{11} \leq -1$, $\det(M) = -1$.

• The subset (1), $L^{\uparrow +}$, is a connected subgroup of $O(1,3)$, which is usually called the *restricted Lorentz group* $SO^+(1,3)$. The elements of $SO^+(1,3)$ are generated by the Lorentz boost in x, y, z directions and the 3d space rotations, where the latter form a $SO(3)$ subgroup of $SO^+(1,3)$.

• Define the parity transformation P =diag$\{1,-1,-1,-1\}$ and the time reversal T =diag$\{-1,1,1,1\}$, we can relate the elements in sets (2)(3)(4) with the elements in the set (1) as:

$$L^{\uparrow -} = P L^{\uparrow +} \tag{5.10}$$

$$L^{\downarrow +} = P T L^{\uparrow +} \tag{5.11}$$

$$L^{\downarrow -} = T L^{\uparrow +} \tag{5.12}$$

• The union of $L^{\uparrow +}$ and $L^{\downarrow +}$ forms the group $SO(1,3)$, which is also disconnected.

• Consider the restricted Lorentz group $SO^+(1,3)$, it is not simply connected (similar to $SO(3)$). The simply-connected matrix group $SL(2,\mathbb{C})$ is the double cover of $SO^+(1,3)$. In other words,

$$SO^+(1,3) \equiv SL(2,\mathbb{C})/\{\pm I\} \,. \tag{5.13}$$

$SL(2,\mathbb{C})$ is also called Spin$(1,3)$, because of its relation with the spinors. Now we discuss the surjective homomorphism $SL(2,\mathbb{C}) \to SO^+(1,3)$ explicitly.

We rewrite the $\mathbb{R}^{1,3}$ coordinates in the form of a 2×2 Hermitian matrix:

$$X = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix} = tI + \vec{x}\cdot\vec{\sigma} \,. \tag{5.14}$$

$\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector of Pauli matrices. Note that

$$\det(X) = t^2 - x^2 - y^2 - z^2 \tag{5.15}$$

is the invariant distance. Now we take a group element $S \in SL(2, \mathbb{C})$, which acts on X as

$$X' = SXS^\dagger \,. \tag{5.16}$$

This group action preserves $\det(X)$ and the Hermitian property of X ($X'^\dagger = X$), hence it corresponds to an isometry of $\mathbb{R}^{1,3}$. We also know that the kernel of $SL(2, \mathbb{C}) \to SO^+(1, 3)$ is $\pm I$, hence $SO^+(1, 3) \equiv SL(2, \mathbb{C})/\{\pm I\}$. We can write down the map $SL(2, \mathbb{C}) \to SO^+(1, 3)$ in the matrix form:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \to \frac{1}{2} \begin{pmatrix} a\bar{a} + b\bar{b} + c\bar{c} + d\bar{d} & a\bar{b} + \bar{a}b + \bar{c}d + c\bar{d} & i(\bar{a}b - a\bar{b} + \bar{c}d - c\bar{d}) & a\bar{a} - b\bar{b} + c\bar{c} - d\bar{d} \\ a\bar{c} + \bar{a}c + b\bar{d} + \bar{b}d & a\bar{d} + \bar{a}d + b\bar{c} + \bar{b}c & i(\bar{a}d - a\bar{d} + b\bar{c} - c\bar{b}) & a\bar{c} + \bar{a}c - b\bar{d} - \bar{b}d \\ i(a\bar{c} - \bar{a}c + b\bar{d} - \bar{b}d) & i(a\bar{d} - \bar{a}d + b\bar{c} - \bar{b}c) & \bar{a}d + a\bar{d} - b\bar{c} - c\bar{b} & i(a\bar{c} - \bar{a}c - b\bar{d} + \bar{b}d) \\ a\bar{a} + b\bar{b} - c\bar{c} - d\bar{d} & a\bar{b} + \bar{a}b - \bar{c}d - c\bar{d} & i(\bar{a}b - a\bar{b} + c\bar{d} - \bar{c}d) & a\bar{a} - b\bar{b} - c\bar{c} + d\bar{d} \end{pmatrix} . \tag{5.17}$$

5.1.2 Representations of the Lorentz group

Note that the associated real Lie algebra of the Lorentz group ($O(1,3)$, $SO(1,3)$, $SO^+(1,3)$, Spin(1,3)) is $\mathfrak{so}(1,3)$.

The generators of $\mathfrak{so}(1,3)$ takes the form of

$$J^{01} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, J^{02} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, J^{03} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$
$$J^{12} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, J^{23} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, J^{31} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \tag{5.18}$$

From these, we define a rank-2 anti-symmetric tensor $J^{\mu\nu} = -J^{\nu\mu}$ ($\mu, \nu = 0, 1, 2, 3$).

They satisfy commutation relations

$$[J^{\mu\nu}, J^{\rho\sigma}] = g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\nu\sigma} J^{\mu\rho} + g^{\mu\sigma} J^{\nu\rho} \,. \tag{5.19}$$

A generic element of $\mathfrak{so}(1,3)$ can be written as

$$\frac{1}{2}\omega_{\mu\nu} J^{\mu\nu} \,, \tag{5.20}$$

where $\omega_{\mu\nu} = -\omega_{\nu\mu}$ is an anti-symmetric 4×4 matrix which parametrize the $\mathfrak{so}(1,3)$ element.

After the complexification, we have $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.

To see this, we define the following linear combinations of generators:

$$\mathcal{C}^+ = \frac{1}{2}(J^{31} + iJ^{23} + iJ^{02} - J^{01}) \,,\ \mathcal{C}^- = \frac{1}{2}(-J^{31} + iJ^{23} - iJ^{02} - J^{01}) \,,\ \mathcal{C}^3 = -iJ^{12} + J^{03}$$
$$\mathcal{D}^+ = \frac{1}{2}(J^{31} + iJ^{23} - iJ^{02} + J^{01}) \,,\ \mathcal{D}^- = \frac{1}{2}(-J^{31} + iJ^{23} + iJ^{02} + J^{01}) \,,\ \mathcal{D}^3 = -iJ^{12} - J^{03} \,. \tag{5.21}$$

They satisfy the commutation relations of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$:

$$\begin{aligned} &[\mathcal{C}^3, \mathcal{C}^\pm] = \pm 2\mathcal{C}^\pm \,,\ [\mathcal{C}^+, \mathcal{C}^-] = \mathcal{C}^3 \\ &[\mathcal{D}^3, \mathcal{D}^\pm] = \pm 2\mathcal{D}^\pm \,,\ [\mathcal{D}^+, \mathcal{D}^-] = \mathcal{D}^3 \\ &[\mathcal{C}, \mathcal{D}] = 0 \,. \end{aligned} \tag{5.22}$$

We first study the representations of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}) \equiv A_1 \oplus A_1$. The irreps are $(\mathbf{m}, \mathbf{n})$. In physics literature, this representation is often denoted

as (s_1, s_2), where

$$s_1 = \frac{m-1}{2} \ , \ s_2 = \frac{n-1}{2} \ . \tag{5.23}$$

Then what is the "spin" of a physical state (such as a particle or field)? To answer this question, we need to consider the $\mathfrak{so}(3)$ subalgebra of $\mathfrak{so}(1,3)$, generated by J^{12}, J^{23}, J^{31}. In other words, this $\mathfrak{so}(3)_{\mathbb{C}}$ subalgebra has generators

$$\mathcal{C}^+ + \mathcal{D}^+ \ , \ \mathcal{C}^- + \mathcal{D}^- \ , \ \mathcal{C}^3 + \mathcal{D}^3 \ . \tag{5.24}$$

The branching rule of $\mathfrak{so}(1,3)$ representation (s_1, s_2) into irreps of $\mathfrak{so}(3)$ is (Weinberg Vol I section 5.6)

$$(s_1, s_2) \to (s_1 + s_2) \oplus (s_1 + s_2 - 1) \oplus \cdots \oplus |s_1 - s_2| \ . \tag{5.25}$$

These numbers which label the irreps of $\mathfrak{so}(3)$ are defined as the "spin" of the corresponding state. For a general representation (s_1, s_2), it may be decomposed into components with different spins. Nonetheless, some of these components are unphysical after imposing the equation of motion.

Now we prove the branching rule (5.25), which coincides with the tensor product decomposition of two irreps of the same A_1 Lie algebra. For the representation (s_1, s_2), we use its definition (tensor product representation ver. 1), and compute the form of each generators $\mathcal{C}$, $\mathcal{D}$:

$$\pi_{(s_1,s_2)}(\mathcal{C}^+) = \pi_{s_1}(X) \otimes I \ , \ \pi_{(s_1,s_2)}(\mathcal{C}^-) = \pi_{s_1}(Y) \otimes I \ , \ \pi_{(s_1,s_2)}(\mathcal{C}^3) = \pi_{s_1}(H) \otimes I \ , \tag{5.26}$$

$$\pi_{(s_1,s_2)}(\mathcal{D}^+) = I \otimes \pi_{s_2}(X) \ , \ \pi_{(s_1,s_2)}(\mathcal{D}^-) = I \otimes \pi_{s_2}(Y) \ , \ \pi_{(s_1,s_2)}(\mathcal{D}^3) = I \otimes \pi_{s_2}(H) \ . \tag{5.27}$$

(H, X, Y) are the standard generators of $A_1 = \mathfrak{sl}(2, \mathbb{C})$. Hence for the generators of the $\mathfrak{so}(3)_{\mathbb{C}}$ subalgebra (5.24), we have

$$\begin{aligned} \pi_{(s_1,s_2)}(\mathcal{C}^+ + \mathcal{D}^+) &= \pi_{s_1}(X) \otimes I + I \otimes \pi_{s_2}(X) \ , \\ \pi_{(s_1,s_2)}(\mathcal{C}^- + \mathcal{D}^-) &= \pi_{s_1}(Y) \otimes I + I \otimes \pi_{s_2}(Y) \ , \\ \pi_{(s_1,s_2)}(\mathcal{C}^3 + \mathcal{D}^3) &= \pi_{s_1}(H) \otimes I + I \otimes \pi_{s_2}(H) \ . \end{aligned} \tag{5.28}$$

The r.h.s. exactly takes the form of the tensor product representation for a single A_1! Hence we expect to see the same result (5.25).

We list a number of commonly used representations in theoretical physics:

• Scalar: $(s_1, s_2) = (0, 0)$, spin $s = 0$.

• Spinor: spin $s = \frac{1}{2}$. In this case, there can be different choice of representations:

(1) $(s_1, s_2) = (\frac{1}{2}, 0)$, called the left-handed Weyl spinor

(2) $(s_1, s_2) = (0, \frac{1}{2})$, called the right-handed Weyl spinor

(3) $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$, the Dirac spinor which is the linear combination of the left-handed Weyl spinor and the right-handed Weyl spinor

• Vector: $(s_1,s_2)=(\frac{1}{2},\frac{1}{2})$, spin $s=1$. As one can check, the representation $(\frac{1}{2},\frac{1}{2})=(\mathbf{2},\mathbf{2})$ is indeed 4-dimensional. Note that after the decomposition into $\mathfrak{so}(3)$ representations, the 4d vector representation can be decomposed into $1\oplus 0$ (in terms of spin). Nonetheless, only the spin-1 part (corresponding to the representation $\mathbf{3}$) is physical.

• Rarita-Schwinger field $(1,\frac{1}{2})\oplus(\frac{1}{2},1)$.

• Graviton (traceless symmetric tensor) $g_{\mu\nu}$: $(s_1,s_2)=(1,1)$.

• $(1,0)\oplus(0,1)$: rank-2 antisymmetric tensor field, e. g. $F_{\mu\nu}$. It corresponds to the adjoint representation of $A_1\oplus A_1$.

Now let us consider the corresponding group representations of $SO^+(1,3)$. There are two subtleties:

(1) Since the group $SO^+(1,3)$ is non-compact, it has no finite-dimensional faithful unitary representation. Hence the mentioned non-trivial representations are all non-unitary.

(2) The representations with $s=$ half integer are projective representations of $SO^+(1,3)$, rather than an actual representation. Namely, such a representation (Π,V) satisfies

$$\Pi(g_1)\Pi(g_2)=\pm\Pi(g_1g_2)\quad(g_1,g_2\in SO^+(1,3))\,. \tag{5.29}$$

(Π,V) is a representation of the double cover $SL(2,\mathbb{C})=$Spin$(1,3)$.

Let us show the Lorentz group action for a few finite-dimensional representations. Let us denote a generic element of the restricted Lorentz group $SO^+(1,3)$ by[3]

$$g=e^{\frac{1}{2}\omega_{\mu\nu}J^{\mu\nu}}\,. \tag{5.30}$$

$(\mu,\nu=0,1,2,3)$.

(1) Scalar field: corresponds to the trivial representation:

$$\phi'=1\cdot\phi\,. \tag{5.31}$$

(2) Vector field: corresponds to the standard representation, after the group transformation:

$$V'^{\mu}=\left(e^{\frac{1}{2}\omega_{\rho\sigma}J^{\rho\sigma}}\right)^{\mu}{}_{\nu}V^{\nu} \tag{5.32}$$

$J^{\rho\sigma}$ here are the 4×4 matrices.

(3) left-handed Weyl spinor (chiral spinor):

$$\psi_\alpha=\left(e^{\frac{1}{2}\omega_{\mu\nu}S_L^{\mu\nu}}\right)_{\alpha}{}^{\beta}\psi_\beta\,. \tag{5.33}$$

[3]One can prove that the exponential map from $\mathfrak{so}(1,3)$ to $SO^+(1,3)$ is surjective.

$S_L^{\mu\nu}$ are 2×2 matrices such that $S_L^{ij} = -\frac{i}{2}\epsilon^{ijk}\sigma^k$ and $S_L^{0i} = -\frac{1}{2}\sigma^i$ for $i,j = 1,2,3$. σ^k are the Pauli matrices. $S_L^{\mu\nu}$ is exactly the left-handed Weyl spinor representation of the Lorentz algebra $\mathfrak{so}(1,3)$.

Note that the spinor representation of $SO^+(1,3)$ is a projective representation. For example let us take a Lorentz algebra element

$$A = 2\pi J^{12}\,. \tag{5.34}$$

It corresponds to the Lorentz group element (in the standard representation)

$$g = e^A = I_4\,. \tag{5.35}$$

But in the spinor representation, we have

$$\pi(g) = e^{2\pi S_L^{12}} = -I_2\,. \tag{5.36}$$

Same arguments can be applied to the right-handed Weyl spinor and Dirac spinor as well.

For the double cover group $SL(2,\mathbb{C})$ =Spin$(1,3)$ group, the spinor representations are well defined representations, since $\pi(g)$ does not equal to the identity element in $SL(2,\mathbb{C})$. A common interpretation is: when we rotate a spinor by $2\pi = 360^\circ$ angle, it does not go back to itself. The spinor only goes back to itself when it is rotated by $4\pi = 720^\circ$ angle!

(4) right-handed Weyl spinor (chiral spinor):

$$\psi_{\dot\alpha} = \left(e^{\frac{1}{2}\omega_{\mu\nu}S_R^{\mu\nu}}\right)_{\dot\alpha}^{\ \dot\beta}\psi_{\dot\beta}\,. \tag{5.37}$$

$S_R^{\mu\nu}$ are 2×2 matrices such that $S_R^{ij} = -\frac{i}{2}\epsilon^{ijk}\sigma^k$ and $S_R^{0i} = \frac{1}{2}\sigma^i$ for $i,j = 1,2,3$. $S_R^{\mu\nu}$ is exactly the right-handed Weyl spinor representation of the Lorentz algebra $\mathfrak{so}(1,3)$.

(5) Dirac spinor:

$$\Psi_a = \left(e^{\frac{1}{2}\omega_{\mu\nu}S^{\mu\nu}}\right)_a^{\ b}\Psi_b\,. \tag{5.38}$$

$S^{\mu\nu} = \frac{1}{4}[\gamma^\mu,\gamma^\nu]$, where γ^μ are the Dirac matrices which satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}I\,. \tag{5.39}$$

Here we use the Weyl basis for Dirac matrices:

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}\ ,\ \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma_i & 0 \end{pmatrix}\,. \tag{5.40}$$

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix}\,. \tag{5.41}$$

Such that the projection from a 4-component Dirac spinor to a left-handed or right-handed spinor takes the form of

$$\psi_L = \frac{1-\gamma_5}{2}\psi = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix} \psi \,, \tag{5.42}$$

$$\psi_R = \frac{1+\gamma_5}{2}\psi = \begin{pmatrix} 0 & 0 \\ 0 & I_2 \end{pmatrix} \psi \,. \tag{5.43}$$

5.1.3 Representations of Poincaré group

In this section, we consider the Poincaré group which is the semi-direct product of $SO^+(1,3)$ with the space-time translation group $\mathbb{R}^4$:

$$ISO^+(1,3) \equiv \mathbb{R}^4 \rtimes SO^+(1,3)\,. \tag{5.44}$$

$\mathbb{R}^4$ is a normal subgroup of the Poincaré group. To show this, let us consider a general element of the translation group $\mathbb{R}^4 \subset ISO^+(1,3)$ of the form

$$g = \left(\begin{array}{c|c} I_{4\times 4} & a \\ \hline 0 & 1 \end{array}\right) \tag{5.45}$$

and a general element of $ISO^+(1,3)$:

$$h = \left(\begin{array}{c|c} M & b \\ \hline 0 & 1 \end{array}\right)\,. \tag{5.46}$$

We can check that

$$hgh^{-1} = \left(\begin{array}{c|c} M & b \\ \hline 0 & 1 \end{array}\right)\left(\begin{array}{c|c} I_{4\times 4} & a \\ \hline 0 & 1 \end{array}\right)\left(\begin{array}{c|c} M^{-1} & -M^{-1}b \\ \hline 0 & 1 \end{array}\right) = \left(\begin{array}{c|c} I_{4\times 4} & Ma \\ \hline 0 & 1 \end{array}\right)\,. \tag{5.47}$$

If we want to define the spinor representations, we can use the double cover of $ISO^+(1,3)$, which is $\mathbb{R}^4 \rtimes Spin(1,3)$. It has the same Lie algebra as $ISO^+(1,3)$.

The associated Lie algebra has generators $J^{\mu\nu} = -J^{\nu\mu}$ and P^μ. They satisfy the commutation relations

$$\begin{aligned} [J^{\mu\nu}, J^{\rho\sigma}] &= g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}\,, \\ [P^\mu, P^\nu] &= 0\,, \\ [P^\mu, J^{\nu\rho}] &= g^{\mu\nu}P^\rho - g^{\mu\rho}P^\nu\,. \end{aligned} \tag{5.48}$$

P^μ and $J^{\mu\nu}$ exactly corresponds to the 4d translation and angular-momentum operator (up to a factor i):

$$P^\mu = \partial^\mu\ ,\ J^{\mu\nu} = x^\mu\partial^\nu - x^\nu\partial^\mu\,. \tag{5.49}$$

In particular, the component $H = iP_0$ is identified as the Hamiltonian operator, $\vec{P} = (iP_1, iP_2, iP_3)$ is identified as the total momentum operator, and $\vec{J} = (iJ^{23}, iJ^{31}, iJ^{12})$ is identified as the angular momentum operator.

To label a representation, one looks at the eigenvalues of the Casimir operators, which are

$$P^2 = -P^\mu P_\mu \tag{5.50}$$

and

$$W^2 = W^\mu W_\mu \quad (W^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} J_{\nu\rho} P_\sigma)\,, \tag{5.51}$$

as well as the eigenvalue of the iP^μ operator.

In terms of the eigenvalue of P^2, we have the following two types of representations that are physically relevant:

1. $P^2 = E^2 - |\vec{p}|^2 = m^2$, which corresponds to a massive particle.

   We can perform a Lorentz transformation and bring iP^μ to the form $(m, \vec{0})$ in a particular reference frame.

   The $iP^\mu = (m, \vec{0})$ is unchanged by a spatial rotation, which forms an $SO(3)$ subgroup of the whole Poincaré group. The $SO(3)$ subgroup is called the *little group* in the massive case.

   The representation of Poincaré group is labeled by m and j, where j is the spin of the particle that also labels the representation under the $SO(3)$ little group, which can be a half-integer.

   In fact, the Casimir operator W^2 has eigenvalue $-m^2 j(j+1)$.

   Note that when the little group is $SO(3)$, the spinor representation is a projective representation. One can replace $SO(3)$ with its double cover $SU(2)$.

2. $P^2 = 0$, which corresponds to a massless particle.

   We can perform a Lorentz transformation and bring iP^μ to the form $(E, 0, 0, E)$.

   $(E, 0, 0, E)$ is unchanged by the spatial rotation in the (x, y)-plane, which forms an $SO(2)$ subgroup of the Poincaré group. $SO(2)$ is the massless little group[4] .

   The representation of the Poincaré group is labeled by $iP^\mu = (E, 0, 0, E)$ and the *helicity* h which labels the representation of $SO(2) \cong U(1)$ little group. h can be a half-integer.

   Examples:

   • A massless scalar can only have $h = 0$.

   • A massless Weyl spinor can have $h = \pm\frac{1}{2}$.

   • A photon can have a helicity $h = \pm 1$. The $\pm$ sign corresponds to two different possible polarizations of the photon.

[4]More precisely, the massless little group is $ISO(2)$, generated by J^{12}, $J^{23} + J^{02}$, $J^{31} - J^{01}$.

• A graviton can have helicity $h = \pm 2$. The $\pm$ sign also corresponds to two polarizations.

5.2 Applications in Particle Physics

Standard model particles We discuss a bit more about the application of Lie groups and Lie algebras in particle physics. We first list the standard model fermions as representations of the standard model gauge group $SU(3)_C \times SU(2)_{EW} \times U(1)_Y$:

$$\begin{aligned} Q: & \begin{pmatrix} u, c, t \\ d, s, b \end{pmatrix}_L (\mathbf{3}, \mathbf{2})_{\frac{1}{6}} \\ U: & (u, c, t)_R \ (\mathbf{3}, \mathbf{1})_{\frac{2}{3}} \\ D: & (d, s, b)_R \ (\mathbf{3}, \mathbf{1})_{-\frac{1}{3}} \\ L: & \begin{pmatrix} \nu_e, \nu_\mu, \nu_\tau \\ e, \mu, \tau \end{pmatrix}_L (\mathbf{1}, \mathbf{2})_{-\frac{1}{2}} \\ E: & (e, \mu, \tau)_R \ (\mathbf{1}, \mathbf{1})_{-1} \,. \end{aligned} \tag{5.52}$$

Note that we use the hypercharge $Y = Q_{em} - T_3$ as the $U(1)$ charge, where Q_{em} is the electromagnetic charge and T_3 is the third component of weak isospin. In the above list, Q is the left-handed quarks, where up type quarks u, c, t have $Q_{em} = \frac{2}{3}$ and $T_3 = \frac{1}{2}$, and down type quarks d, s, b have $Q_{em} = -\frac{1}{3}$ and $T_3 = -\frac{1}{2}$. U is the right-handed up type quarks with $Q_{em} = \frac{2}{3}$ and $T_3 = 0$. D is the right-handed down type quarks with $Q_{em} = -\frac{1}{3}$ and $T_3 = 0$. L is the left-handed leptons, where the neutrinos ν_e, ν_μ, ν_τ have $Q_{em} = 0$ and $T_3 = \frac{1}{2}$, and e, μ, τ has $Q_{em} = -1$ and $T_3 = -\frac{1}{2}$. Finally E is the right-handed e, μ, τ with $Q_{em} = -1$ and $T_3 = 0$.

Besides the standard model fermions, we also have the standard model gauge bosons (before the electroweak symmetry breaking):

$$\begin{aligned} g &: (\mathbf{8}, \mathbf{1})_0 \\ W^\pm, W^0 &: (\mathbf{1}, \mathbf{3})_0 \\ B &: (\mathbf{1}, \mathbf{1})_0 \,. \end{aligned} \tag{5.53}$$

Finally, there is the Higgs boson

$$H: \ (\mathbf{1}, \mathbf{2})_{\frac{1}{2}} \,. \tag{5.54}$$

The antiparticles live the conjugate (dual) representations, for instance $\overline{Q}$ lives in the representation $(\overline{\mathbf{3}}, \mathbf{2})_{-\frac{1}{6}}$.

Hence from the representations under $SU(3)_C$ and $SU(2)_{EW}$, we see that the quarks Q, U, D and the gluon g participate in the strong interaction. The left-handed quarks Q, left-handed leptons L, the W-bosons $W^\pm$, W^0 and the Higgs boson H participate in the weak interaction.

Standard model Lagrangian To construct the standard model Lagrangian, each term in the Lagrangian should be in the singlet (trivial representation) of $SU(3)_C \times SU(2)_{EW} \times U(1)_Y$. Hence for each term in the Lagrangian, the tensor product decomposition of representations should contain a trivial representation.

On the other hand, each term in the Lagrangian should also be a singlet under the Lorentz (Poincaré) algebra $A_1 \oplus A_1$. Such a condition would provide addition constraints. In particular, let us consider the irreducible spinor representations: left-handed Weyl spinor $\psi_L : (\frac{1}{2}, 0)$ and right-handed Weyl spinor $\psi_R : (0, \frac{1}{2})$.

The 4-component Dirac spinor can be written as $\psi = (\psi_L, \psi_R)$. Using it we can write down the Lorentz scalar

$$\bar{\psi}\psi = \psi^\dagger \gamma^0 \psi \,. \tag{5.55}$$

Since in the Weyl basis we have

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} , \tag{5.56}$$

actually

$$\bar{\psi}\psi = (\psi_L)^\dagger \psi_R + (\psi_R)^\dagger \psi_L = \bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L \,. \tag{5.57}$$

Hence the mass term for standard model fermion actually combines the left-handed and right-handed Weyl spinors. We can attempt to write down terms such as $\overline{Q}U$, $\overline{Q}D$ or $\overline{L}E$, however such terms are not a singlet under $SU(2)_{EW}$! Hence to construct actual gauge singlets, we should write down instead the following *Yukawa terms* with the Higgs boson:

$$\overline{HQ}U \ , \ H\overline{Q}D, H\overline{L}E \,. \tag{5.58}$$

For example, one can check that for $\overline{HQ}U$, the tensor product decomposition

$$(\mathbf{1}, \mathbf{2})_{-\frac{1}{2}} \otimes (\bar{\mathbf{3}}, \mathbf{2})_{-\frac{1}{6}} \otimes (\mathbf{3}, \mathbf{1})_{\frac{2}{3}} = (\mathbf{1}, \mathbf{1})_0 \oplus (\mathbf{1}, \mathbf{3})_0 \oplus (\mathbf{8}, \mathbf{1})_0 \oplus (\mathbf{8}, \mathbf{3})_0 \tag{5.59}$$

indeed contains a gauge singlet $(\mathbf{1}, \mathbf{1})_0$.

Besides these terms, we also have the kinetic terms for spinors which are also Lorentz scalars:

$$\bar{\psi}\gamma^\mu D_\mu \psi = \bar{\psi}_L \gamma^\mu D_\mu \psi_L + \bar{\psi}_R \gamma^\mu D_\mu \psi_R \,, \tag{5.60}$$

where $D_\mu = \partial_\mu + A_\mu$ is the covariant derivative. Here A_μ lives in the adjoint representation of $SU(3)_C \times SU(2)_{EW} \times U(1)_Y$, which includes all the gauge bosons g, $W^\pm$, W^0, B. One can check that for each standard model fermion

$QUDLE$, (5.60) always contain a gauge singlet $(\mathbf{1},\mathbf{1})_0$ after the tensor product decomposition.

We also have the kinetic terms for the gauge bosons that are gauge and Lorentz singlets

$$F_{\mu\nu}F^{\mu\nu}\ ,\ F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]\,. \tag{5.61}$$

Finally, we have the Higgs kinetic terms

$$(D^\mu H)^\dagger D_\mu H \tag{5.62}$$

and the Higgs potential

$$V(H^\dagger H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2 \tag{5.63}$$

which are all gauge and Lorentz singlets.

Electroweak symmetry breaking (EWSB) To go down to lower energy scales, an important physical process is the electroweak symmetry breaking, where we give a non-zero vev (vacuum expectation value) to the Higgs field[5].
,

$$\langle H\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}\,. \tag{5.64}$$

As a consequence, the electroweak gauge bosons acquire mass through the Higgs kinetic term $(D^\mu H)^\dagger D_\mu H$, and the electroweak gauge group $SU(2)_{EW}\times U(1)_Y$ is broken to the electromagnetic subgroup $U(1)_{em}$ (see e.g. Peskin Quantum Field Theory). The electric charge under $U(1)_{em}$ is expressed as

$$Q_{em} = T_3 + Y\,. \tag{5.65}$$

T_3 corresponds to the charge under the Cartan subalgebra $\mathfrak{u}(1)\subset\mathfrak{su}(2)_{EW}$. For each weight $\lambda = (\langle\lambda, H_\alpha\rangle)$ of a representation (π, V) of $\mathfrak{su}(2)_{EW}$, it equals to

$$T_3 = \frac{1}{2}\langle\lambda, H_\alpha\rangle\,. \tag{5.66}$$

For example, for the left-handed leptons in the fundamental representation $\mathbf{2}$ of $\mathfrak{su}(2)_{EW}$, the weight (1) corresponding to ν_e, ν_μ, ν_τ has $T_3 = \frac{1}{2}$, the weight (-1) corresponding to e, μ, τ has $T_3 = -\frac{1}{2}$.

[5]Strictly speaking this is not a good description since one attempts to give a non-zero vev to a non-gauge-invariant quantity.

Taking into account of the different gauge coupling constants of $SU(2)_{EW}$ and $U(1)_Y$, the W^0 and B vector fields are mixed into the γ photon and Z-boson by the *Weinberg angle* (see Peskin for details):

$$\begin{pmatrix} \gamma \\ Z \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B \\ W^0 \end{pmatrix} . \tag{5.67}$$

After we assign $\langle H \rangle = \frac{1}{\sqrt{2}}(0, v)^T$, the Yukawa terms would give non-zero mass to the standard model fermions.

Flavor physics Up to this step, we have not incorporated the difference between three different families of standard model fermions. Before the EWSB, we can treat (u, c, t), (d, s, b), $(\nu_e, \nu_\mu, \nu_\tau)$, (e, μ, τ) as 3d vectors under different $SU(3)$ flavor symmetry. However, after the EWSB, the Yukawa terms give the mass term for quarks

$$L = -Y^u_{ij} \bar{u}^i_L u^j_R - Y^d_{ij} \bar{d}^i_L d^j_R \tag{5.68}$$

where $u^i = (u, c, t)$ and $d^i = (d, s, b)$, with non-trivial mixing matrices Y^u, Y^d. As a consequence, the weak eigenstates of quarks are different from the mass eigenstates!

We can diagonalize

$$U_{u_L} Y^u U^\dagger_{u_R} = diag(y_u, y_c, y_t) \ , \ U_{d_L} Y^d U^\dagger_{d_R} = diag(y_d, y_s, y_b) \, . \tag{5.69}$$

The coupling between quarks and the charged W-boson is

$$L \sim (\bar{u}_L, \bar{c}_L, \bar{t}_L) U_{CKM} \gamma^\mu W^+_\mu \begin{pmatrix} d_L \\ s_L \\ b_L \end{pmatrix} , \tag{5.70}$$

where $U_{CKM} = U_{u_L} U^\dagger_{d_L}$ is called the *CKM matrix*.

Similarly we can repeat the discussion for leptons, and there is the analogous *PMNS matrix*. However, the nature of neutrinos are more mysterious, since there is no right-handed neutrino in the standard model. Right-handed neutrino in beyond standard model? ν_R has to be a complete singlet $(\mathbf{1}, \mathbf{1})_0$ under $SU(3)_C \times SU(2)_{EW} \times U(1)_Y$. It can be very heavy (seesaw mechanism).